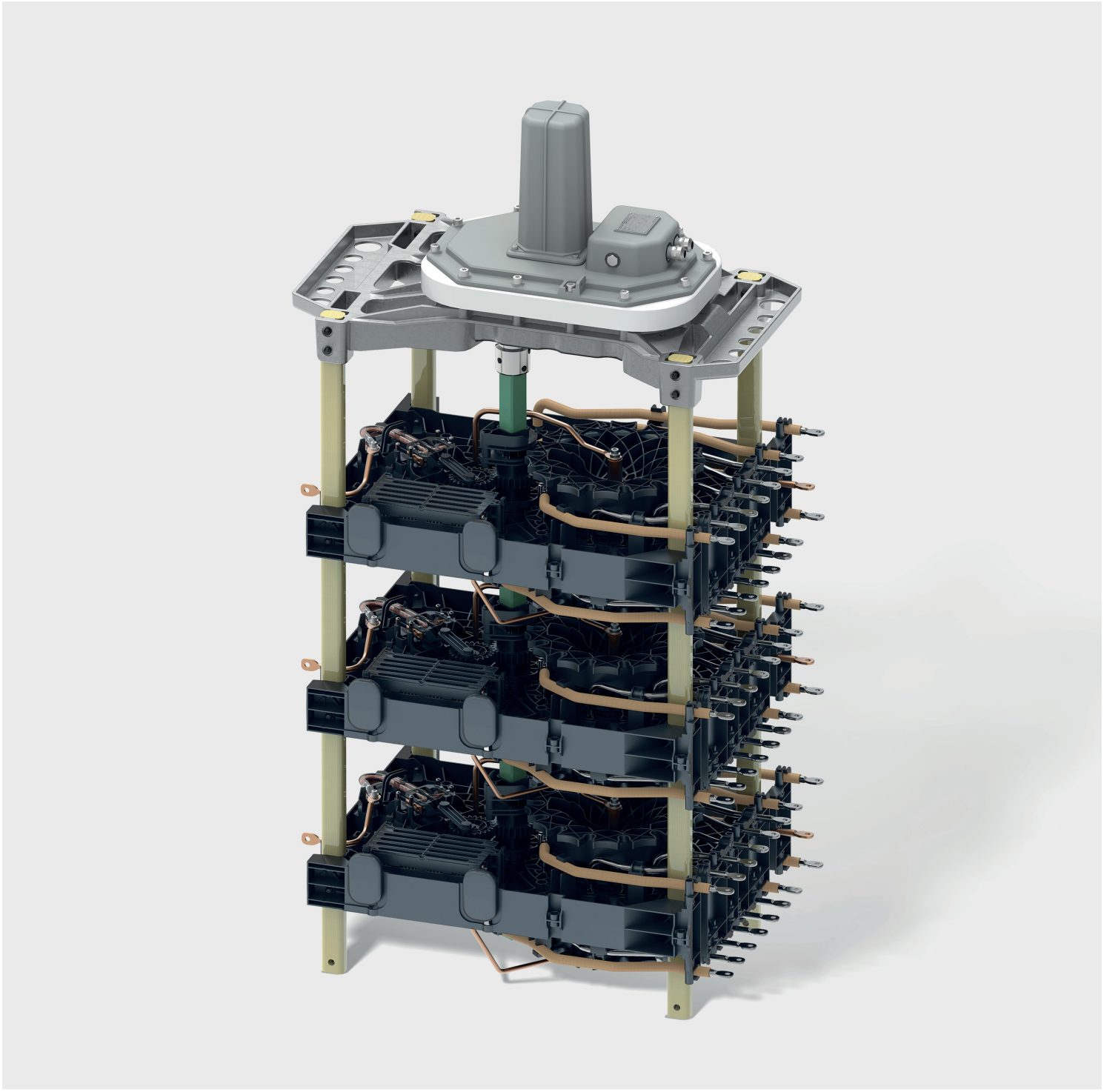
THE POWER BEHIND POWER.

Installation and commissioning instructions VACUTAP® VI. On-load tap-changer

12219500/01 EN



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Infringements will result in liability for compensation. All rights reserved in the event of the granting of patents, utility models or designs.
The product may have been altered since this document was published.
We reserve the right to change the technical data, design and scope of supply.
Generally the information provided and agreements made when processing the individual quotations and orders are binding.
The product is delivered in accordance with MR's technical specifications, which are based on information provided by the customer. The customer has a duty of care to ensure the compatibility of the specified product with the customer's planned scope of application.
The original operating instructions were written in German.

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1 Introduction

This technical file contains detailed descriptions of the safe and proper installation, connection, and commissioning of the product.

It also includes safety instructions and general information about the product.

Information about operation can be found in the operating instructions.

This technical file is intended solely for specially trained and authorized personnel.

1.1 Manufacturer

Maschinenfabrik Reinhausen GmbH
Falkensteinstraße 8
93059 Regensburg
Deutschland
+49 941 4090-0
sales@reinhausen.com
reinhausen.com

Customer portal

You can find additional information on this product in our customer portal at portal.reinhausen.com.



1.2 Completeness

This technical file is incomplete without the supporting documents.

The following documents are considered supporting documents:
- Unpacking instructions
- Supplement
- Routine test report
- Connection diagrams
- Dimensional drawings
- Order confirmation

1.3 Safekeeping

Keep this technical file and all supporting documents ready at hand and accessible for future use at all times.

1.4 Notation conventions

1.4.1 Hazard communication system

Warnings in this technical file are displayed as follows.

1.4.1.1 Warning relating to section

Warnings relating to sections refer to entire chapters or sections, sub-sections or several paragraphs within this technical file. Warnings relating to sections use the following format:

⚠ WARNING



Type of danger!
Source of the danger and outcome.
› Action
› Action

1.4.1.2 Embedded warning information

Embedded warnings refer to a particular part within a section. These warnings apply to smaller units of information than the warnings relating to sections. Embedded warnings use the following format:

⚠ DANGER! Instruction for avoiding a dangerous situation.

1.4.1.3 Signal words and pictograms

The following signal words are used:

Signal word	Definition
DANGER	Indicates a hazardous situation which, if not avoided, will result in death or serious injury.
WARNING	Indicates a hazardous situation which, if not avoided, could result in death or serious injury.
CAUTION	Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.
NOTICE	Indicates measures to be taken to prevent damage to property.

Table 1: Signal words in warning notices

Pictograms warn of dangers:

Pictogram	Definition
	Warning of a danger point
	Warning of dangerous electrical voltage
	Warning of combustible substances
	Warning of danger of tipping
	Warning of danger of crushing

Table 2: Pictograms used in warning notices

1.4.2 Information system

Information is designed to simplify and improve understanding of particular procedures. In this technical file it is laid out as follows:

Important information.

1.4.3 Instruction system

This technical file contains single-step and multi-step instructions.

Single-step instructions

Instructions which consist of only a single process step are structured as follows:

Aim of action
✓ Requirements (optional).
› Step 1 of 1.
  » Result of step (optional).
» Result of action (optional).

Multi-step instructions

Instructions which consist of several process steps are structured as follows:

Aim of action

✓ Requirements (optional).

1. Step 1.
   » Result of step (optional).
2. Step 2.
   » Result of step (optional).

» Result of action (optional).

2 Safety

- Read this technical file through to familiarize yourself with the product.
- This technical file is a part of the product.
- Read and observe the safety instructions provided in this chapter.
- Read and observe the warnings in this technical file in order to avoid function-related dangers.
- The product is manufactured on the basis of state-of-the-art technology. Nevertheless, risks to life and limb for the user or impairment of the product and other material assets due to the function may arise in the event of improper use.

2.1 Appropriate use

The product is an on-load tap-changer and adjusts the transmission ratio of transformers without interrupting the load flow. The product is designed solely for use in electrical energy systems and facilities. If used as intended and in compliance with the requirements and conditions specified in this technical file as well as the warning notices in this technical file and attached to the product, then the product does not present any danger to people, property or the environment. This applies throughout the service life of the product, from delivery, installation and operation to removal and disposal.

The following is considered intended use:

- Never actuate the on-load tap-changer with the hand crank if the transformer is energized. Only actuate the on-load tap-changer with the hand crank in emergency operation [▶Section 7.2, Page 73]. De-energize the transformer before using the hand crank.
- Only use the product with the transformer specified in the order.
- The serial numbers of the on-load tap-changer and control cabinet must match if the on-load tap-changer and control cabinet are supplied as a set for one order.
- You will find the standard valid for the product and the year of issue on the nameplate.
- Operate the product in accordance with this provided technical file, the agreed delivery conditions and the technical data.
- Ensure that all necessary work is performed by qualified personnel only.
- Only use the equipment and special tools included in the scope of delivery for the intended purpose and in accordance with the specifications of this technical document.
- The on-load tap-changer is not intended to be used with an oil filter unit.

Permitted electrical operating conditions

In addition to the design data in accordance with the order confirmation, observe the following limits for the through-current and the step voltage:

In the standard version, the on-load tap-changer is designed for sinusoidal 50/60 Hz alternating current with a curve form symmetrical to the zero axis and can switch 2 times the rated through-current I_r at its rated step voltage U_{ir}.

Exceeding the rated step voltage U_{ir} by up to 10% for a short period is permitted as long as the rated step capacity P_{StN} permissible for this step voltage is not exceeded.

2.2 Inappropriate use

Use is considered inappropriate if the product is used in a way other than as described in the "Appropriate use" section. In addition, observe the following:

Prohibited electrical operating conditions

All operating conditions that do not comply with the design data in accordance with the order confirmation are prohibited.

Prohibited operating conditions may arise due to short circuits as well as due to inrush current impulses when energizing transformers or other electrical machines. This applies to the affected transformer itself just as it does to transformers electrically connected in parallel or serially or other electrical machines.

Higher voltages may occur due to transformer overexcitation following load shedding, for example.

Operations outside of the permitted operating conditions can lead to injury to persons and damage to the product.

- Prevent any such operations outside of the permitted operating conditions by taking suitable measures.

2.3 Fundamental safety instructions

To prevent accidents, malfunctions and damage as well as unacceptable adverse effects on the environment, those responsible for transport, installation, operation, maintenance and disposal of the product or parts of the product must ensure the following:

Personal protective equipment

Loosely worn or unsuitable clothing increases the danger of becoming trapped or caught up in rotating parts and the danger of getting caught on protruding parts. This poses a danger to life and limb.

- Wear appropriate personal protective equipment such as a helmet, work gloves, etc. for the respective activity.
- Never wear damaged personal protective equipment.
- Never wear rings, necklaces, or other jewelry.
- If you have long hair, wear a hairnet.

Work area

Untidy and poorly lit work areas can lead to accidents.

- Keep the work area clean and tidy.
- Make sure that the work area is well lit.
- Observe the applicable laws for accident prevention in the relevant country.

Working during operation

The product may only be operated in a sound, operational condition. Otherwise it poses a danger to life and limb.

- Regularly check the operational reliability of safety equipment.
- Comply with the inspection work, maintenance work and maintenance intervals described in this technical file.

Explosion protection

Highly flammable or explosive gases, vapors and dusts can cause serious explosions and fire. This increases the danger to life and limb.

- Do not install, operate or perform maintenance work on the product in areas where a risk of explosion is present.

Safety markings

Warning signs and safety information plates are safety markings on the product. They are an important aspect of the safety concept.

- Observe all safety markings on the product.
- Make sure all safety markings on the product remain intact and legible.
- Replace safety markings that are damaged or missing.

Ambient conditions

To ensure reliable and safe operation, the product must only be operated under the ambient conditions specified in the technical data.

- Observe the specified operating conditions and requirements for the installation location.

Auxiliary materials and operating materials

Auxiliary materials and operating materials not approved by the manufacturer can lead to personal injury, damage to property and malfunctions of the product.

- Use insulating fluids that meet the requirements in accordance with IEC 60296.
- Only use conductive and grounded hoses, pipes, and pump equipment that are approved for flammable liquids.
- Only use lubricants and auxiliary materials approved by the manufacturer.
- Contact the manufacturer.

Modifications and conversions

Unauthorized or inappropriate changes to the product may lead to personal injury, material damage and operational faults.

- Only modify the product after consultation with Maschinenfabrik Reinhausen GmbH.

Spare parts

Spare parts not approved by Maschinenfabrik Reinhausen GmbH may lead to physical injury, damage to the product and malfunctions.

- Only use spare parts that have been approved by Maschinenfabrik Reinhausen GmbH.
- Contact Maschinenfabrik Reinhausen GmbH.

2.4 Personnel qualification

The person responsible for assembly, commissioning, operation, maintenance and inspection must ensure that the personnel are sufficiently qualified.

Electrically skilled person

The electrically skilled person has a technical qualification and therefore has the required knowledge and experience, and is also conversant with the applicable standards and regulations. The electrically skilled person is also proficient in the following:

- Can identify potential dangers independently and is able to avoid them.
- Is able to perform work on electrical systems.
- Is specially trained for the working environment in which (s)he works.

– Must satisfy the requirements of the applicable statutory regulations for accident prevention.

Electrically trained persons

An electrically trained person receives instruction and guidance from an electrically skilled person in relation to the tasks undertaken and the potential dangers in the event of inappropriate handling as well as the protective devices and safety measures. The electrically trained person works exclusively under the guidance and supervision of an electrically skilled person.

Operator

The operator uses and operates the product in line with this technical file. The operating company provides the operator with instruction and training on the specific tasks and the associated potential dangers arising from improper handling.

Technical Service

We strongly recommend having maintenance, repairs and retrofitting carried out by our Technical Service department. This ensures that all work is performed correctly. If maintenance is not carried out by our Technical Service department, please ensure that the personnel who carry out the maintenance are trained and authorized by Maschinenfabrik Reinhausen GmbH to carry out the work.

Authorized personnel

Authorized personnel are trained by Maschinenfabrik Reinhausen GmbH to carry out special maintenance.

2.5 Personal protective equipment

Personal protective equipment must be worn during work to minimize risks to health.

– Always wear the personal protective equipment required for the job at hand.
– Never wear damaged personal protective equipment.
– Observe information about personal protective equipment provided in the work area.

Protective clothing	Close-fitting work clothing with a low tearing strength, with tight sleeves and with no protruding parts. It mainly serves to protect the wearer against being caught by moving machine parts.
Safety shoes	To protect against falling heavy objects and slipping on slippery surfaces.
Safety glasses	To protect the eyes from flying parts and splashing liquids.
Visor	To protect the face from flying parts and splashing liquids or other dangerous substances.
Hard hat	To protect against falling and flying parts and materials.
Hearing protection	To protect against hearing damage.
Protective gloves	To protect against mechanical, thermal, and electrical hazards.

Table 3: Personal protective equipment

3 Product description

3.1 Scope of delivery

You can choose between ETOS® ID and Tap Changer Control for the control cabinet.

The product is packaged with protection against moisture and is usually delivered as follows:
- On-load tap-changer with installed sealing module and guard plate
- Motor cover with control cables
- ETOS® ID or Tap Changer Control control cabinet
- Technical documents

Please refer to the delivery slip for full details of the scope of delivery.
Note the following information:
- Check the shipment for completeness on the basis of the shipping documents.
- Store the parts in a dry place until installation
- The product must remain in its airtight, protective wrapping and may only be removed immediately before installation

You will find more information in the "Packaging, transport, and storage" chapter.

3.2 On-load tap-changer

3.2.1 Function description

On-load tap-changers are used to adjust the transmission ratio of transformers without interrupting the load flow. This makes it possible to compensate for aspects such as fluctuations in voltage occurring in the power transmission grid. For this purpose, on-load tap-changers are fitted in transformers and connected to the active part of the transformer.

A motor-drive unit which receives a control impulse from an electronic regulator changes the operating position of the on-load tap-changer, which adapts the transformer's transmission ratio to the prevailing operating requirements.

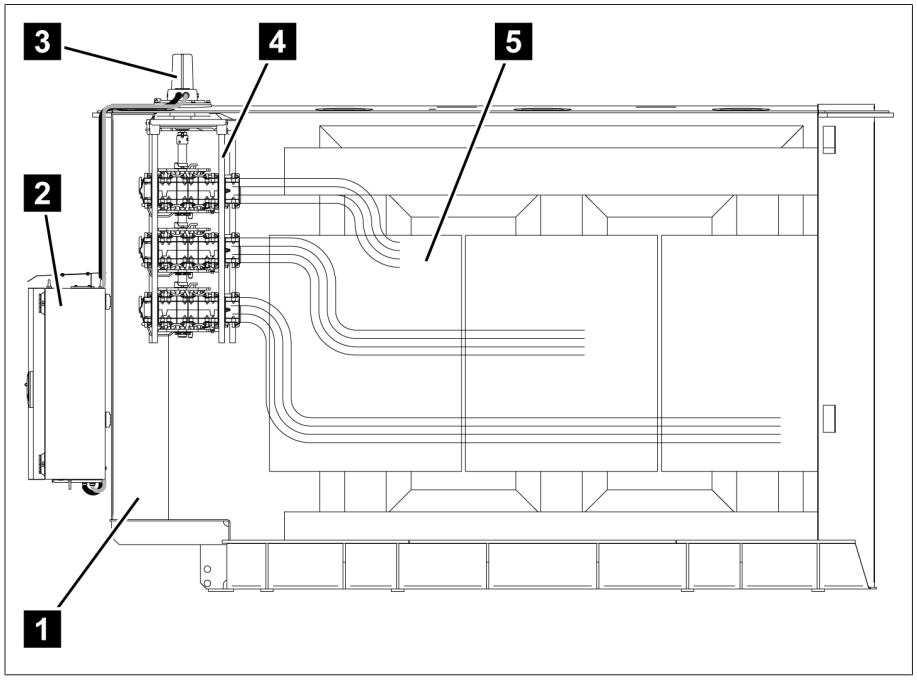


Figure 1: System overview of on-load tap-changer, drive system and transformer

1	Transformer tank	4	On-load tap-changer
2	Control cabinet	5	Active part
3	Motor		

3.2.2 Setup/models

The following drawings show the main components of the on-load tap-changer.

You will find a detailed drawing of the on-load tap-changer in the "Drawings [▶Section 10, Page 88]" chapter.

On-load tap-changer

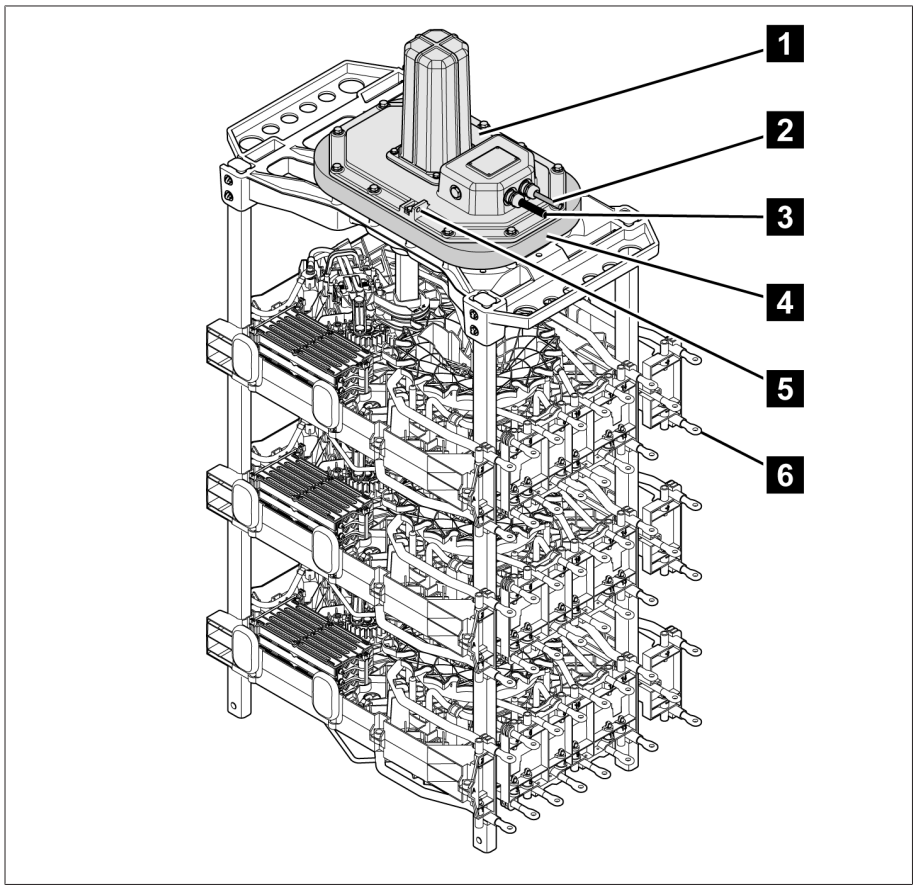


Figure 2: On-load tap-changer

1	Motor cover	2	Motor cable
3	Sensor cable	4	Mounting flange
5	Security lug	6	Connection contacts

On-load tap-changer with AutoSync function (can be ordered as an option)

Homing is not necessary with on-load tap-changers that have the AutoSync function. The control system can automatically detect the position of the on-load tap-changer via the built-in sensor.

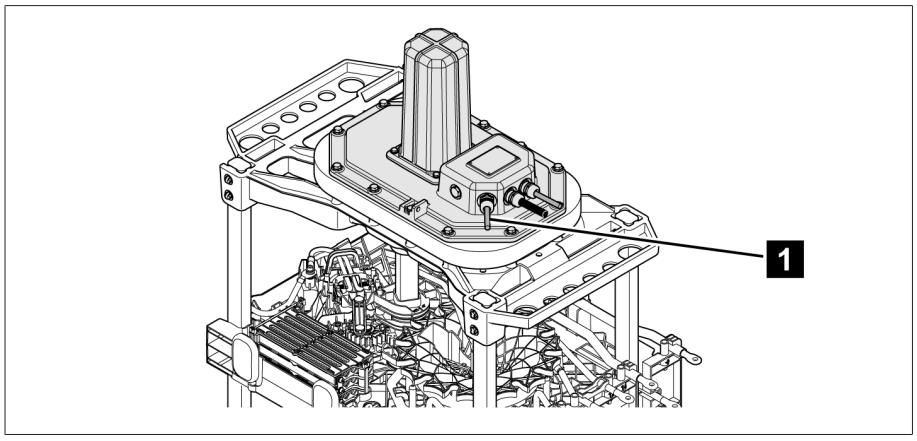


Figure 3: Overview with AutoSync function

1	Sensor cable, AutoSync function

3.2.3 Safety markings

The name plate with serial number is located on the motor cover.

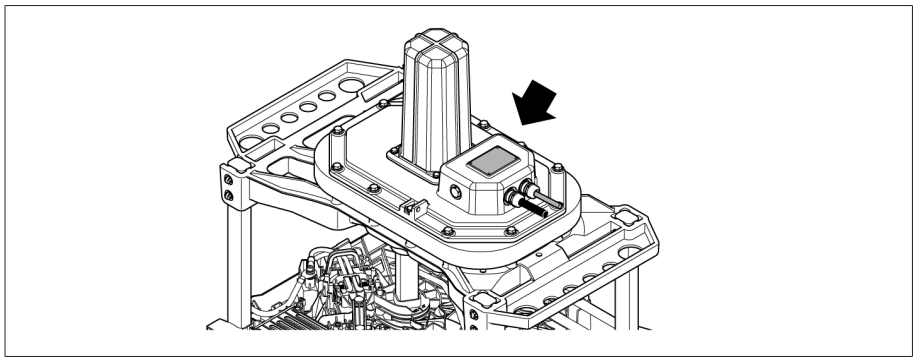


Figure 4: Name plate

The following safety markings are used on the product:

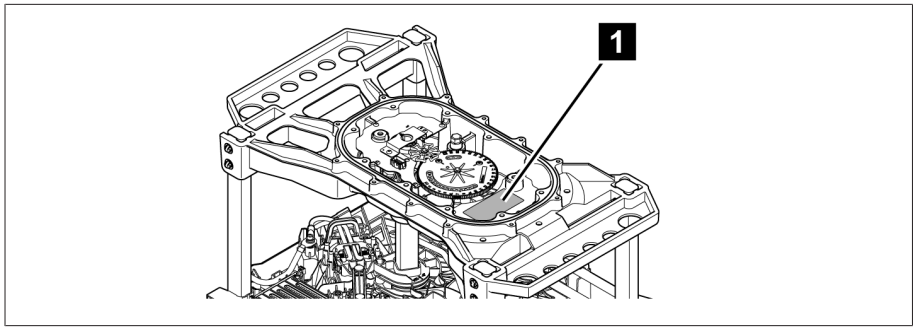


Figure 5: Sealing module warning

1	Warning

3.2.4 Protective devices

The on-load tap-changer is situated in the oil room of the transformer without it's own oil compartment.

This eliminates the classic on-load tap-changer protection and the transformer protective devices take over the protection of the on-load tap-changer. The transformer manufacturer is responsible for the selection and dimensioning of the transformer protective devices. In this respect, it must be ensured that these protective devices are also effective in the event of on-load tap-changer errors and that no false tripping occurs.

To avoid false tripping, the response value must be greater than 0.65 m/s when using a Buchholz relay.

3.2.4.1 Security lug

The motor cover of the on-load tap-changer is fitted with a security lug for securing the motor cover with a padlock to prevent unauthorized access.

4 Packaging, transport and storage

4.1 Packaging

The products are sometimes supplied with sealed packaging and sometimes in a dry state, depending on requirements.

Sealed packaging surrounds the packaged goods with plastic foil on all sides.

Products that have also been dried are identified by a yellow label on the sealed packaging. In the dry state, delivery is also possible in a transport container.

The information in the following sections should be applied as appropriate.

4.1.1 Suitability

NOTICE

Property damage due to incorrectly stacked crates!

Stacking the crates incorrectly can lead to damage to the packaged goods.

- The outer marking on the packaging states if, for example, the on-load tap-changer or selector has been packed upright. Never stack these crates.
- General rule: Do not stack crates above a height of 1.5 m.
- For other crates: Only stack up to 2 equally sized crates on top of one another.

The packaging is suitable to ensure undamaged and fully functional means of transportation in compliance with local transportation laws and regulations.

The packaged goods are packed in a sturdy crate. This crate ensures that, when in the intended transportation position, the packaged goods are stabilized to prevent impermissible changes in position, and that none of the parts touch the loading surface of the means of transport or touch the ground after unloading.

Sealed packaging surrounds the packaged goods with plastic foil on all sides. The packaged goods are protected from humidity using a desiccant. The plastic foil was bonded after the desiccant is added.

4.1.2 Markings

The packaging bears a signature with instructions for safe transport and correct storage. The following symbols apply to the shipment of non-hazardous goods. Adherence to these symbols is mandatory.

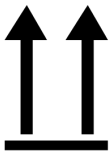
				
Protect against moisture	Top	Fragile	Attach lifting gear here	Center of mass

Table 4: Shipping pictograms

4.2 Transportation, receipt and handling of shipments

⚠ WARNING



Danger of death or severe injury!

Danger of death or serious injuries due to tipping or falling load.

- Only transport the crate when closed.
- Do not remove the securing material used in the crate during transport.
- If the product is delivered on a pallet, secure it sufficiently.
- Only trained and authorized persons may select the sling gear and secure the load.
- Do not walk under the suspended load.
- Use means of transport and lifting gear with a sufficient carrying capacity in accordance with the weight stated on the delivery slip.

In addition to oscillation stress, jolts must also be expected during transportation. In order to prevent possible damage, avoid dropping, tipping, knocking over and colliding with the product.

If a crate tips over, falls from a certain height (e.g. when slings tear) or experiences an unbroken fall, damage must be expected regardless of the weight.

Every delivered shipment must be checked for the following by the recipient before acceptance (acknowledgment of receipt):

- Completeness based on the delivery slip
- External damage of any type

The checks must take place after unloading when the crate or transport container can be accessed from all sides.

Visible damage

If external transport damage is found upon receipt of the shipment, proceed as follows:

- Immediately record the identified transport damage in the shipping documents and have this countersigned by the carrier.
- In the event of severe damage, total loss or high damage costs, immediately notify the manufacturer and the relevant insurance company.
- After identifying damage, do not modify the condition of the shipment further and retain the packaging material until an inspection decision has been made by the transport company or the insurance company.
- Record the details of the damage immediately on site together with the carrier involved. This is essential for any claim for damages.

- Photograph damage to packaging and packaged goods. This also applies to signs of corrosion on the packaged goods due to moisture inside the packaging (rain, snow, condensation).
- *NOTICE!* Damage to packaged goods due to damaged sealed packaging. If the product is delivered in sealed packaging, check the sealed packaging immediately. If the sealed packaging is damaged, do not under any circumstances install or commission the packaged goods. Either re-dry the dried packaged goods as per the operating instructions, or contact the manufacturer to agree on how to proceed.
- Identify the damaged parts.

Hidden damage

When damages are not determined until unpacking after receipt of the shipment (hidden damage), proceed as follows:

- Make the party responsible for the damage liable as soon as possible by telephone and in writing, and prepare a damage report.
- Observe the time periods applicable to such actions in the respective country. Inquire about these in good time.

With hidden damage, it is very hard to make the transportation company (or other responsible party) liable. Any insurance claims for such damages can only be successful if relevant provisions are expressly included in the insurance terms and conditions.

4.3 Storage of shipments

Packaged goods dried by Maschinenfabrik Reinhausen

Upon receipt of the shipment, immediately remove the packaged goods dried by Maschinenfabrik Reinhausen from the sealed packaging and store air-tight in dry insulating fluid until used if the packaged goods were not supplied in insulating fluid.

Non-dried packaged goods

Non-dried packaged goods but with a functional sealed packaging can be stored outdoors when the following conditions are complied with.

When selecting and setting up the storage location, ensure the following:

- Protect stored goods against moisture (flooding, water from melting snow and ice), dirt, pests such as rats, mice, termites and so on, and against unauthorized access.
- Store the crates on timber beams and planks as a protection against rising damp and for better ventilation.
- Ensure sufficient carrying capacity of the ground.
- Keep entrance paths free.
- Check stored goods at regular intervals. Also take appropriate action after storms, heavy rain or snow and so on.

Protect the packaging foil from direct sunlight so that it does not disintegrate under the influence of UV rays, which would cause the packaging to lose its sealing function.

If the product is installed more than 6 months after delivery, suitable measures must be taken without delay. The following measures can be used:

- Correctly regenerate the drying agent and restore the sealed packaging.
- Unpack the packed goods and store in a suitable storage space (well ventilated, as dust-free as possible, humidity < 50% where possible).

4.4 Unpacking shipments and checking for transportation damages

- *NOTICE!* Damage to packaged goods due to ineffectively sealed packaging. Transport the packaged crate to the place where the packaged goods are to be installed. Do not open the sealed packaging until just before installation.
- ⚠ WARNING! When unpacking, check the condition of the packaged goods.Place the packaged goods in an upright crate and protect them from tipping out. Otherwise, the packaged goods may become damaged and serious injuries may result.
- Check the completeness of the accessories kit based on the delivery slip.

Attachment points for on-load tap-changer lifting gear

⚠ WARNING



Danger of death and damage to property!

Danger of death and damage to property due to tipping or falling load!

- Only trained and authorized persons may select the sling gear and secure the load.
- Do not walk under the suspended load.
- Use means of transport and lifting gear with a sufficient carrying capacity in accordance with the weight stated in the Technical data section.

- Hang all 4, but at least 2, diagonally opposite lifting straps into the crane harness. Observe the weight information in the on-load tap-changer characteristics [▶Section 9.1, Page 79] in the "Technical data" section.

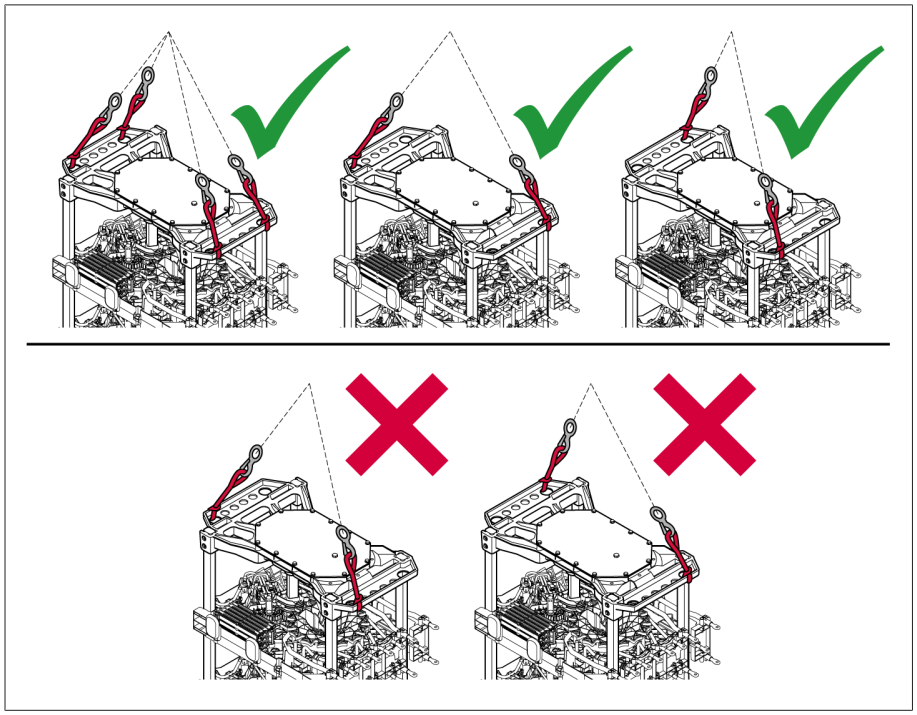


Figure 6: On-load tap-changer lifting points

- ⚠ WARNING! Serious injuries due to the on-load tap-changer tipping. Secure the on-load tap-changer against tipping until the on-load tap-changer has been fully installed in the transformer.

5 Mounting

The on-load tap-changer is suitable for bell-type tank installation and normal installation. Please contact the Maschinenfabrik Reinhausen GmbH Technical Service department if the intended installation deviates from the installation described in this chapter.

For reasons of clarity, the following chapter only shows illustrations of the installation of the on-load tap-changer without AutoSync function.

5.1 Preparatory work

Perform the work stated below before installing the on-load tap-changer in the transformer.

5.1.1 Fitting mounting flange on transformer cover

A mounting flange is required for mounting the on-load tap-changer on the transformer cover. This can be supplied as an option or can be produced by the customer. Mounting flanges made by the customer must comply with the installation drawings in the appendix.

1. *NOTICE!* Position the mounting flange so that it is central to the transformer cover. If mounted on top of the transformer cover, it may collide with the sealing flange.

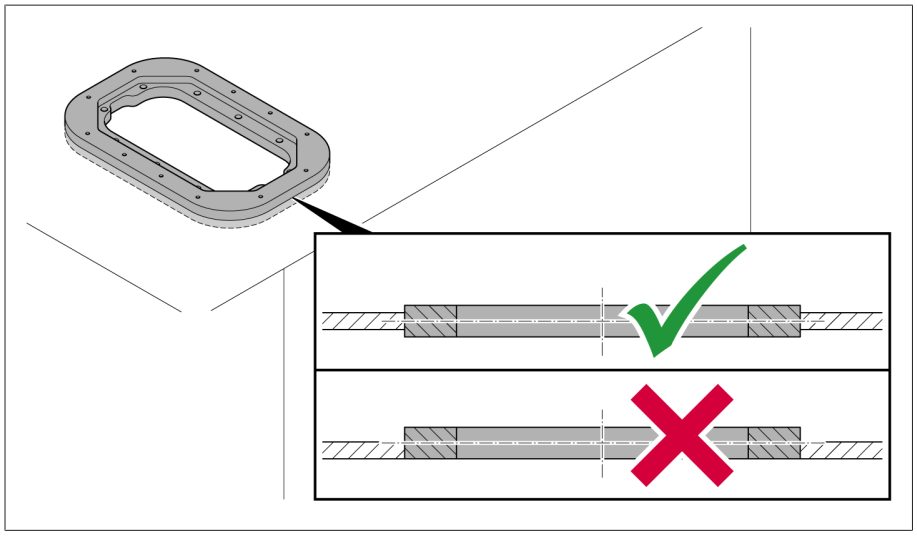


Figure 7: Correct mounting flange position

2. NOTICE! Fit the mounting flange on the transformer cover (pressure-tight). Ensure that the mounting flange is level so that the tap changer does not exceed an inclined position of 1° when installed. Ensure that the sealing surfaces are free of damage.

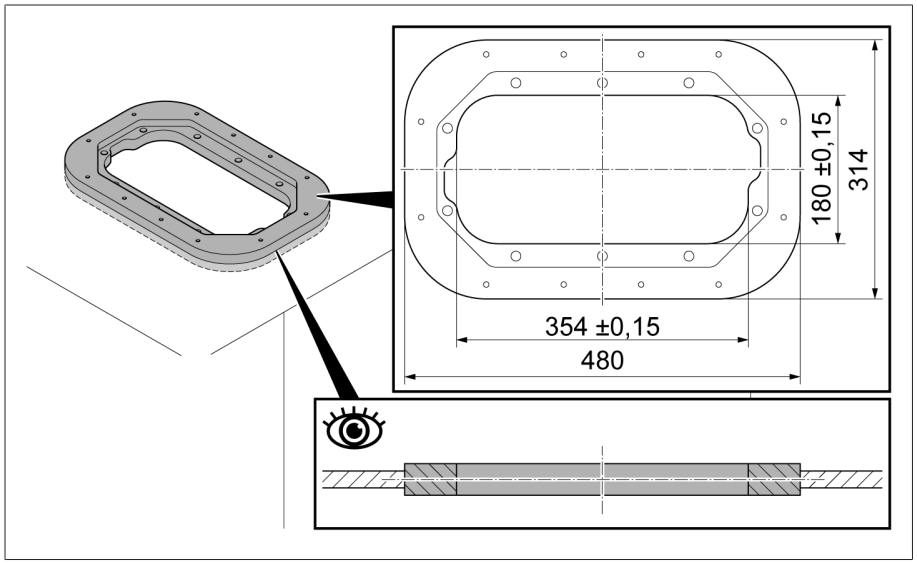


Figure 8: Mounting flange dimensions

⚠ CAUTION! A mounting flange that was not designed in accordance with MR specifications can lead to damage to the on-load tap-changer during installation.

The geometric specifications listed in the dimensional drawing [▶10162405] must be strictly adhered to.

> The mounting flange supplied by Maschinenfabrik Reinhausen is equipped with two lifting eye bolts for better handling. Remove the lifting eye bolts from the mounting flange after welding.

5.1.2 Access opening

> To ensure that the on-load tap-changer can be accessed for inspections, an access opening on the front of the transformer is recommended. The minimum dimensions depend on the version of the on-load tap-changer and the dimensional drawing [▶10162434].

5.2 Installing the on-load tap-changer in the transformer (standard version)

5.2.1 Fastening on-load tap-changer to transformer cover

1. ⚠ CAUTION! An unstably positioned on-load tap-changer may tip, resulting in serious injuries or property damage. Place the on-load tap-changer on a level surface and secure it against tipping.
2. Remove the red-colored packaging material and the transport material from the on-load tap-changer.

3. Unscrew the guard plate screws and remove the guard plate from the sealing module. Store the screws, washers and guard plate safely. These parts will be needed again in a later mounting step. *NOTICE!* In order to prevent soiling and damage to the mechanics, do not leave the sealing module open for long periods of time.

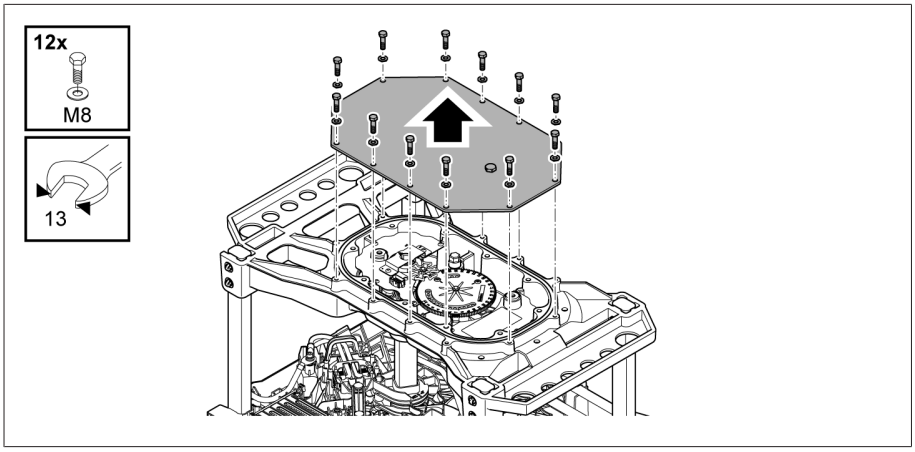


Figure 9: Removing the guard plate

4. *NOTICE!* Incorrect installation can lead to damage to the on-load tap-changer. Only install the on-load tap-changer using the original MR mounting device and lifting eye bolts. Place the mounting device on the sealing module on the studs provided for this purpose. Screw the mounting device to the sealing module hand-tight using two lifting eye bolts. Ensure that the mounting device is level and that the lifting eye bolts are fully screwed in.

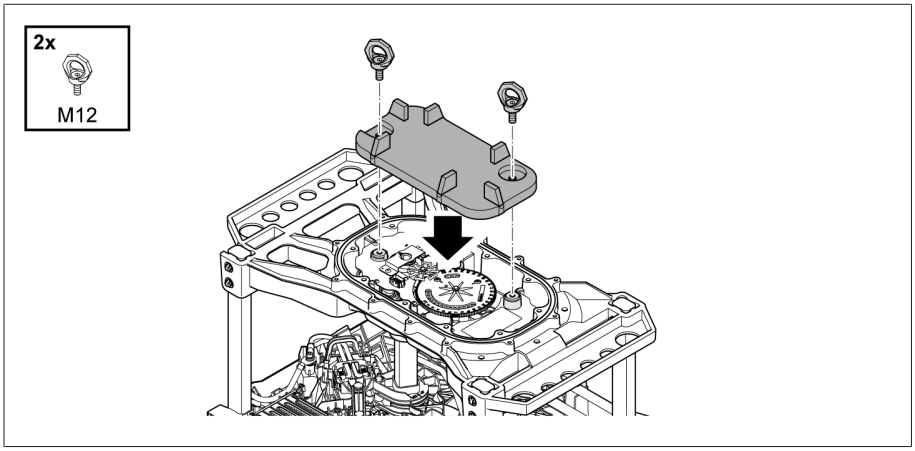


Figure 10: Mounting device on the sealing module

5. *NOTICE!* Unsuitable gaskets may lead to oil escaping, and therefore to damage to the on-load tap-changer. Only use the original o-ring gasket supplied by Maschinenfabrik Reinhausen. Clean the sealing surfaces on the mounting flange and sealing module of the on-load tap-changer.

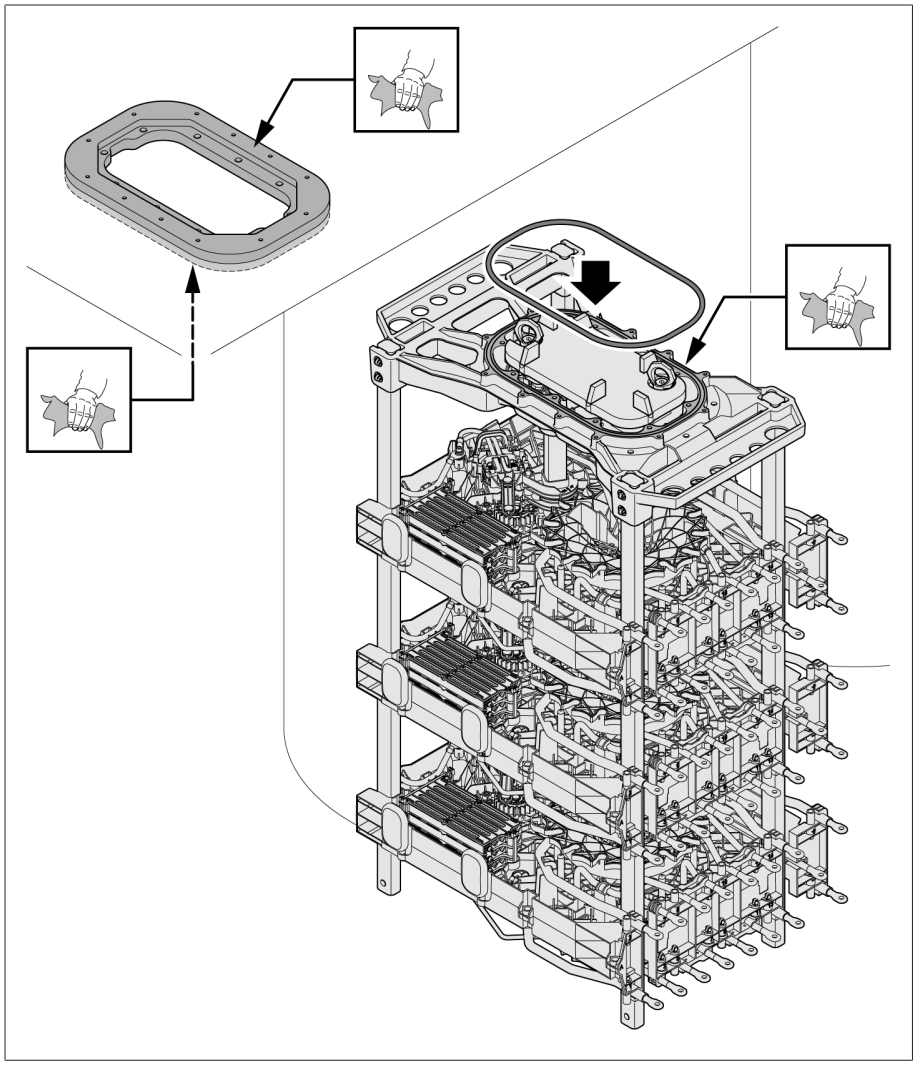


Figure 11: Cleaning the sealing surfaces

6. Ensure that the key adapter on the lifting eye bolt is aligned inwards when screwed in to avoid jamming when lifting.

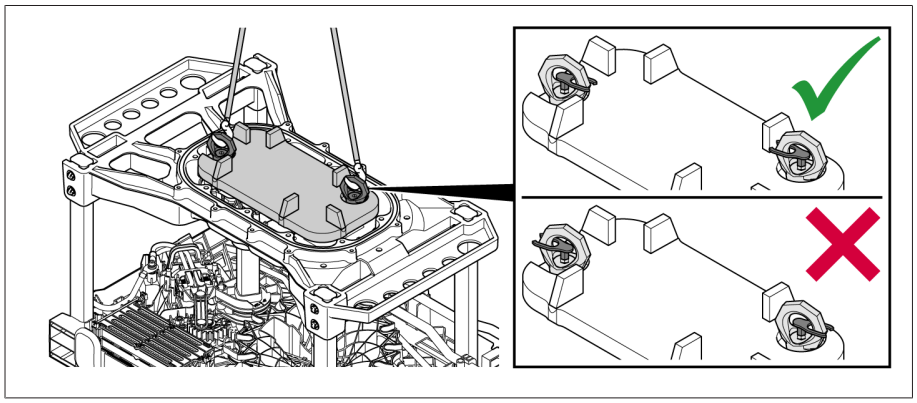


Figure 12: Aligning the key adapters

7. Position the on-load tap-changer under the mounting flange in the transformer cover and align it approximately. Observe the specified constructional installation position.

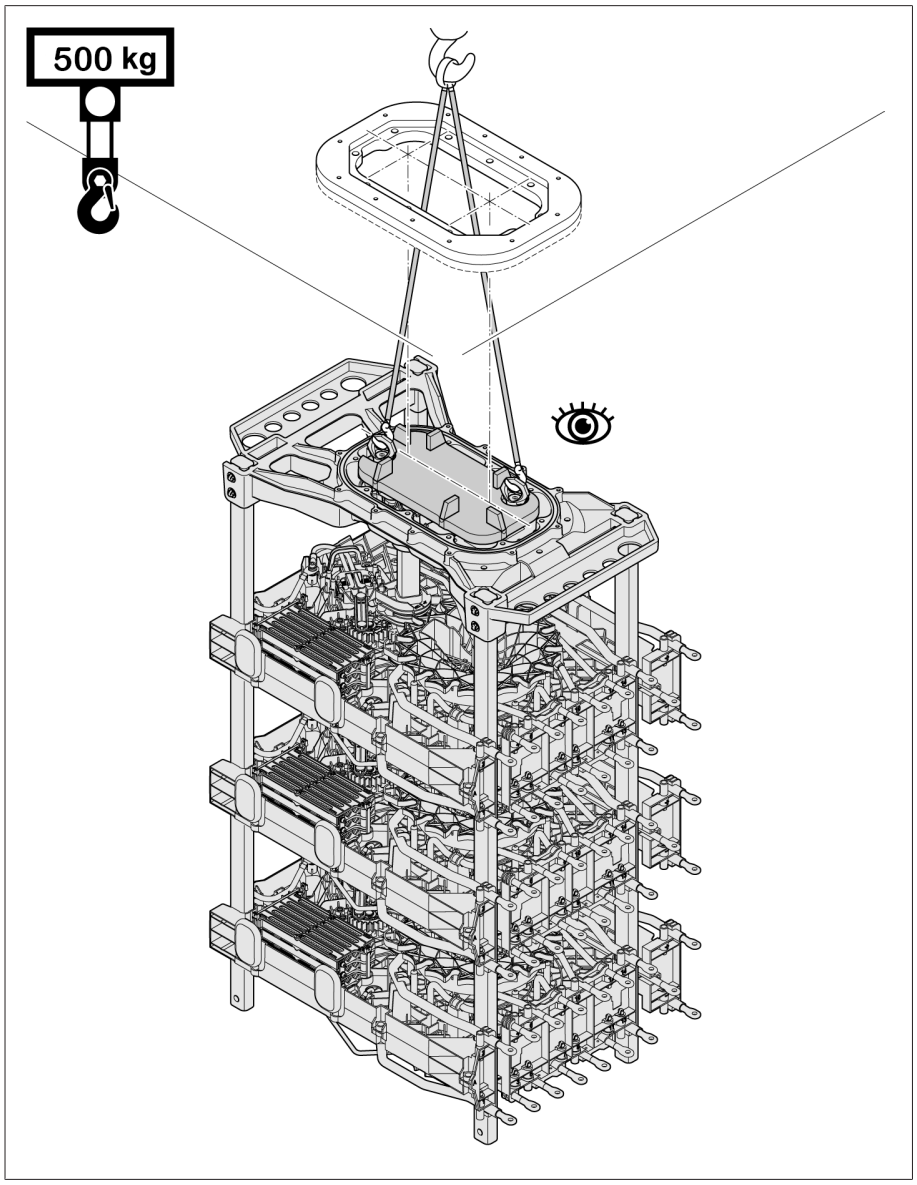


Figure 13: Aligning the on-load tap-changer to the mounting flange

8. *NOTICE!* Lifting the on-load tap-changer without due care can result in the on-load tap-changer colliding with the transformer cover and becoming damaged. Lift the on-load tap-changer using the lifting eye bolts from below up against the mounting flange in the transformer cover. The mounting device aligns the on-load tap-changer automatically.

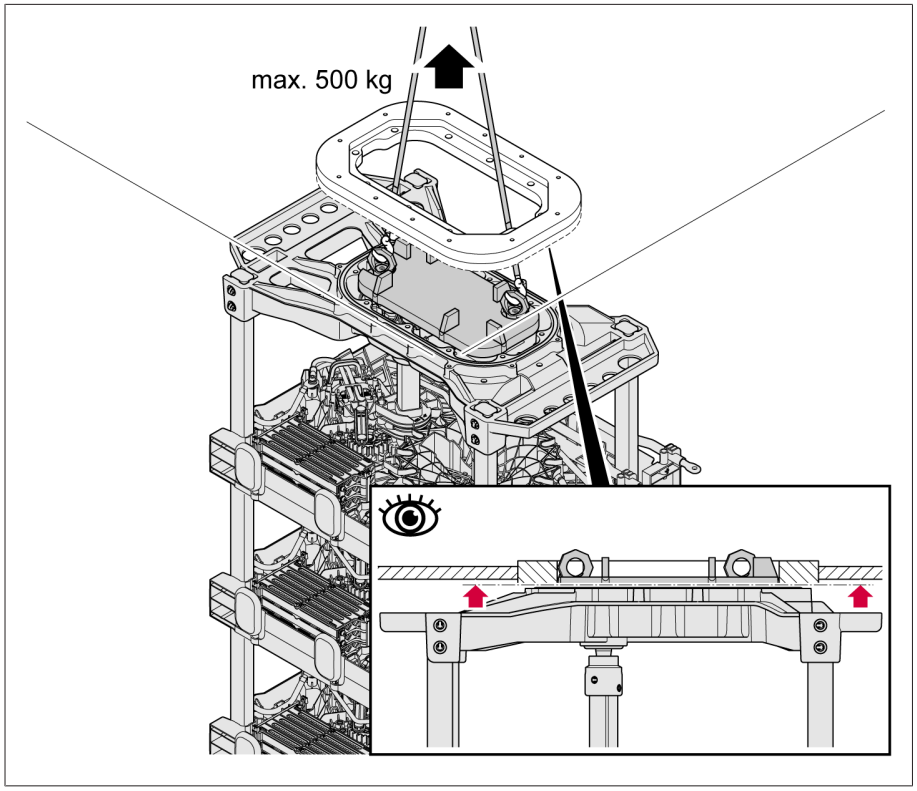


Figure 14: Lifting the on-load tap-changer

9. Screw the on-load tap-changer sealing module to the mounting flange. To do so, use the screws and washers provided by Maschinenfabrik Reinhausen GmbH. *NOTICE!* Improper screw connection will result in damage to the on-load tap-changer. Gradually tighten screws crosswise as described below without warping or deforming.

- Evenly tighten screws crosswise by hand.
- Check the seat of the flange and the seal for evenness.

- Tighten screws crosswise with the full tightening torque of 35 Nm.
- Tighten screws again with full tightening torque, working clockwise and tightening one screw after another as far as each screw will go.

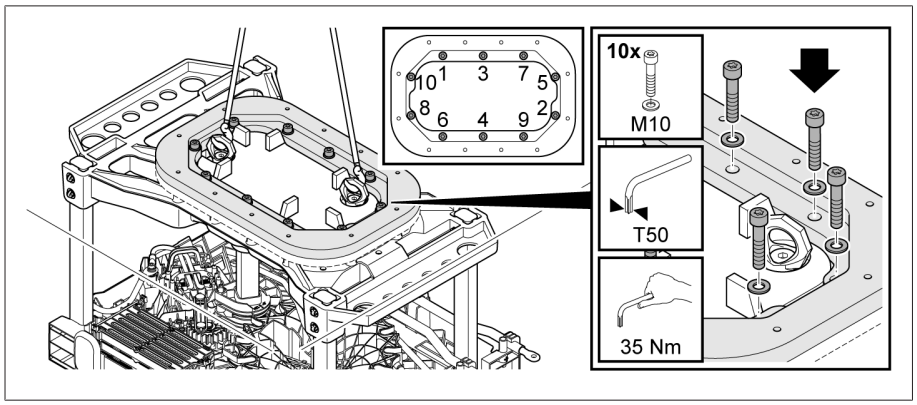


Figure 15: Screwing the on-load tap-changer into place

› ⚠ WARNING! Carefully remove the mounting device and lifting eye bolts, as serious injuries can occur when the mounting device is lifted out with the crane. Ensure that the mounting device is pulled out evenly. If the mounting device jams in the mounting flange, loosen the jam by tapping it lightly with a mallet.

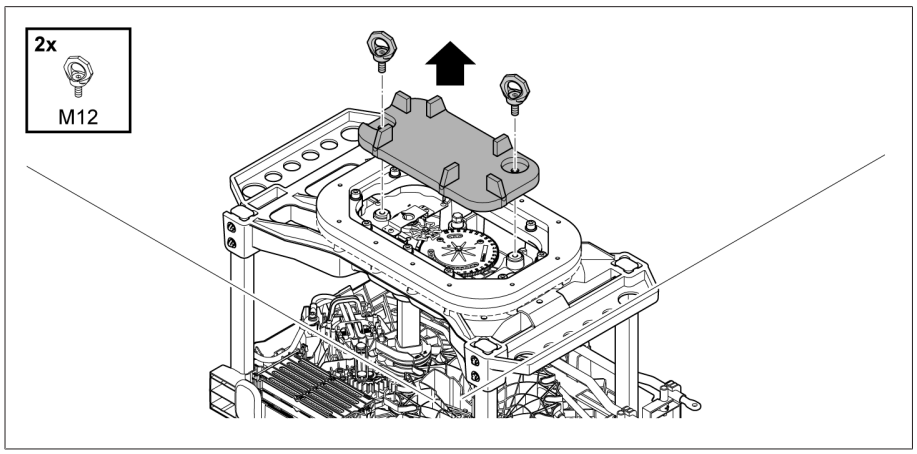


Figure 16: Removing the mounting device

5.2.2 Connecting the tap winding and on-load tap-changer take-off lead

NOTICE

Damage to the on-load tap-changer!

Connecting leads which place mechanical strain on the on-load tap-changer will damage the on-load tap-changer!

› Establish and secure connections with care.
› Do not twist connection contacts.
› Connect connecting leads without warping or deforming.
› If necessary use an expansion loop for connecting leads.

Connect the tap winding and on-load tap-changer take-off lead in accordance with the connection diagram included with the delivery.

Tap selector contacts

The tap selector connection contacts are provided with a vertical through-hole for M8 screws.

Bridges can be ordered as an option and are not included in the delivery. The following illustrations show examples of bridges that can be ordered from Maschinenfabrik Reinhausen GmbH.

1. Fasten the connection bridges to the tap selector in accordance with the connection diagram provided. For this purpose, use either your own bridges and screening caps or, as an option, those ordered from Maschinenfabrik Reinhausen (see dimensional drawings [▶10162430] and [▶10186369]). The fastening materials are not included in the scope of delivery. Take suitable measures (for example by using clamping washers) to ensure that each screw connection cannot come loose or settle.

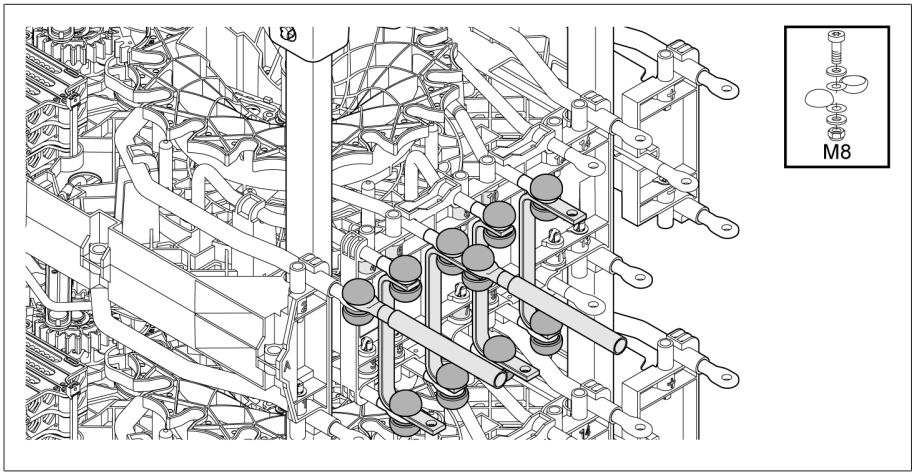


Figure 17: Tap selector bridges (example)

2. Fasten the tap-winding connecting leads to the tap selector in accordance with the connection diagram provided. The cable shoes and fastening materials are not included in the scope of delivery. Take suitable measures (e.g. by using clamping washers) to ensure that each screw connection cannot come loose or settle.

Change-over selector contacts

Connection cables for version 3W are not included in the scope of delivery and must be manufactured by the customer.
Bridges can be ordered as an option and are not included in the delivery. The illustrations show examples of bridges that can be ordered from Maschinenfabrik Reinhausen GmbH.

1. Fasten the connection bridges to the change-over selector in accordance with the connection diagram provided. For this purpose, use either your own bridges and screening caps or, as an option, those ordered from Maschinenfabrik Reinhausen (see dimensional drawings [▶10186372] and

[►10186373]). The fastening materials are not included in the scope of delivery. Take suitable measures (for example by using clamping washers) to ensure that each screw connection cannot come loose or settle.

2. Fasten the tap-winding connecting leads to the change-over selector in accordance with the connection diagram provided. The cable shoes and fastening materials are not included in the scope of delivery. Take suitable measures (e.g. by using clamping washers) to ensure that each screw connection cannot come loose or settle.

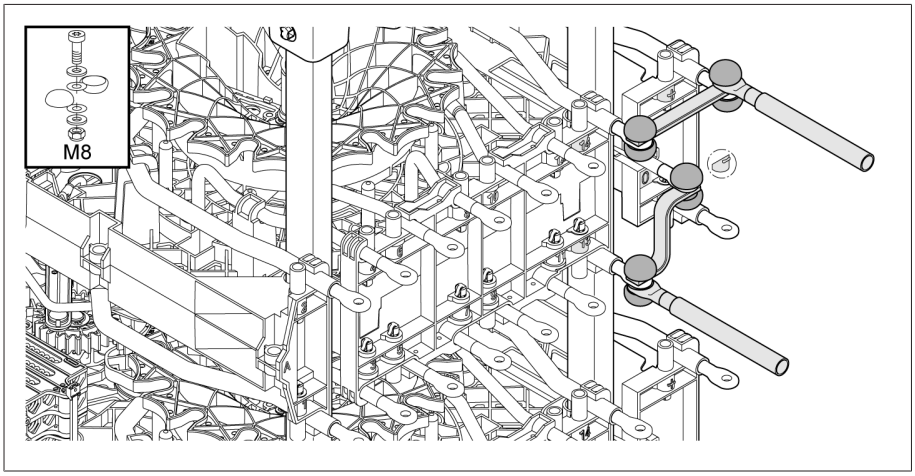


Figure 18: Change-over selector bridges, coarse tap connection version (example)

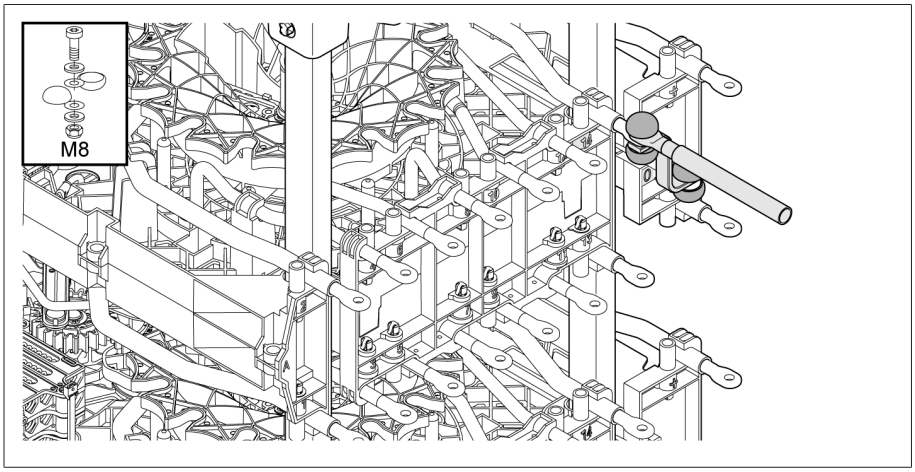


Figure 19: Change-over selector bridges, reversing change-over selector version (example)

On-load tap-changer take-off lead

› Fasten the on-load tap-changer take-off lead in accordance with the connection diagram provided. The cable shoes and fastening materials are not included in the scope of delivery. Take suitable measures (e.g. by using clamping washers) to ensure that each screw connection cannot come loose or settle.

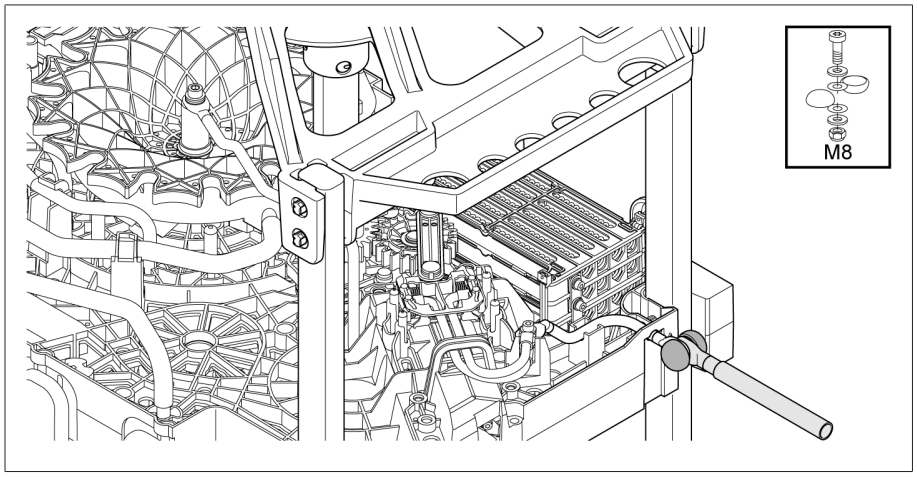


Figure 20: On-load tap-changer take-off lead

> For certain applications, it can be advantageous to pull the on-load tap-changer take-off lead to the front. A special version can be ordered as an option for this purpose (see dimensional drawing [▶10162441]).

5.2.3 Mounting the motor cover and control cabinet

5.2.3.1 Checking adjustment position

› Ensure that the on-load tap-changer is in the adjustment position.

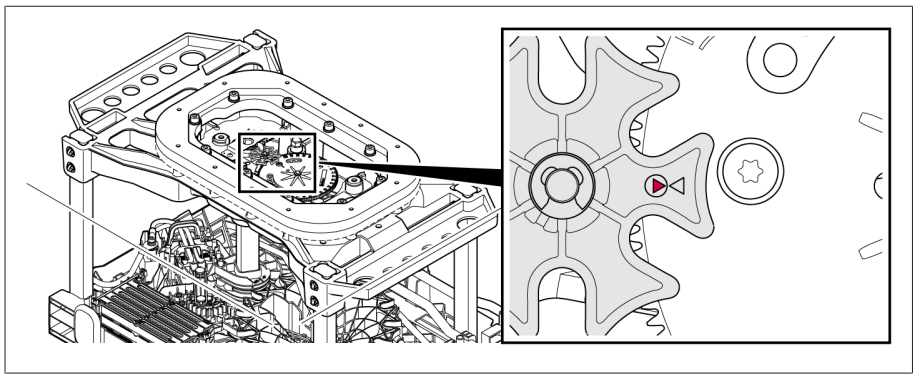


Figure 21: Checking the adjustment position

If the on-load tap-changer is not in the adjustment position, it must be switched into the adjustment position using the hand crank adapter:

1. Remove the hand crank adapter installed in the sealing module and insert it into the hand crank aperture.

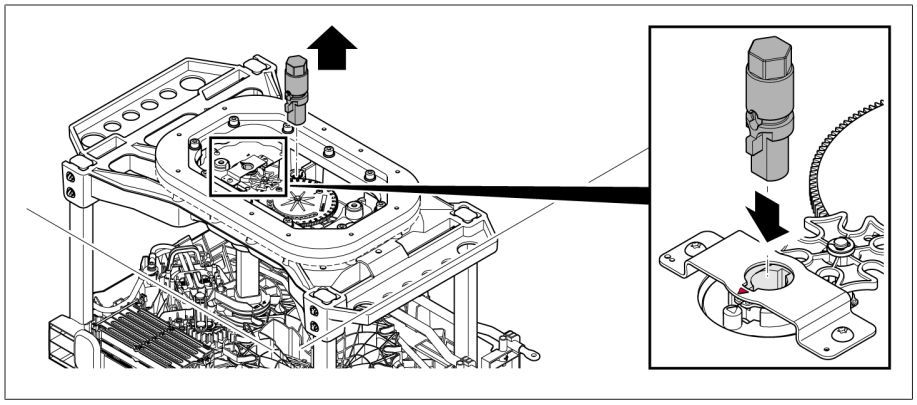


Figure 22: Hand crank adapter

2. Turn the hand crank adapter using a suitable tool (wrench size 17) until the adjustment position is reached.
3. Remove the hand crank adapter and replace it in the mounting bracket.

5.2.3.2 Mounting the motor cover

1. Ensure that the on-load tap-changer is in the adjustment position.

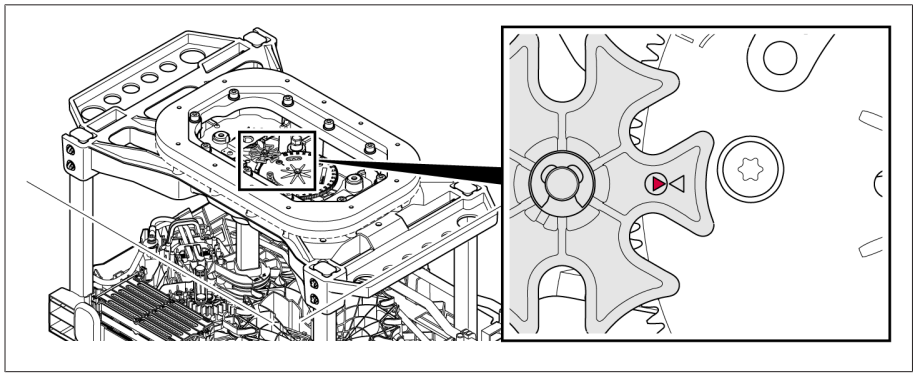


Figure 23: Checking the adjustment position

2. Clean the sealing surface of the motor cover and the mounting flange. Check that the o-ring gasket sits cleanly in the motor cover.

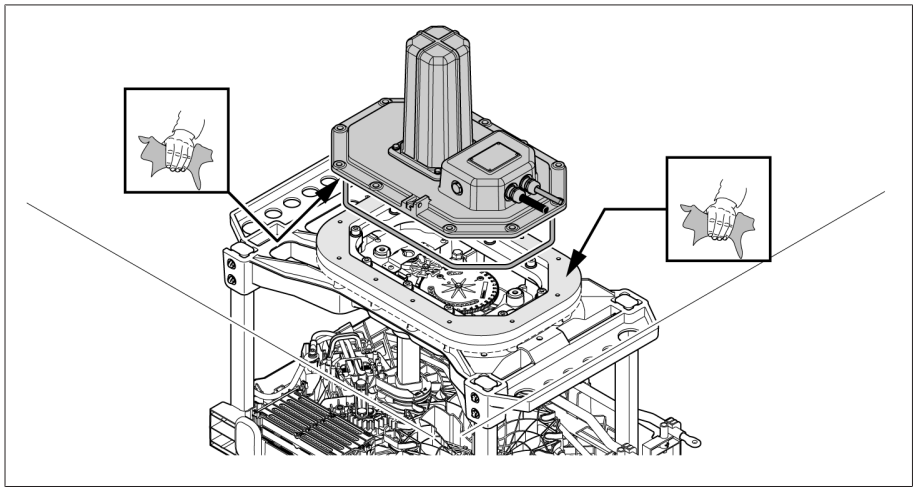


Figure 24: Cleaning sealing surfaces

3. Align the motor shaft so that the feather key is in line with the triangle mark on the motor plate.

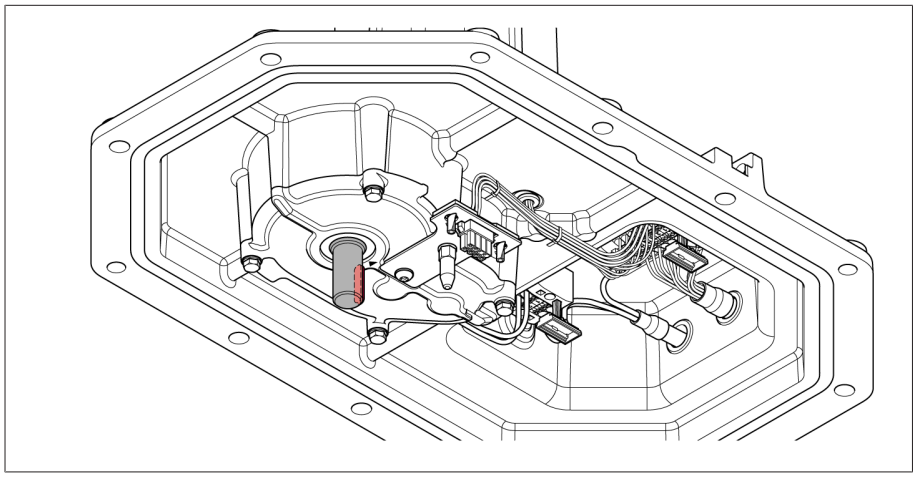


Figure 25: Triangle mark

4. When doing so, ensure that the shaft does not become damaged or scratched during alignment.

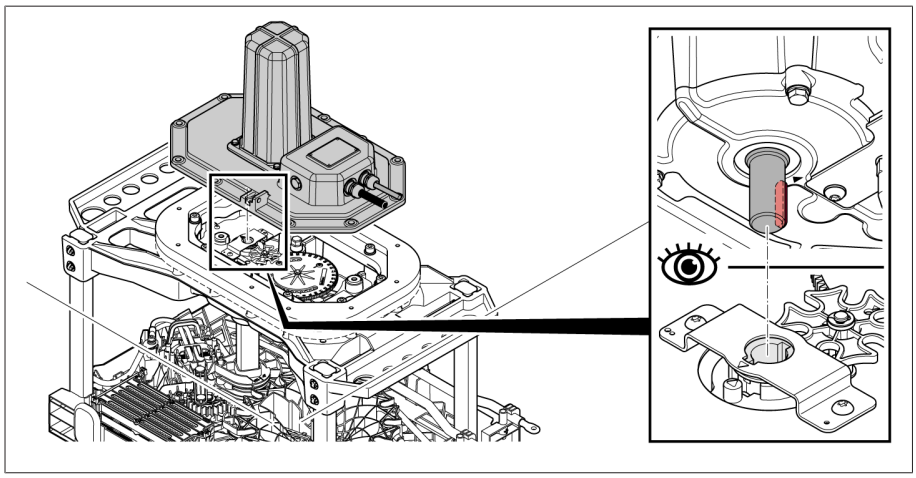


Figure 26: Aligning the motor shaft

5. Put the motor cover on so that the feather key on the drive shaft of the motor-drive unit slides into the keyway on the drive shaft of the on-load tap-changer in the sealing module and that at the same time the centering bolt in the motor plate engages in the locating hole provided for this in the sealing module. This ensures that the plugs of the micro-switch wiring in the motor cover and the sealing module are aligned.

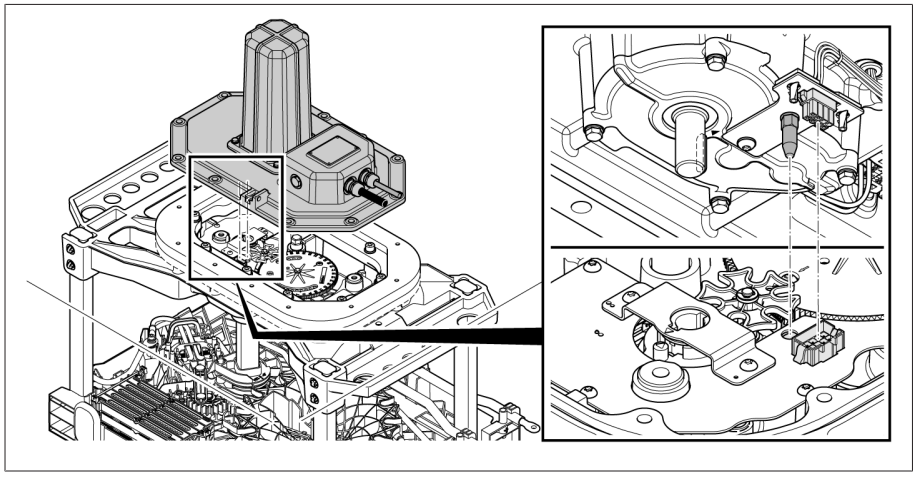


Figure 27: Centering

6. Screw the motor cover to the mounting flange, tightening the screws crosswise. To do so, use the screws and washers provided by Maschinenfabrik Reinhausen.

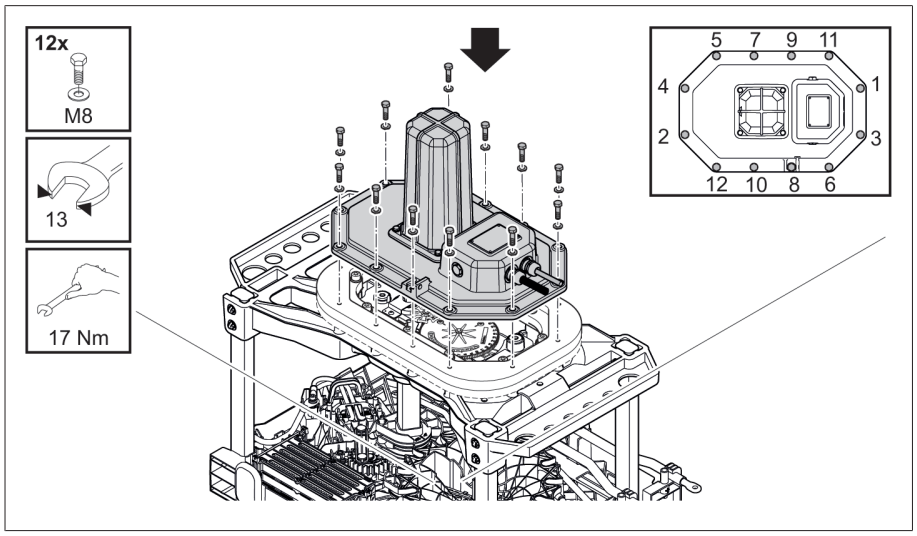


Figure 28: Screwing the motor cover on

5.2.3.3 Fitting the control cabinet

› Fit the ETOS® ID or Tap Changer Control control cabinet to the transformer in accordance with the corresponding Maschinenfabrik Reinhausen operating instructions.

5.2.3.4 Connecting the on-load tap-changer to the control cabinet

› Connect the control cabinet in accordance with the corresponding Maschinenfabrik Reinhausen ETOS® ID or Tap Changer Control operating instructions.

5.2.3.5 Electrically connecting the control cabinet and motor cover

› Connect the motor cover and ETOS® ID or Tap Changer Control control cabinet electrically in accordance with the corresponding Maschinenfabrik Reinhausen operating instructions.

5.2.3.6 Performing homing

› Perform a homing operation in accordance with the ETOS® ID or the Tap Changer Control user guide.

> Homing is not necessary with on-load tap-changers that have the AutoSync function. The control system can automatically detect the position of the on-load tap-changer via the built-in sensor.

NOTICE

Damage to the on-load tap-changer.

Without homing, the on-load tap-changer may become damaged.

› After homing, do not separate the motor cover from the sealing module again.

› Repeat homing if the motor cover is separated from the sealing module.

5.2.3.7 Driving to the pass-through position

For the transformer ratio test, DC resistance measurement or high-voltage tests, it may be necessary to drive to the pass-through positions individually.

Further information on this is available in the chapter "Setup mode" in the Maschinenfabrik Reinhausen ETOS® ID or Tap Changer Control control cabinet operating instructions.

5.2.4 Taking measurements

Before taking measurements, observe the section "Fitting the motor cover and control cabinet".

NOTICE

Damage to the on-load tap-changer.

Switching the on-load tap-changer via the hand crank can lead to damage to the on-load tap-changer.

- Only switch the on-load tap-changer via the control cabinet.
- Do not remove the motor cover.

NOTICE

Damage to the on-load tap-changer.

Dry tap-change operations lead to damage to the on-load tap-changer.

- If the on-load tap-changer is not yet in the oil-filled transformer tank, moisten all movable and fixed on-load tap-changer contacts with transformer oil before taking measurements.

5.2.4.1 Performing the transformer ratio test

1. Switch the on-load tap-changer into the desired operating position.
2. Perform the transformer ratio test.
3. Repeat the transformer ratio test in all operating positions.
4. Switch the on-load tap-changer into the adjustment position (see the accompanying connection diagram for the on-load tap-changer).

5.2.4.2 Performing DC resistance measurement on the transformer prior to drying

> The measured DC current is normally restricted to 10% of the rated current of the measured transformer winding in order to prevent the winding from overheating.

Observe the maximum permitted measured currents for the on-load tap-changer during the DC resistance measurement on the transformer.

For the DC resistance measurement, carry out the tasks as per those listed in the previous chapter, "Performing the transformer ratio test".

Transformer measurement status	Without interruption of the measured current	With interruption (measured current = 0 A before change of operating position)
Transformer measurement, empty	Maximum 10 A DC	Maximum 50 A DC
Transformer measurement, filled with insulating fluid	Maximum 50 A DC	Maximum 50 A DC

Table 5: Maximum permitted measured currents when performing DC resistance measurement on transformer

5.2.5 Drying the on-load tap-changer in the autoclave or transformer tank

NOTICE

Damage to the on-load tap-changer, motor cover and motor control unit due to drying.

The motor cover and motor control unit will become damaged if they are dried.

- Never leave the motor cover or motor control unit mounted when drying the device.
- Affix the guard plate in accordance with the following description prior to drying to protect the on-load tap-changer.

To prepare for drying, proceed as follows:

1. Remove the screws from the motor cover with mounting flange.

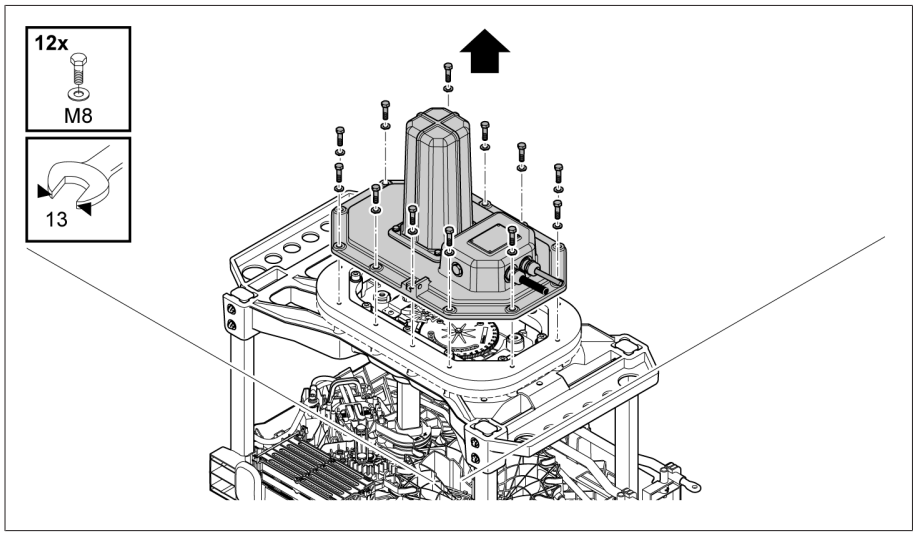


Figure 29: Removing the motor cover

2. Detach and remove the motor cover. To simplify removal of the motor cover, four tapped holes are provided in the motor cover. By screwing in screws, the motor can be pulled out evenly.

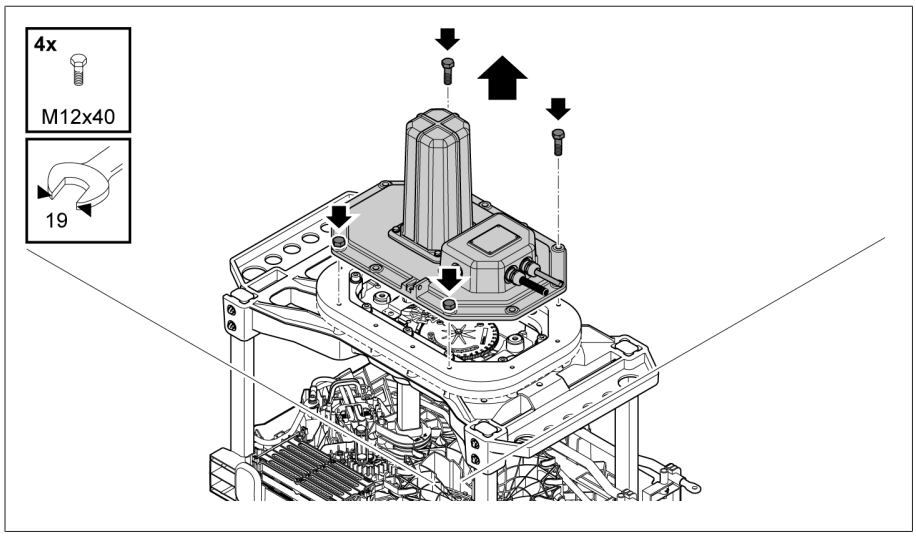


Figure 30: Pulling the motor cover off

3. Remove the control cabinet in accordance with the ETOS® ID or Tap Changer Control operating instructions.
4. Clean the sealing surfaces on the mounting flange and guard plate.

5. Attach and fasten the guard plate prior to drying. Ensure that the flat gasket is positioned correctly.

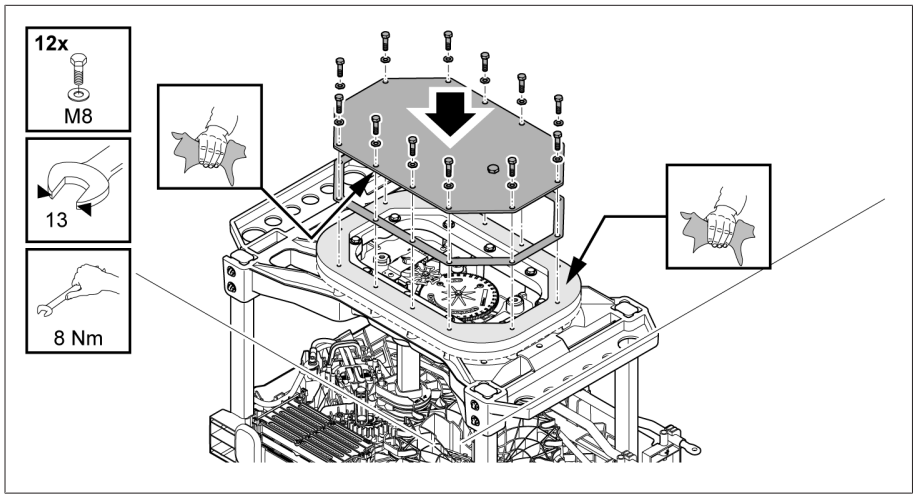


Figure 31: Mounting the guard plate

You can dry the on-load tap-changer using one of the following methods:
- Vacuum-drying – in an autoclave or in the transformer tank
- Vapor-phase drying – in an autoclave or in the transformer tank

5.2.5.1 Vacuum-drying

1. Heat up the on-load tap-changer in air at atmospheric pressure with a temperature increase of approximately 10 °C/h to a final temperature of maximum 110 °C.
2. Pre-dry the on-load tap-changer in circulating air at a maximum temperature of 110 °C for a period of at least 20 hours.
3. Vacuum-dry the on-load tap-changer at between 105°C and maximum 125°C for at least 50 hours.
4. Residual pressure of no more than 10^{-3} bar.

5.2.5.2 Vapor-phase drying

1. Supply kerosene vapor at a temperature of around 90 °C. Keep this temperature constant for 3 to 4 hours.
2. Increase the kerosene vapor temperature by approx. 10°C/hour to the desired final temperature of max. 125°C at the on-load tap-changer.
3. Vacuum-dry the on-load tap-changer at between 105°C and maximum 125°C for at least 50 hours.
4. Residual pressure of no more than 10^{-3} bar.

5.2.6 Filling transformer with oil

NOTICE

Damage to the on-load tap-changer.

Unsuitable insulating fluids cause damage to the on-load tap-changer.

› Only use insulating fluids [▶Section 9.2, Page 80] that have been approved by Maschinenfabrik Reinhausen GmbH.

For the oil filling of the transformer, use new mineral insulating oil for transformers as per IEC 60296 (Specification of unused mineral insulating oils for transformers and switchgear).

If approved by the transformer manufacturer, synthetic ester liquids as per IEC 61099 (Specification for unused synthetic organic esters for electrical purposes) or natural ester liquids as per IEC 62770 (Specification for unused natural esters for transformers and similar electrical equipment) can be used as alternatives.

Contact Maschinenfabrik Reinhausen GmbH if you want to use an alternative insulating fluid.

Observe the permitted temperature range of the transformer oil in the chapter Permissible ambient conditions [▶Section 9.2, Page 80].

NOTICE

Damage to the on-load tap-changer.

If a transformer is not completely filled with oil, the on-load tap-changer may be damaged.

› Ensure that the transformer is completely filled with transformer oil.

✓ Before commissioning the on-load tap-changer, completely fill the transformer with oil.

1. Completely fill the transformer with oil under vacuum.
2. Take an oil sample from the transformer.
3. Record the temperature of the oil sample just after the sample is taken.
4. Determine dielectric strength and water content at an oil temperature of 20 °C ±5 °C. The limit values listed in chapter Limit values for dielectric strength and water content of insulating fluids [▶Section 9.3, Page 81] must be complied with.

5.2.7 Mounting the motor cover and control cabinet

5.2.7.1 Checking adjustment position

1. Remove the guard plate and flat gasket from the on-load tap-changer sealing module. To do so, remove the screws. Note: Store the screws, washers, guard plate and flat gasket safely. These parts will be needed again in a later mounting step.
   NOTICE! Only use the flat gasket in combination with the guard plate! The motor cover with mounting flange is sealed using an o-ring. In order to prevent soiling and damage to the mechanics, do not leave the sealing module open for long periods of time.

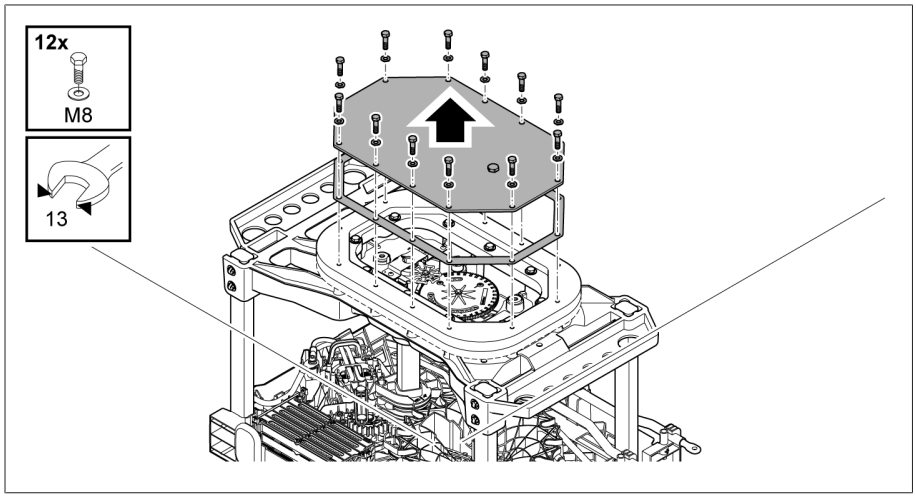


Figure 32: Removing the guard plate and flat gasket

2. Ensure that the on-load tap-changer is in the adjustment position.

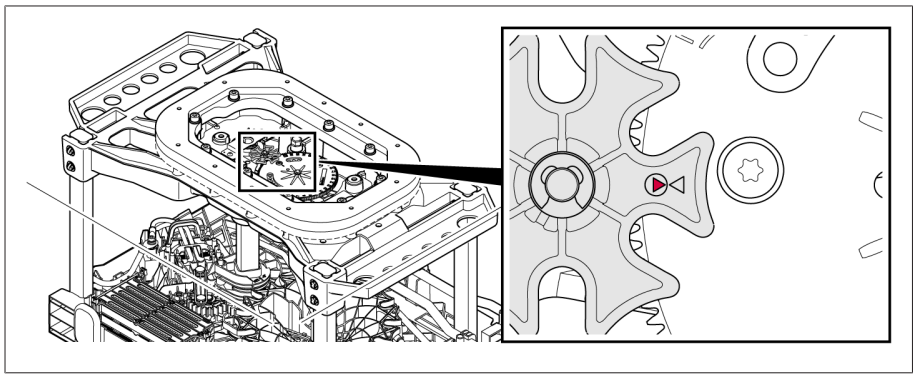


Figure 33: Checking the adjustment position

If the on-load tap-changer is not in the adjustment position, it must be switched into the adjustment position using the hand crank adapter:

1. Remove the hand crank adapter installed in the sealing module and insert it into the hand crank aperture.

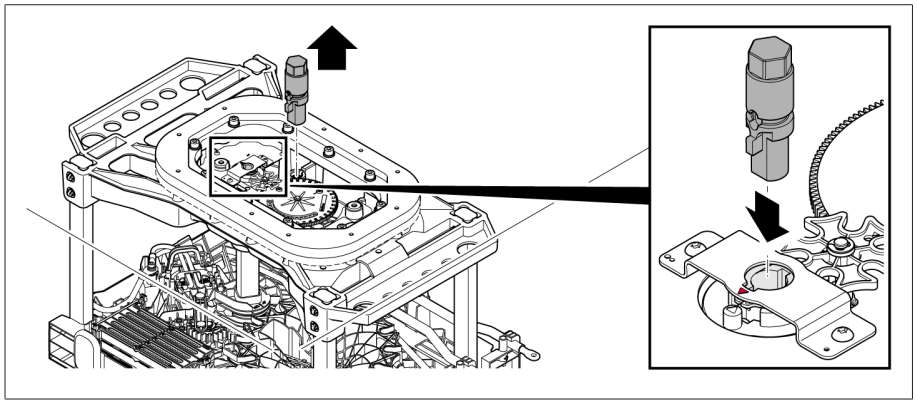


Figure 34: Hand crank adapter

2. Turn the hand crank adapter using a suitable tool (wrench size 17) until the adjustment position is reached.
3. Remove the hand crank adapter and replace it in the mounting bracket.

5.2.7.2 Mounting the motor cover

1. Ensure that the on-load tap-changer is in the adjustment position.

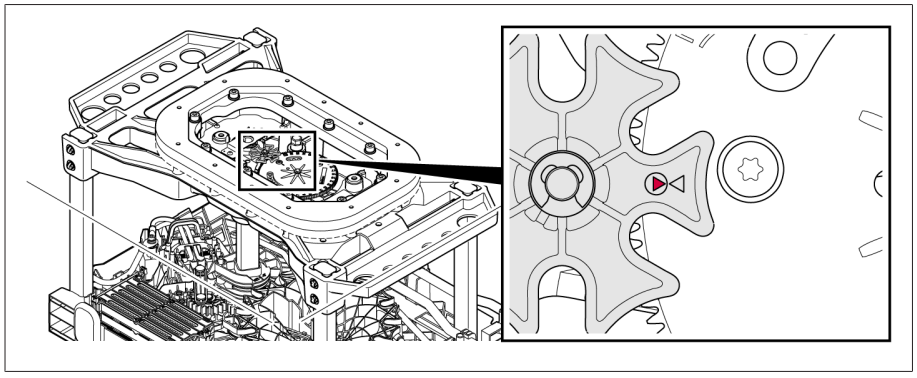


Figure 35: Checking the adjustment position

2. Clean the sealing surface of the motor cover and the mounting flange. Check that the o-ring gasket sits cleanly in the motor cover.

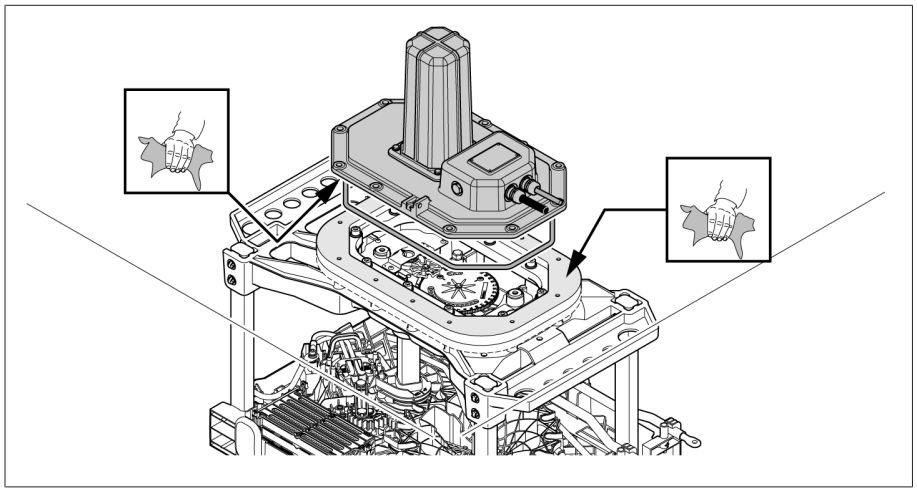


Figure 36: Cleaning sealing surfaces

3. Align the motor shaft so that the feather key is in line with the triangle mark on the motor plate.

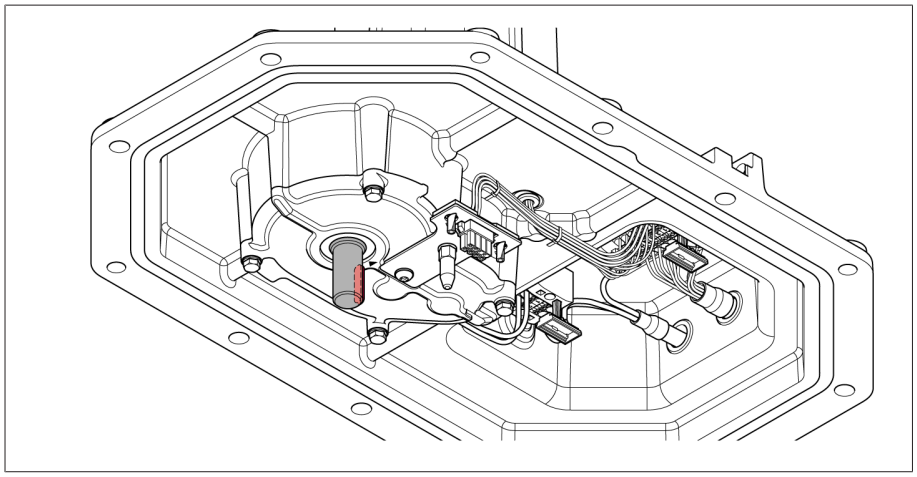


Figure 37: Triangle mark

4. When doing so, ensure that the shaft does not become damaged or scratched during alignment.

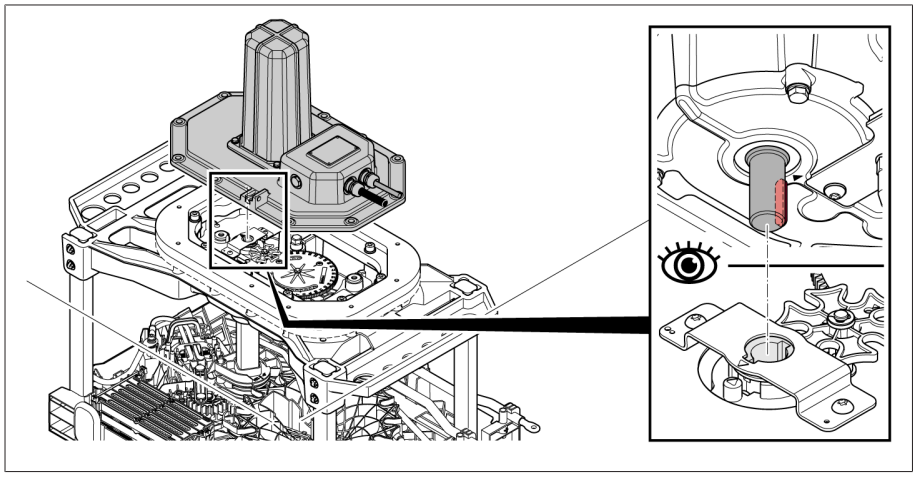


Figure 38: Aligning the motor shaft

5. Put the motor cover on so that the feather key on the drive shaft of the motor-drive unit slides into the keyway on the drive shaft of the on-load tap-changer in the sealing module and that at the same time the centering bolt in

the motor plate engages in the locating hole provided for this in the sealing module. This ensures that the plugs of the micro-switch wiring in the motor cover and the sealing module are aligned.

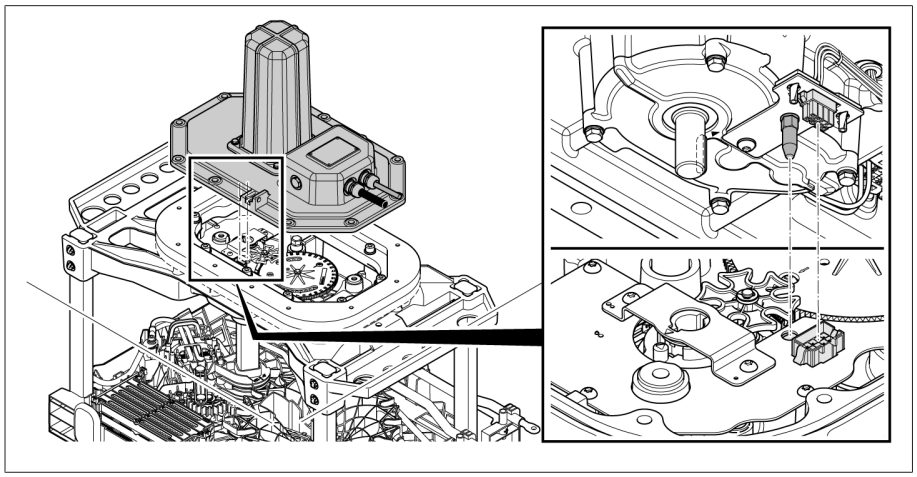


Figure 39: Centering

6. Screw the motor cover to the mounting flange, tightening the screws crosswise. To do so, use the screws and washers provided by Maschinenfabrik Reinhausen.

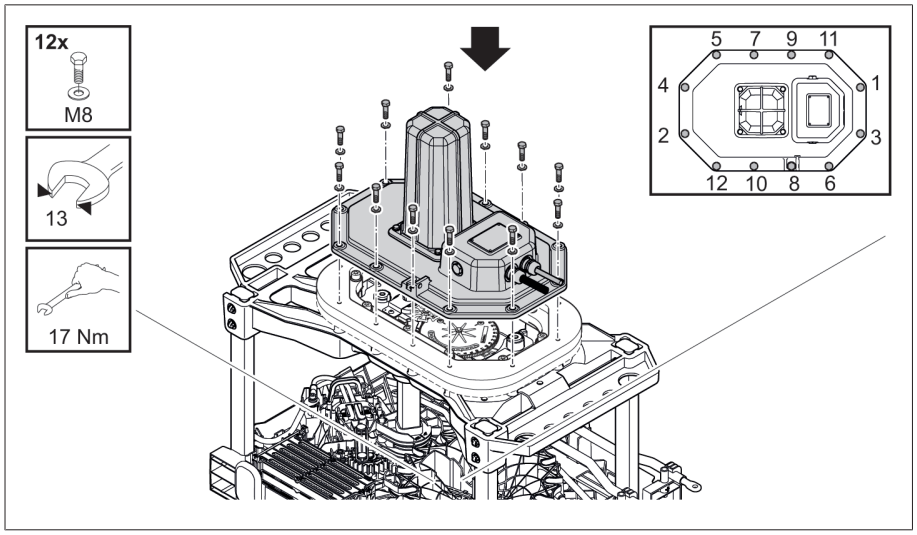


Figure 40: Screwing the motor cover on

5.2.7.3 Fitting the control cabinet

› Fit the ETOS® ID or Tap Changer Control control cabinet to the transformer in accordance with the corresponding Maschinenfabrik Reinhausen operating instructions.

5.2.7.4 Connecting the on-load tap-changer to the control cabinet

› Connect the control cabinet in accordance with the corresponding Maschinenfabrik Reinhausen ETOS® ID or Tap Changer Control operating instructions.

5.2.7.5 Electrically connecting the control cabinet and motor cover

› Connect the motor cover and ETOS® ID or Tap Changer Control control cabinet electrically in accordance with the corresponding Maschinenfabrik Reinhausen operating instructions.

5.2.7.6 Performing homing

› Perform a homing operation in accordance with the ETOS® ID or the Tap Changer Control user guide.

Homing is not necessary with on-load tap-changers that have the AutoSync function. The control system can automatically detect the position of the on-load tap-changer via the built-in sensor.

NOTICE

Damage to the on-load tap-changer.

Without homing, the on-load tap-changer may become damaged.

- After homing, do not separate the motor cover from the sealing module again.
- Repeat homing if the motor cover is separated from the sealing module.

5.2.7.7 Driving to the pass-through position

For the transformer ratio test, DC resistance measurement or high-voltage tests, it may be necessary to drive to the pass-through positions individually.

Further information on this is available in the chapter "Setup mode" in the Maschinenfabrik Reinhausen ETOS® ID or Tap Changer Control control cabinet operating instructions.

5.2.8 Performing the transformer ratio test

Before taking measurements, observe the section "Fitting the motor cover and control cabinet".

NOTICE

Damage to the on-load tap-changer.

Switching the on-load tap-changer via the hand crank can lead to damage to the on-load tap-changer.

- Only switch the on-load tap-changer via the control cabinet.
- Do not remove the motor cover.

NOTICE

Damage to the on-load tap-changer.

Dry tap-change operations lead to damage to the on-load tap-changer.

- If the on-load tap-changer is not yet in the oil-filled transformer tank, moisten all movable and fixed on-load tap-changer contacts with transformer oil before taking measurements.

1. Switch the on-load tap-changer into the desired operating position.
2. Perform the transformer ratio test.
3. Repeat the transformer ratio test in all operating positions.
4. Switch the on-load tap-changer into the adjustment position (see the accompanying connection diagram for the on-load tap-changer).

5.3 Installing on-load tap-changer in transformer (bell-type tank version)

5.3.1 Inserting on-load tap-changer into supporting structure

1. Remove the red-colored packaging material and the transport material from the on-load tap-changer.
2. *NOTICE!* Tensile forces can lead to damage and malfunctions on the on-load tap-changer. Using spacers, insert the on-load tap-changer vertically into the supporting structure (maximum 1° deviation from the vertical) so that the on-load tap-changer reaches its final installation height and only has to be raised a maximum of 15...30 mm once the step of the tap winding and the on-load tap-changer take-off lead have been connected and the bell-type tank fitted (see dimensional drawing [▶10186367]).

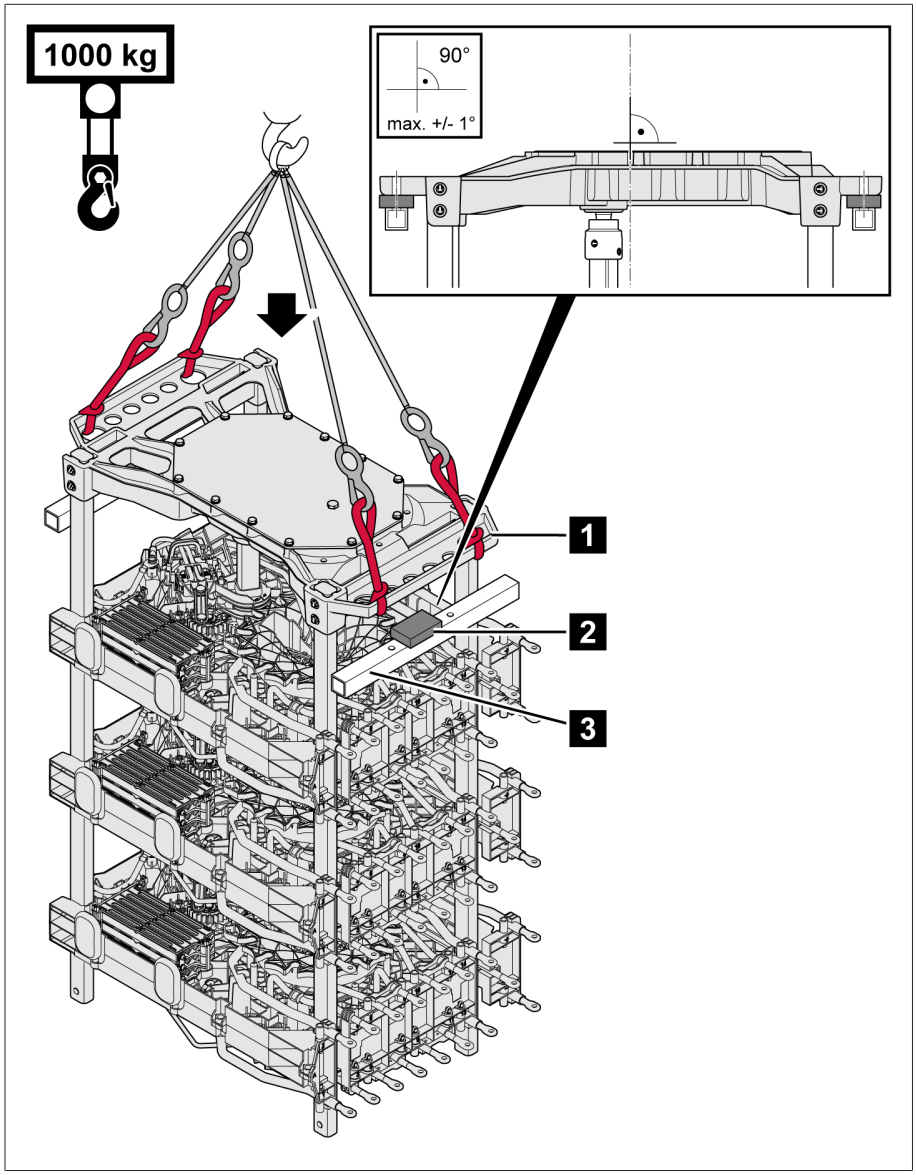


Figure 41: On-load tap-changer in the supporting structure

1	Sealing module	2	Spacer
3	Supporting structure		

3. Temporarily fasten the on-load tap-changer to the supporting structure. The sealing module has drill holes for this purpose.

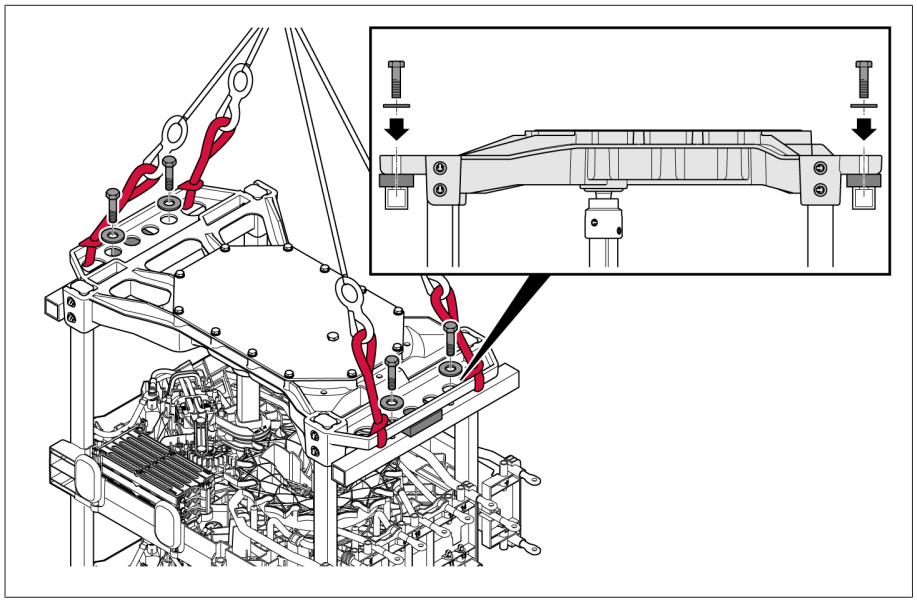


Figure 42: Temporary fastening

5.3.2 Connecting the tap winding and on-load tap-changer take-off lead

NOTICE

Damage to the on-load tap-changer!

Connecting leads which place mechanical strain on the on-load tap-changer will damage the on-load tap-changer!

- Establish and secure connections with care.
- Do not twist connection contacts.
- Connect connecting leads without warping or deforming.
- If necessary use an expansion loop for connecting leads.

Connect the tap winding and on-load tap-changer take-off lead in accordance with the connection diagram included with the delivery.

Tap selector contacts

The tap selector connection contacts are provided with a vertical through-hole for M8 screws.

Bridges can be ordered as an option and are not included in the delivery. The following illustrations show examples of bridges that can be ordered from Maschinenfabrik Reinhausen GmbH.

1. Fasten the connection bridges to the tap selector in accordance with the connection diagram provided. For this purpose, use either your own bridges and screening caps or, as an option, those ordered from Maschinenfabrik Reinhausen (see dimensional drawings [▶10162430] and [▶10186369]). The

fastening materials are not included in the scope of delivery. Take suitable measures (for example by using clamping washers) to ensure that each screw connection cannot come loose or settle.

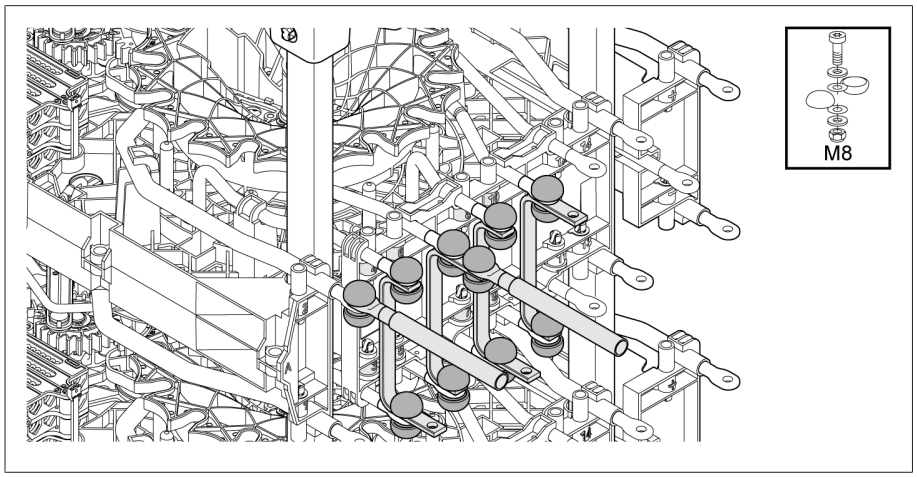


Figure 43: Tap selector bridges (example)

2. Fasten the tap-winding connecting leads to the tap selector in accordance with the connection diagram provided. The cable shoes and fastening materials are not included in the scope of delivery. Take suitable measures (e.g. by using clamping washers) to ensure that each screw connection cannot come loose or settle.

Change-over selector contacts

Connection cables for version 3W are not included in the scope of delivery and must be manufactured by the customer.
Bridges can be ordered as an option and are not included in the delivery. The illustrations show examples of bridges that can be ordered from Maschinenfabrik Reinhausen GmbH.

1. Fasten the connection bridges to the change-over selector in accordance with the connection diagram provided. For this purpose, use either your own bridges and screening caps or, as an option, those ordered from Maschinenfabrik Reinhausen (see dimensional drawings [▶10186372] and

[►10186373]). The fastening materials are not included in the scope of delivery. Take suitable measures (for example by using clamping washers) to ensure that each screw connection cannot come loose or settle.

2. Fasten the tap-winding connecting leads to the change-over selector in accordance with the connection diagram provided. The cable shoes and fastening materials are not included in the scope of delivery. Take suitable measures (e.g. by using clamping washers) to ensure that each screw connection cannot come loose or settle.

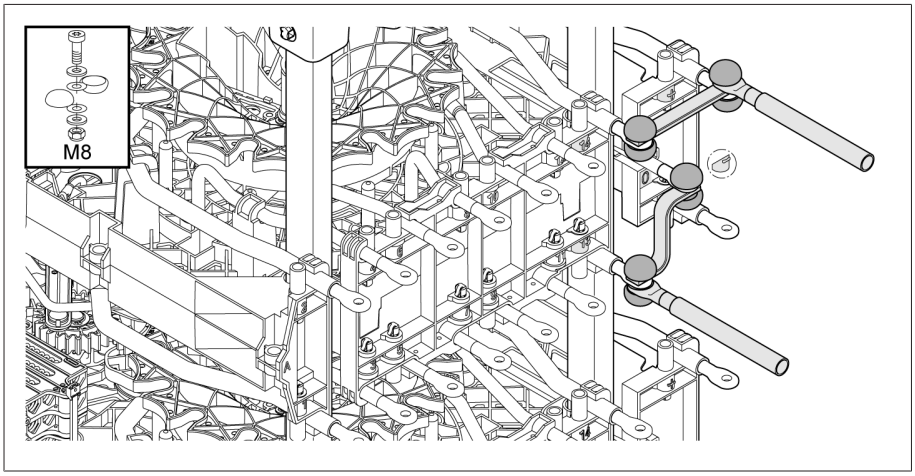


Figure 44: Change-over selector bridges, coarse tap connection version (example)

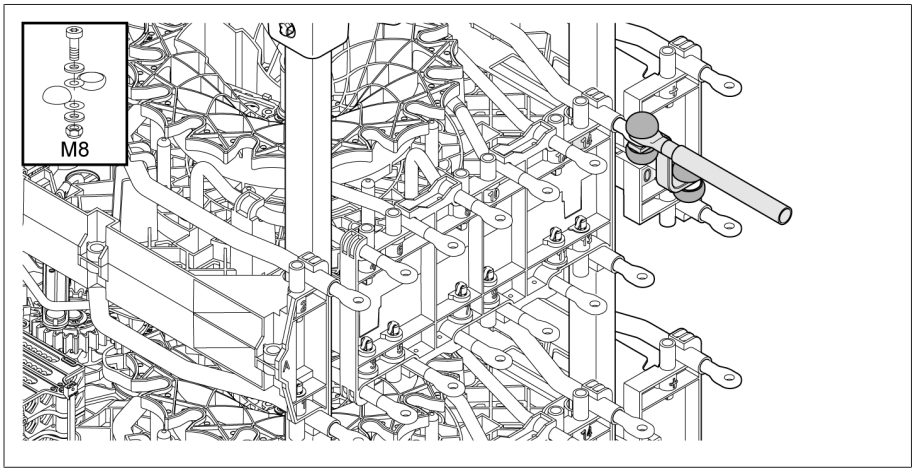


Figure 45: Change-over selector bridges, reversing change-over selector version (example)

On-load tap-changer take-off lead

› Fasten the on-load tap-changer take-off lead in accordance with the connection diagram provided. The cable shoes and fastening materials are not included in the scope of delivery. Take suitable measures (e.g. by using clamping washers) to ensure that each screw connection cannot come loose or settle.

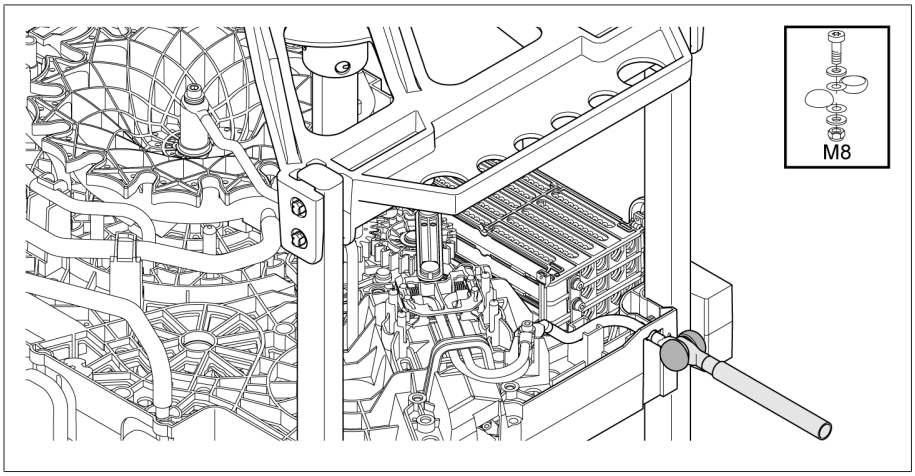


Figure 46: On-load tap-changer take-off lead

> For certain applications, it can be advantageous to pull the on-load tap-changer take-off lead to the front. A special version can be ordered as an option for this purpose (see dimensional drawing [▶10162441]).

5.3.3 Mounting the motor cover and control cabinet

5.3.3.1 Checking adjustment position

1. Remove the guard plate from the on-load tap-changer sealing module. To do so, remove the screws. Note: Store the screws, washers and guard plate safely. These parts will be needed again in a later mounting step. In order to prevent soiling and damage to the mechanics, do not leave the sealing module open for long periods of time.

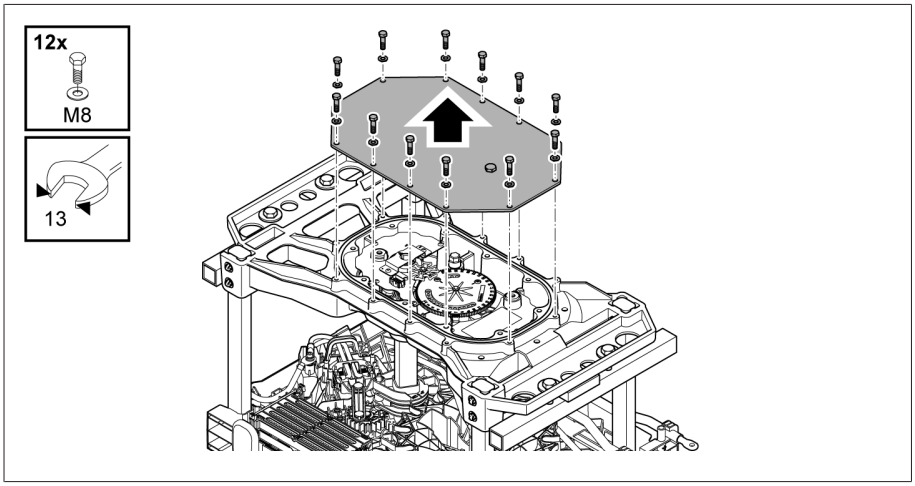


Figure 47: Removing the guard plate

2. Ensure that the on-load tap-changer is in the adjustment position.

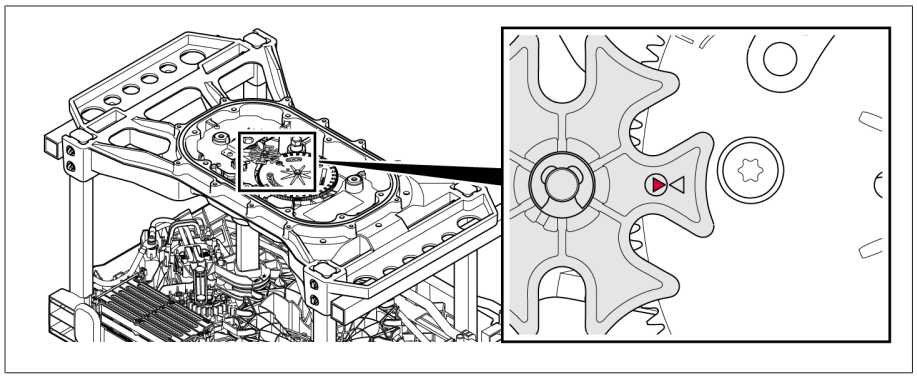


Figure 48: Checking the adjustment position

If the on-load tap-changer is not in the adjustment position, it must be switched into the adjustment position using the hand crank adapter:

1. Remove the hand crank adapter installed in the sealing module and insert it into the hand crank aperture.

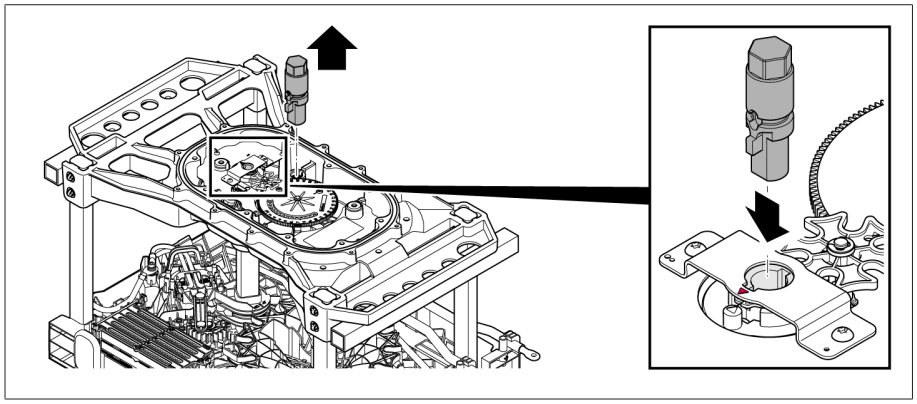


Figure 49: Hand crank adapter

2. Turn the hand crank adapter using a suitable tool (wrench size 17) until the adjustment position is reached.
3. Remove the hand crank adapter and replace it in the mounting bracket.

5.3.3.2 Mounting the motor cover

1. Mount the mounting flange on the sealing module. The mounting flange is available as an option or it can be made by the customer itself. If the customer intends to produce the mounting flange, it is to be produced in accordance with the dimensional drawing [▶10162405].

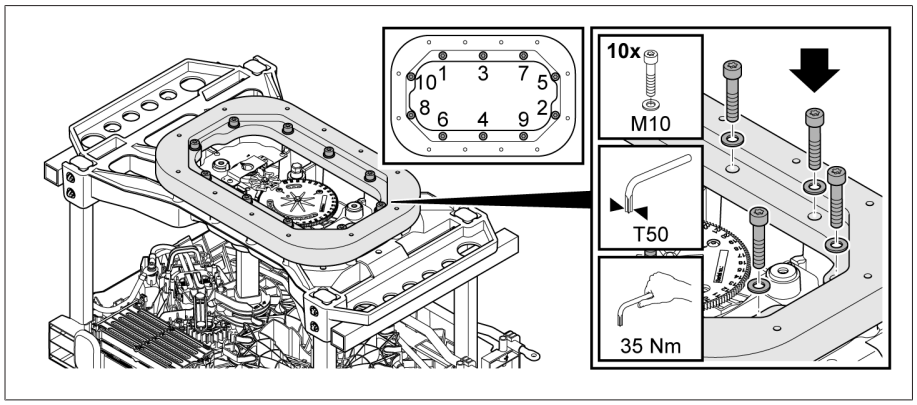


Figure 50: Screwing on the mounting flange

2. Ensure that the on-load tap-changer is in the adjustment position.

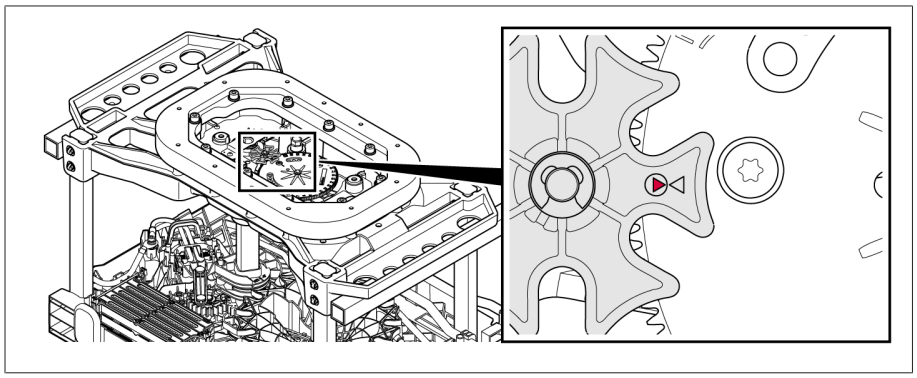


Figure 51: Checking the adjustment position

3. Clean the sealing surface of the motor cover and the mounting flange. Check that the o-ring gasket sits cleanly in the motor cover.

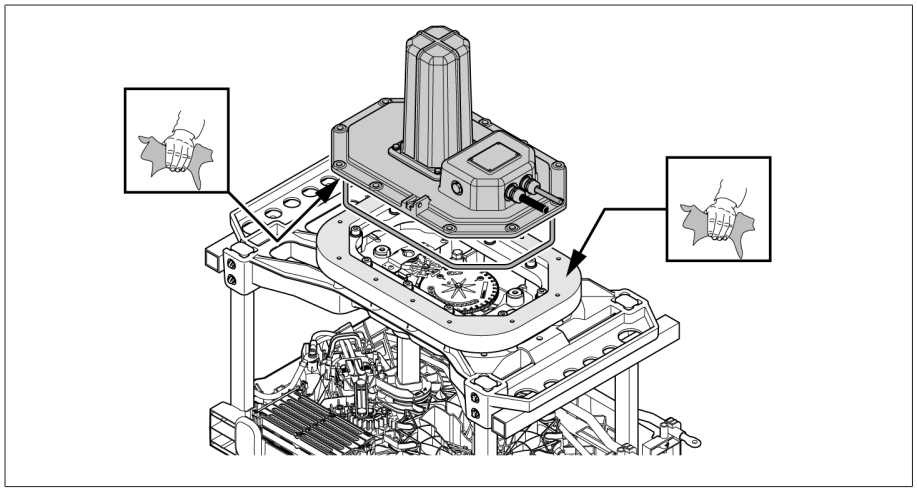


Figure 52: Cleaning the sealing surface

4. Align the motor shaft so that the feather key is in line with the triangle mark on the motor plate.

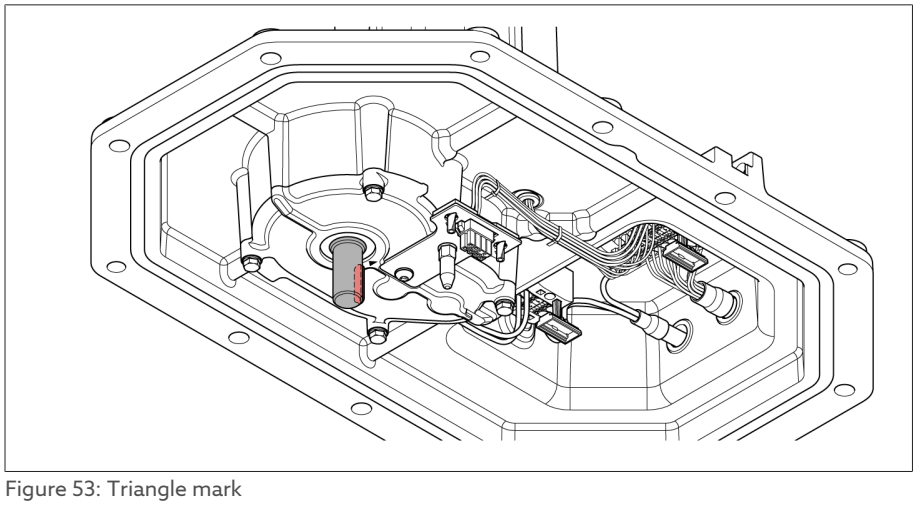


Figure 53: Triangle mark

5. When doing so, ensure that the shaft does not become damaged or scratched during alignment.

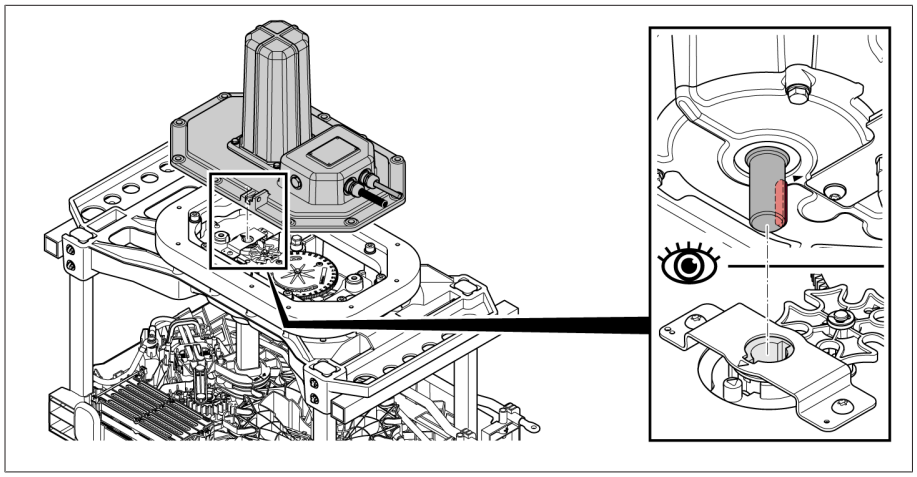


Figure 54: Aligning the motor shaft

6. Put the motor cover on so that the feather key on the drive shaft of the motor-drive unit slides into the keyway on the drive shaft of the on-load tap-changer in the sealing module and that at the same time the centering bolt in the motor plate engages in the locating hole provided for this in the sealing module. This ensures that the plugs of the micro-switch wiring in the motor cover and the sealing module are aligned.

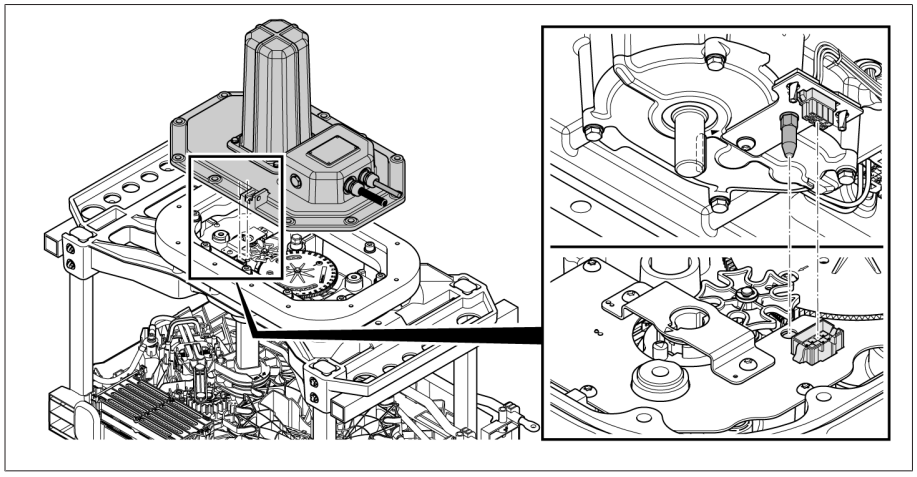


Figure 55: Centering

7. Screw the motor cover to the mounting flange, tightening the screws crosswise. To do so, use the screws and washers provided by Maschinenfabrik Reinhausen.

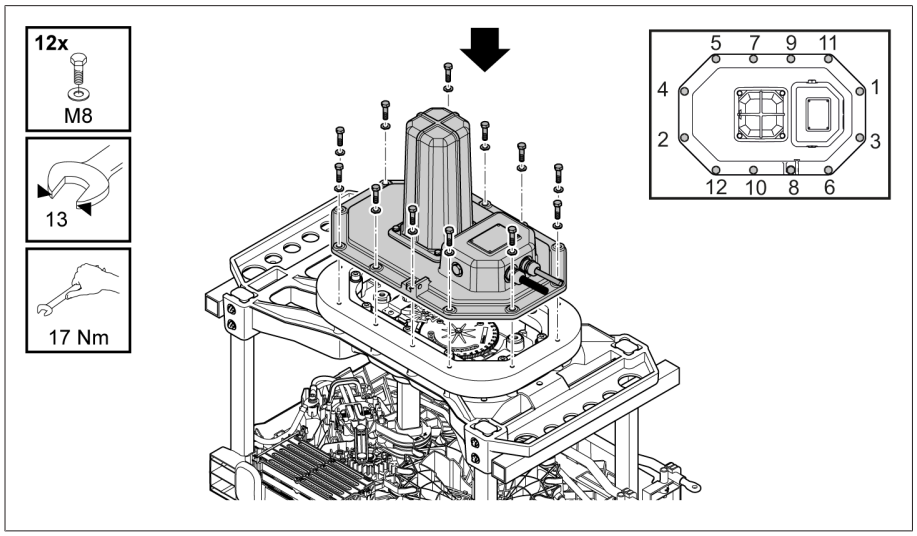


Figure 56: Screwing the motor cover on

5.3.3.3 Fitting the control cabinet

› Fit the ETOS® ID or Tap Changer Control control cabinet to the transformer in accordance with the corresponding Maschinenfabrik Reinhausen operating instructions.

5.3.3.4 Connecting the on-load tap-changer to the control cabinet

› Connect the control cabinet in accordance with the corresponding Maschinenfabrik Reinhausen ETOS® ID or Tap Changer Control operating instructions.

5.3.3.5 Electrically connecting the control cabinet and motor cover

› Connect the motor cover and ETOS® ID or Tap Changer Control control cabinet electrically in accordance with the corresponding Maschinenfabrik Reinhausen operating instructions.

5.3.3.6 Driving to the pass-through position

For the transformer ratio test, DC resistance measurement or high-voltage tests, it may be necessary to drive to the pass-through positions individually.

Further information on this is available in the chapter "Setup mode" in the Maschinenfabrik Reinhausen ETOS® ID or Tap Changer Control control cabinet operating instructions.

5.3.4 Taking measurements

Before taking measurements, observe the section "Fitting the motor cover and control cabinet".

NOTICE

Damage to the on-load tap-changer.

Switching the on-load tap-changer via the hand crank can lead to damage to the on-load tap-changer.

› Only switch the on-load tap-changer via the control cabinet.
› Do not remove the motor cover.

NOTICE

Damage to the on-load tap-changer.

Dry tap-change operations lead to damage to the on-load tap-changer.

› If the on-load tap-changer is not yet in the oil-filled transformer tank, moisten all movable and fixed on-load tap-changer contacts with transformer oil before taking measurements.

5.3.4.1 Performing the transformer ratio test

1. Switch the on-load tap-changer into the desired operating position.
2. Perform the transformer ratio test.
3. Repeat the transformer ratio test in all operating positions.
4. Switch the on-load tap-changer into the adjustment position (see the accompanying connection diagram for the on-load tap-changer).

5.3.4.2 Performing DC resistance measurement on the transformer prior to drying

> The measured DC current is normally restricted to 10% of the rated current of the measured transformer winding in order to prevent the winding from overheating.

Observe the maximum permitted measured currents for the on-load tap-changer during the DC resistance measurement on the transformer.

For the DC resistance measurement, carry out the tasks as per those listed in the previous chapter, "Performing the transformer ratio test".

Transformer measurement status	Without interruption of the measured current	With interruption (measured current = 0 A before change of operating position)
Transformer measurement, empty	Maximum 10 A DC	Maximum 50 A DC
Transformer measurement, filled with insulating fluid	Maximum 50 A DC	Maximum 50 A DC

Table 6: Maximum permitted measured currents when performing DC resistance measurement on transformer

5.3.5 Drying the on-load tap-changer in an autoclave

NOTICE

Damage to the on-load tap-changer, motor cover and motor control unit due to drying.

The motor cover and motor control unit will become damaged if they are dried.

› Never leave the motor cover or motor control unit mounted when drying the device.

› Affix the guard plate in accordance with the following description prior to drying to protect the on-load tap-changer.

To prepare for drying, proceed as follows:

1. Remove the screws from the motor cover and mounting flange.

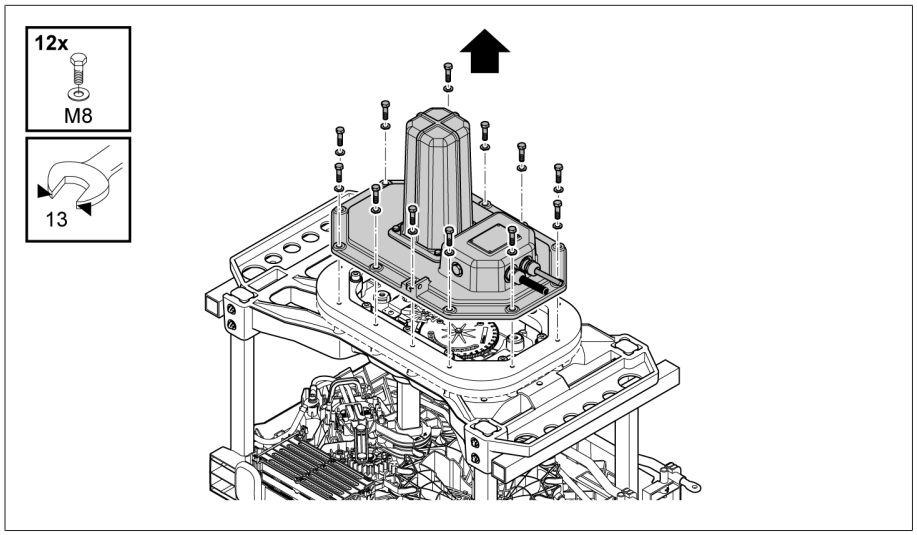


Figure 57: Removing the motor cover

2. Detach and remove the motor cover. To simplify removal of the motor cover, 4 tapped holes are provided in the motor cover. By screwing in screws, the motor can be pulled out evenly.

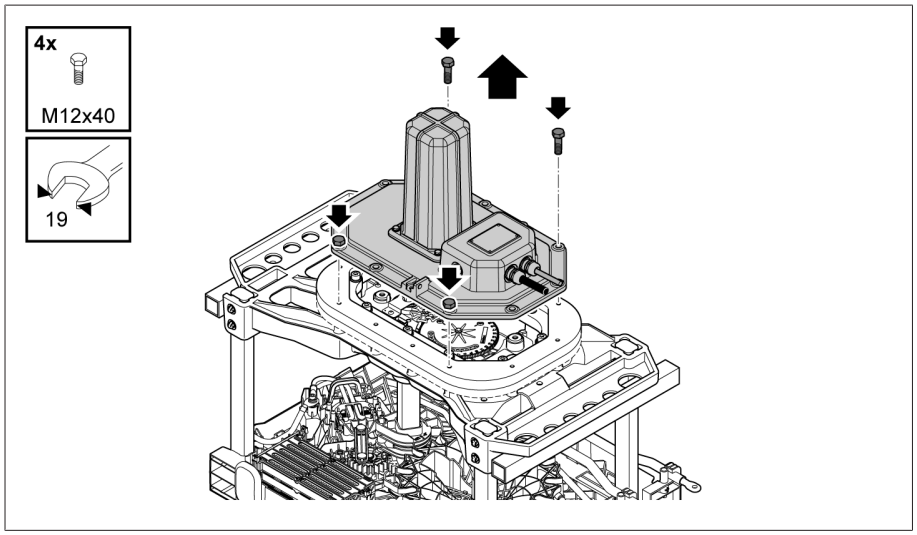


Figure 58: Pulling the motor cover off

3. Remove the screws attaching the adapter flange to the sealing module.

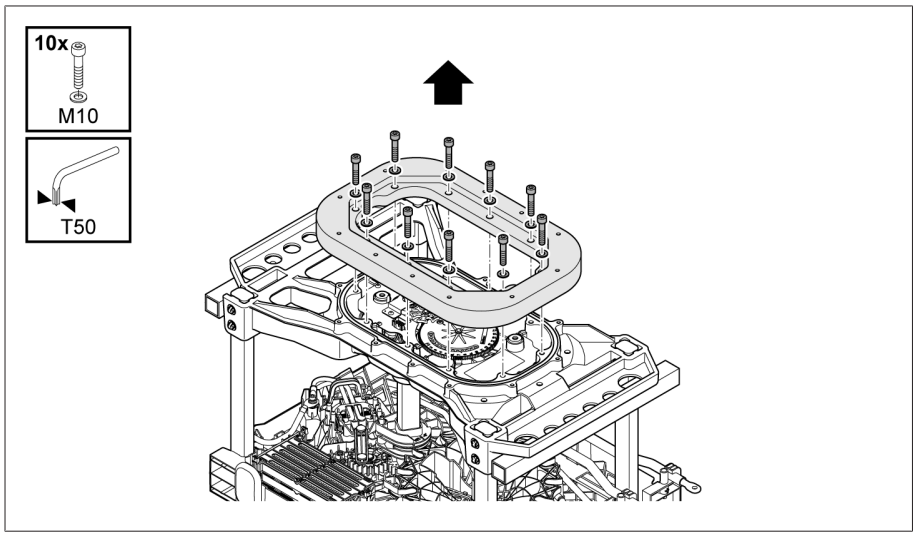


Figure 59: Removing the mounting flange

4. Detach and remove the mounting flange.
5. Remove the control cabinet in accordance with the ETOS® ID or the Tap Changer Control user guide.
6. Clean the sealing surfaces on the sealing module and guard plate.
7. Attach and fasten the guard plate. Check that the o-ring gasket sits cleanly in the sealing flange.

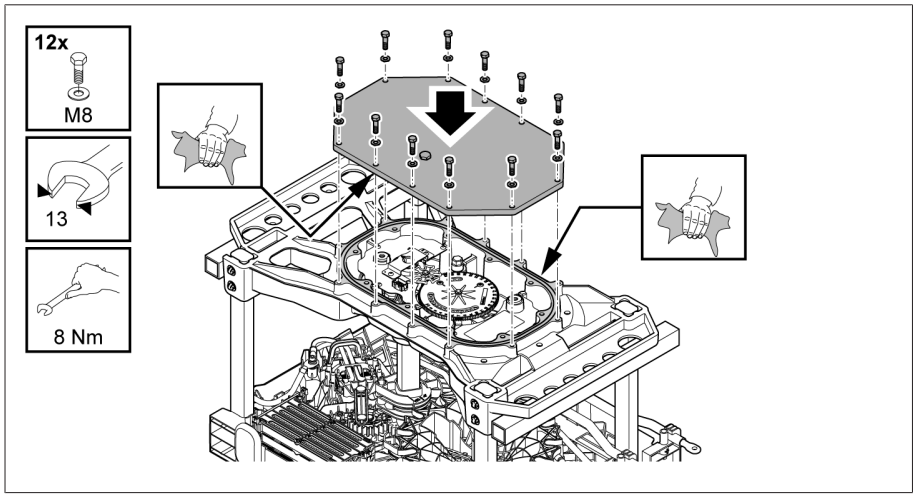


Figure 60: Mounting the guard plate

You can dry the on-load tap-changer using one of the following methods:
- Vacuum-drying – in an autoclave or in the transformer tank
- Vapor-phase drying – in an autoclave or in the transformer tank

5.3.5.1 Vacuum-drying

1. Heat up the on-load tap-changer in air at atmospheric pressure with a temperature increase of approximately 10 °C/h to a final temperature of maximum 110 °C.
2. Pre-dry the on-load tap-changer in circulating air at a maximum temperature of 110 °C for a period of at least 20 hours.
3. Vacuum-dry the on-load tap-changer at between 105°C and maximum 125°C for at least 50 hours.
4. Residual pressure of no more than 10^{-3} bar.

5.3.5.2 Vapor-phase drying

1. Supply kerosene vapor at a temperature of around 90 °C. Keep this temperature constant for 3 to 4 hours.
2. Increase the kerosene vapor temperature by approx. 10°C/hour to the desired final temperature of max. 125°C at the on-load tap-changer.
3. Vacuum-dry the on-load tap-changer at between 105°C and maximum 125°C for at least 50 hours.
4. Residual pressure of no more than 10^{-3} bar.

5.3.6 Removing temporary fasteners and spacers

1. Remove the guard plate from the on-load tap-changer sealing module. To do so, remove the screws.

> Store the screws, washers and guard plate safely. These parts will be needed again in a later mounting step. In order to prevent soiling and damage to the mechanics, do not leave the sealing module open for long periods.

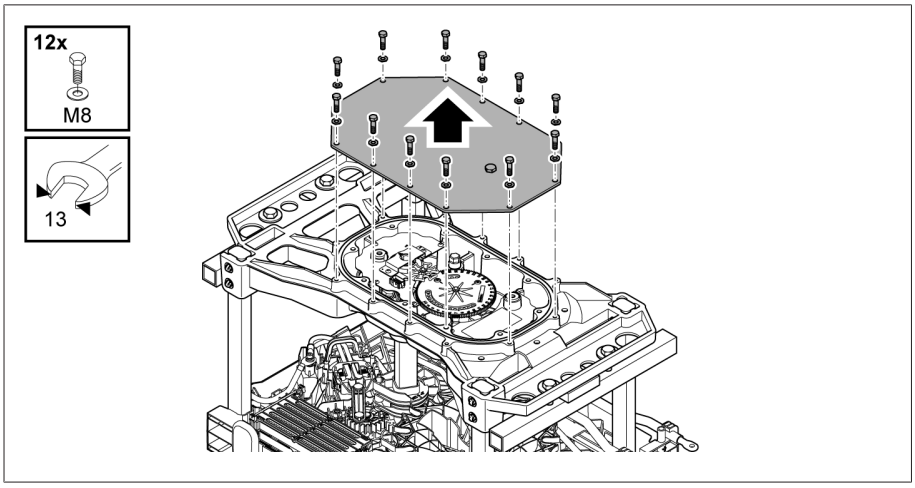


Figure 61: Remove the guard plate

2. Tighten the supplied lifting eye bolts hand-tight.

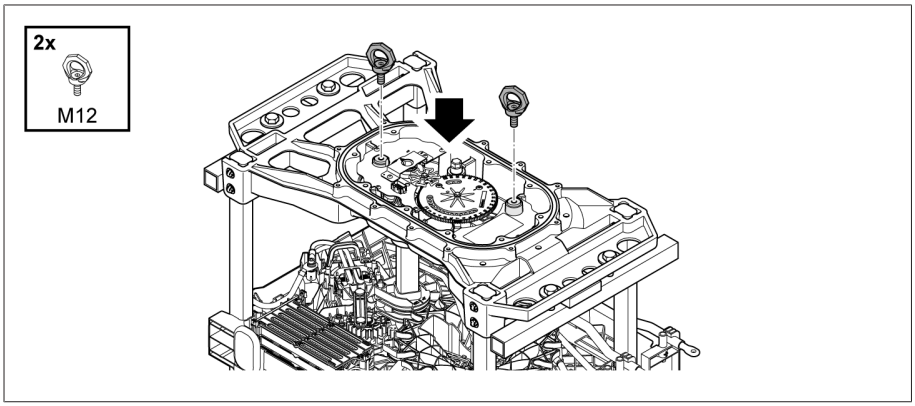


Figure 62: Inserting the lifting eye bolts

3. Lifting the diverter switch using lifting eye bolts

4. Remove temporary fastening and spacers and carefully lower the on-load tap-changer.

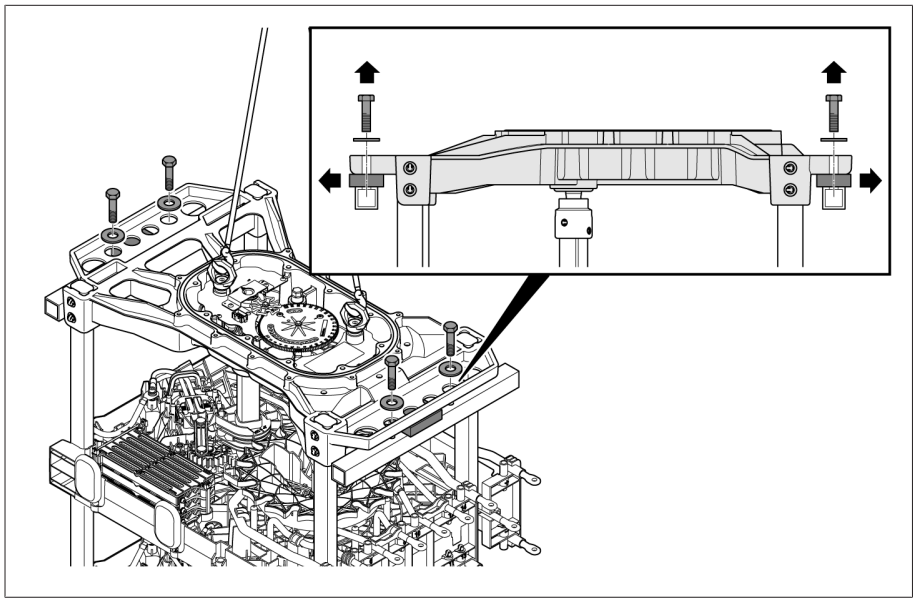


Figure 63: Removing the temporary fasteners

5. Remove the lifting eye bolts.

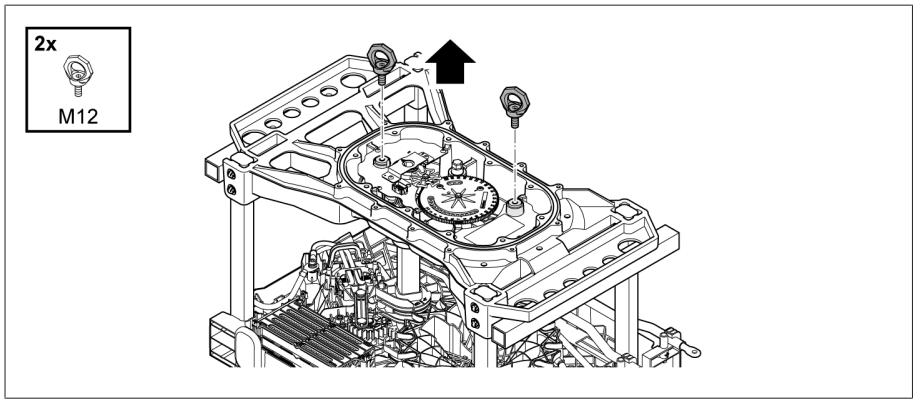


Figure 64: Removing the lifting eye bolts

5.3.7 Positioning the bell-type tank and connecting the on-load tap-changer sealing module to the mounting flange

5.3.7.1 Attaching bell-type tank

1. Clean the sealing surface on the sealing flange and mounting flange. Check that the o-ring gasket sits cleanly in the sealing flange.

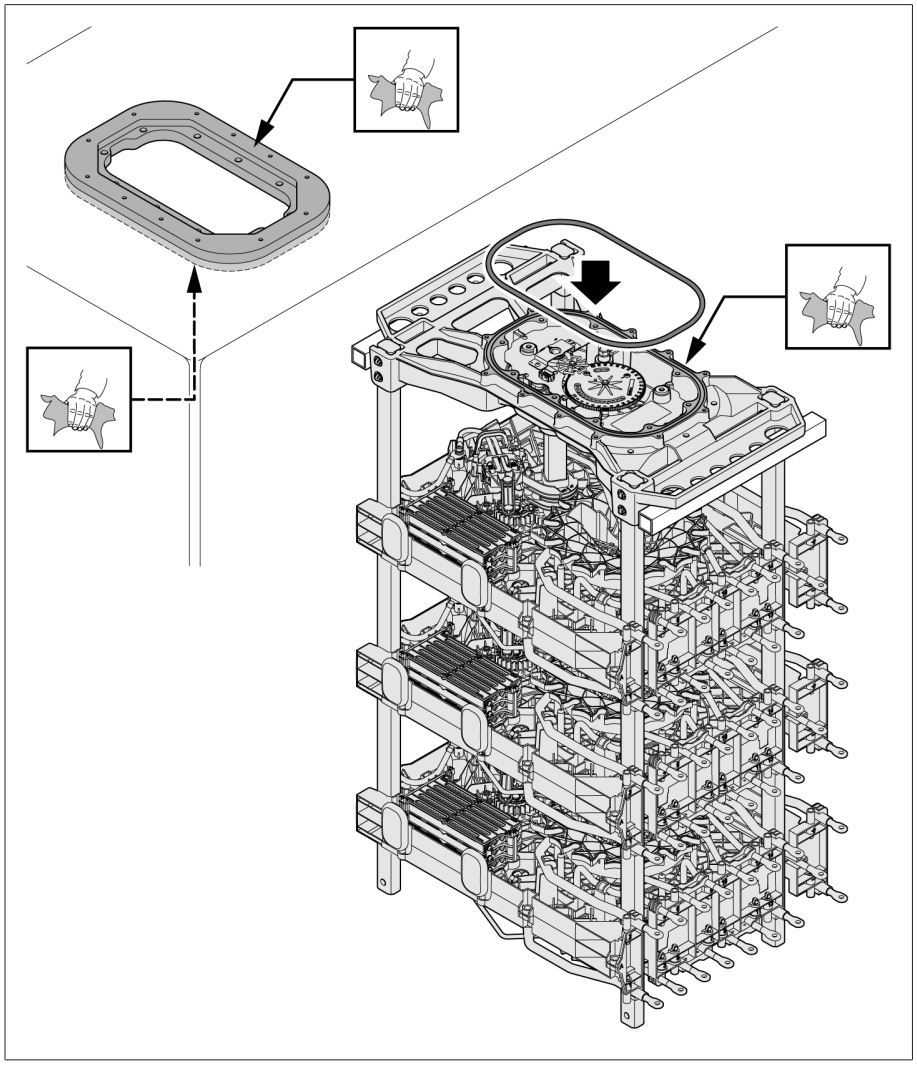


Figure 65: Cleaning the sealing surfaces

NOTICE

Damage to the on-load tap-changer and transformer

Too much misalignment between the mounting flange and the sealing module can lead to damage when the on-load tap-changer is lifted or to not being able to install the mounting device.

› Ensure that the on-load tap-changer sealing module is in line with the bell-type tank mounting flange.

2. Lift the bell-type tank over the active part of the transformer.

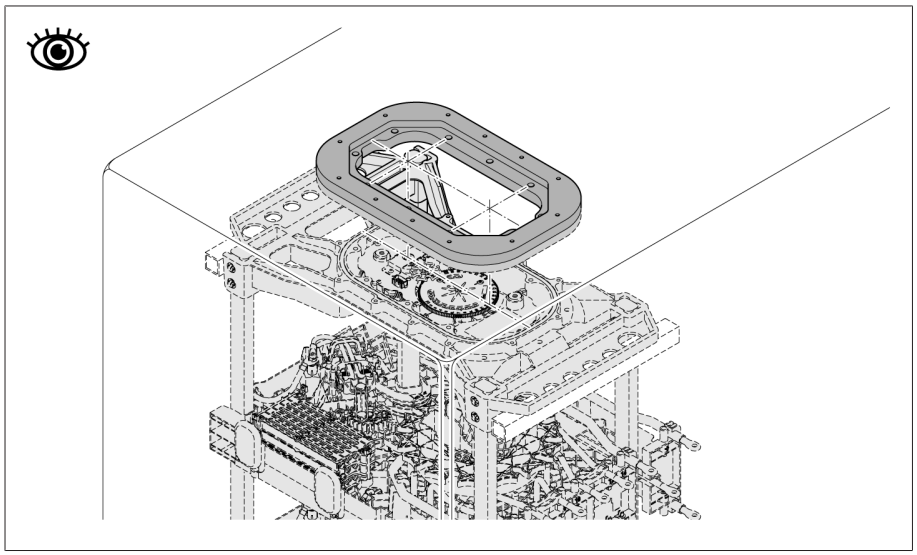


Figure 66: Positioning the bell-type tank

5.3.7.2 Installing the mounting device

⚠ CAUTION! Incorrect installation can lead to damage to the on-load tap-changer. Only install the on-load tap-changer using the original MR mounting device and lifting eye bolts.

1. Place the mounting device on the sealing module on the studs provided for this purpose.

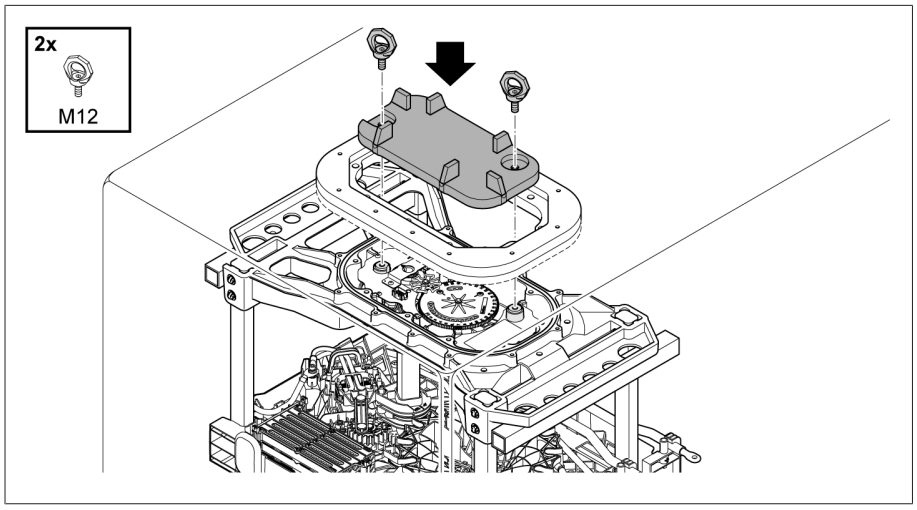


Figure 67: Inserting the mounting device

2. Screw the mounting device to the sealing module hand-tight using two lifting eye bolts. Ensure that the mounting device is level and that the lifting eye bolts are fully screwed in.

5.3.7.3 Connecting the on-load tap-changer to the mounting flange

NOTICE

Damage to the on-load tap-changer.

If the on-load tap-changer is not lifted by the lifting eye bolts provided for this purpose on the mounting device, the on-load tap-changer may be damaged.

› Only lift the on-load tap-changer using the lifting eye bolts on the mounting device.

1. Ensure that the key adapter on the lifting eye bolt is aligned inwards when screwed in to avoid jamming when lifting.
2. *NOTICE!* Inaccurate alignment of the sealing module to the mounting flange will damage the on-load tap-changer when it is lifted. Affix two ropes of the same length to the lifting eye bolts on the mounting device. Slowly lift the on-load tap-changer from below towards the mounting flange. Ensure that the centering studs on the mounting device engage in the recess on the mounting flange. The on-load tap-changer centers itself via the device.

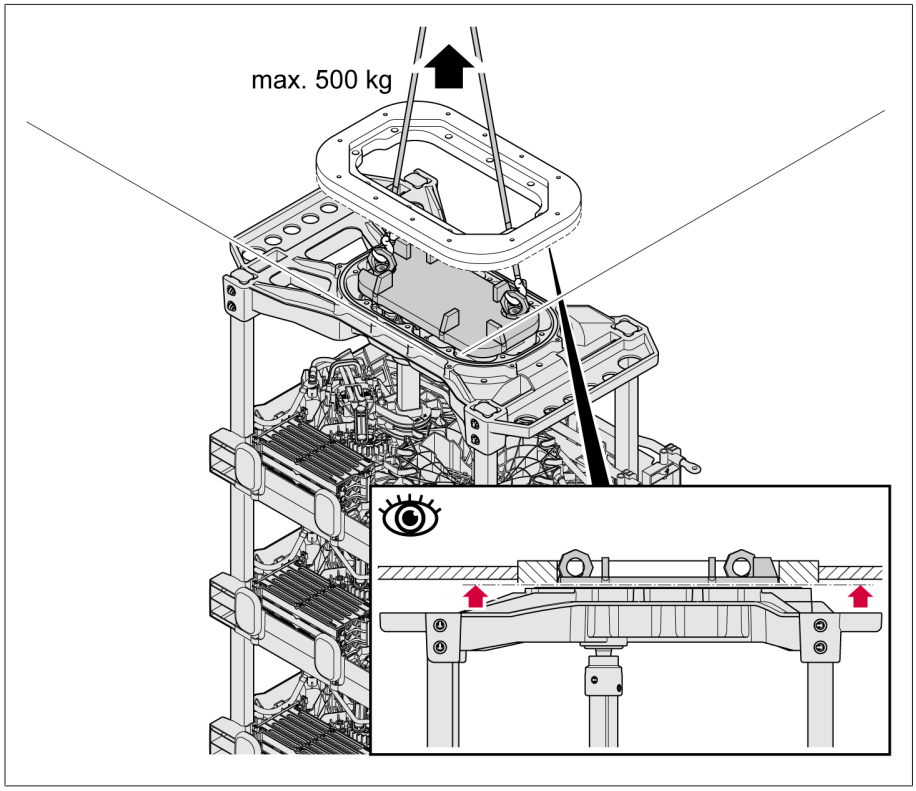


Figure 68: Lifting the on-load tap-changer

3. Screw the on-load tap-changer sealing module to the mounting flange.
4. ⚠ CAUTION! Gradually tighten screws crosswise as described below without warping or deforming. Improper screw connection will result in damage to the on-load tap-changer.

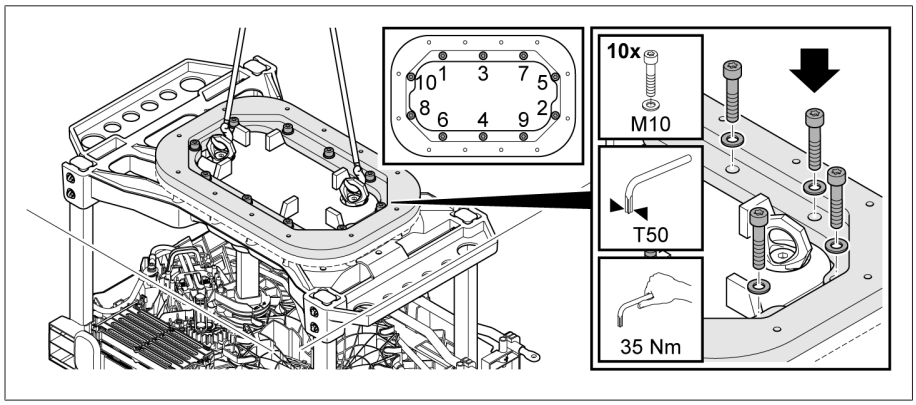


Figure 69: Screwing the on-load tap-changer into place

5. Evenly tighten screws crosswise by hand.
6. Check seat of flange and seal for regularity.
7. Tighten screws crosswise with the full tightening torque of 35 Nm.
8. Tighten screws again with full tightening torque, working clockwise and tightening one screw after another as far as each screw will go.

9. ⚠ WARNING! Carefully remove the mounting device and lifting eye bolts, as serious injuries can occur when the mounting device is lifted out with the crane. Ensure that the mounting device is pulled out evenly. If the mounting device jams in the mounting flange, loosen the jam by tapping it lightly with a mallet.

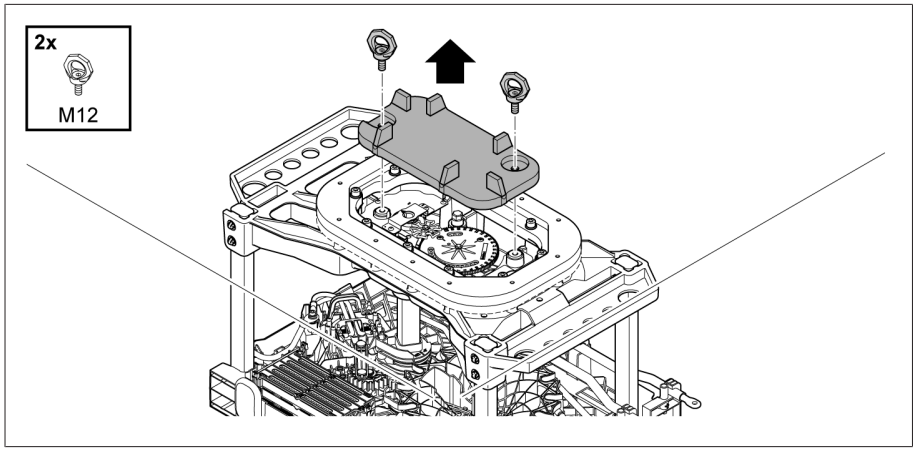


Figure 70: Removing the mounting device

5.3.8 Drying on-load tap-changer in transformer tank

NOTICE

Damage to the on-load tap-changer, motor cover and motor control unit due to drying.

The motor cover and motor control unit will become damaged if they are dried.

- Never leave the motor cover or motor control unit mounted when drying the device.
- Affix the guard plate in accordance with the following description prior to drying to protect the on-load tap-changer.

1. Clean the sealing surfaces on the mounting flange and guard plate.
2. Attach and fasten the guard plate prior to drying. Ensure that the flat gasket is positioned correctly.

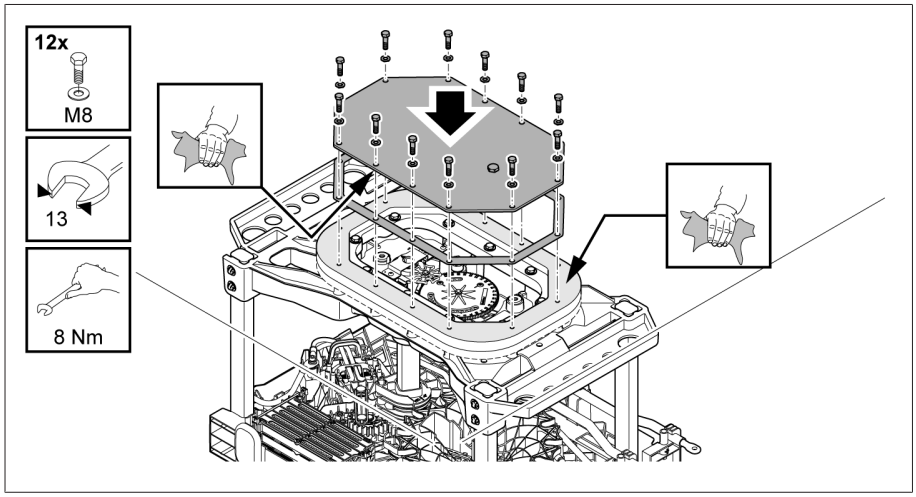


Figure 71: Mounting the guard plate

You can dry the on-load tap-changer using one of the following methods:

- Vacuum-drying – in an autoclave or in the transformer tank
- Vapor-phase drying – in an autoclave or in the transformer tank

5.3.8.1 Vacuum-drying

1. Heat up the on-load tap-changer in air at atmospheric pressure with a temperature increase of approximately 10 °C/h to a final temperature of maximum 110 °C.
2. Pre-dry the on-load tap-changer in circulating air at a maximum temperature of 110 °C for a period of at least 20 hours.
3. Vacuum-dry the on-load tap-changer at between 105°C and maximum 125°C for at least 50 hours.
4. Residual pressure of no more than 10^{-3} bar.

5.3.8.2 Vapor-phase drying

1. Supply kerosene vapor at a temperature of around 90 °C. Keep this temperature constant for 3 to 4 hours.
2. Increase the kerosene vapor temperature by approx. 10°C/hour to the desired final temperature of max. 125°C at the on-load tap-changer.
3. Vacuum-dry the on-load tap-changer at between 105°C and maximum 125°C for at least 50 hours.
4. Residual pressure of no more than 10^{-3} bar.

5.3.9 Filling transformer with oil

NOTICE

Damage to the on-load tap-changer.

Unsuitable insulating fluids cause damage to the on-load tap-changer.

› Only use insulating fluids [▶Section 9.2, Page 80] that have been approved by Maschinenfabrik Reinhausen GmbH.

For the oil filling of the transformer, use new mineral insulating oil for transformers as per IEC 60296 (Specification of unused mineral insulating oils for transformers and switchgear).

If approved by the transformer manufacturer, synthetic ester liquids as per IEC 61099 (Specification for unused synthetic organic esters for electrical purposes) or natural ester liquids as per IEC 62770 (Specification for unused natural esters for transformers and similar electrical equipment) can be used as alternatives.

Contact Maschinenfabrik Reinhausen GmbH if you want to use an alternative insulating fluid.

Observe the permitted temperature range of the transformer oil in the chapter Permissible ambient conditions [▶Section 9.2, Page 80].

NOTICE

Damage to the on-load tap-changer.

If a transformer is not completely filled with oil, the on-load tap-changer may be damaged.

› Ensure that the transformer is completely filled with transformer oil.

✓ Before commissioning the on-load tap-changer, completely fill the transformer with oil.

1. Completely fill the transformer with oil under vacuum.
2. Take an oil sample from the transformer.
3. Record the temperature of the oil sample just after the sample is taken.
4. Determine dielectric strength and water content at an oil temperature of 20 °C ±5 °C. The limit values listed in chapter Limit values for dielectric strength and water content of insulating fluids [▶Section 9.3, Page 81] must be complied with.

5.3.10 Mounting the motor cover and control cabinet

5.3.10.1 Checking adjustment position

1. Remove the guard plate and flat gasket from the on-load tap-changer sealing module. To do so, remove the screws. Note: Store the screws, washers, guard plate and flat gasket safely. These parts will be needed again in a later mounting step.
   NOTICE! Only use the flat gasket in combination with the guard plate! The motor cover with mounting flange is sealed using an o-ring. In order to prevent soiling and damage to the mechanics, do not leave the sealing module open for long periods of time.

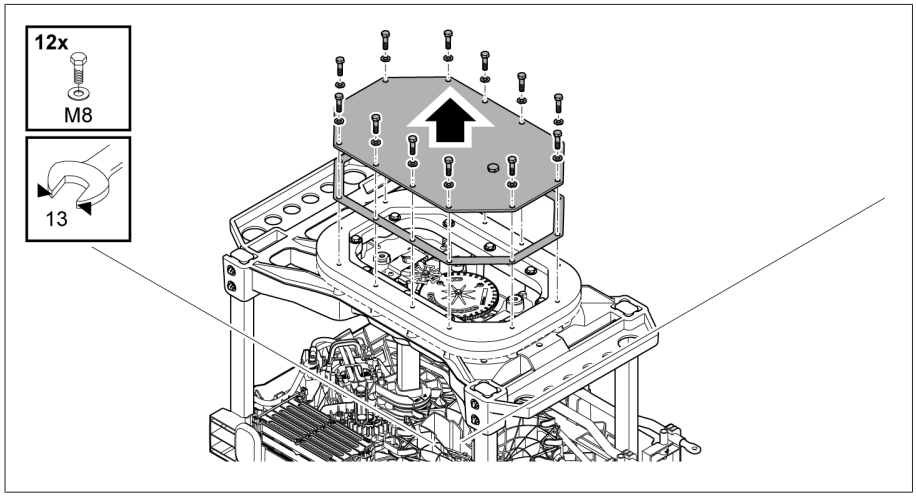


Figure 72: Removing the guard plate and flat gasket

2. Ensure that the on-load tap-changer is in the adjustment position.

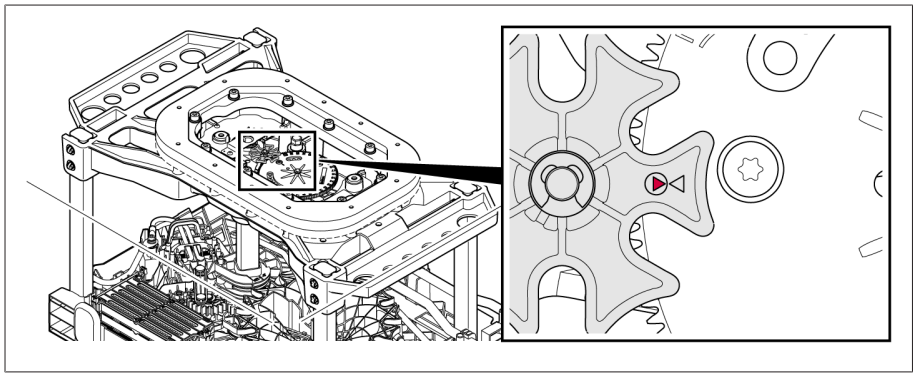


Figure 73: Checking the adjustment position

If the on-load tap-changer is not in the adjustment position, it must be switched into the adjustment position using the hand crank adapter:

1. Remove the hand crank adapter installed in the sealing module and insert it into the hand crank aperture.

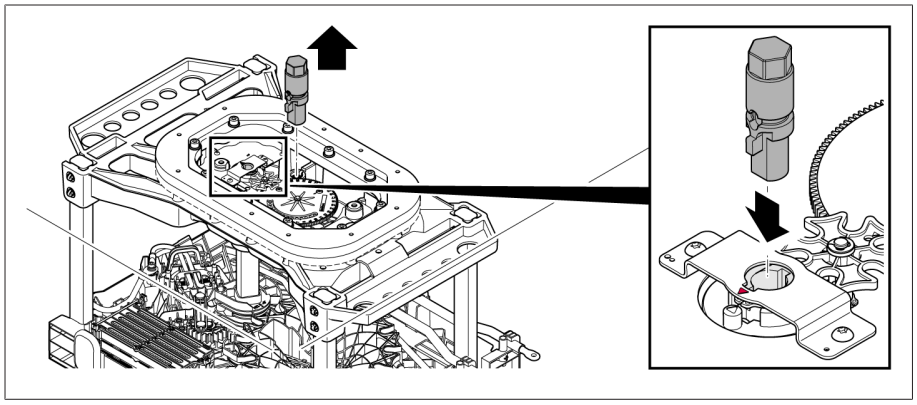


Figure 74: Hand crank adapter

2. Turn the hand crank adapter using a suitable tool (wrench size 17) until the adjustment position is reached.
3. Remove the hand crank adapter and replace it in the mounting bracket.

5.3.10.2 Mounting the motor cover

1. Ensure that the on-load tap-changer is in the adjustment position.

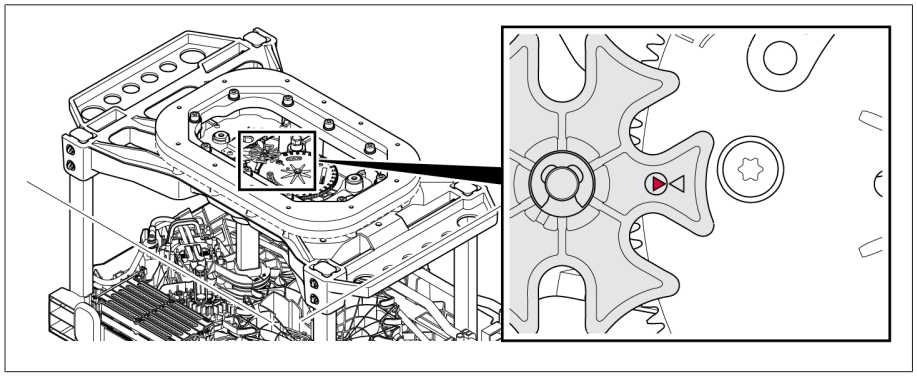


Figure 75: Checking the adjustment position

2. Clean the sealing surface of the motor cover and the mounting flange. Check that the o-ring gasket sits cleanly in the motor cover.

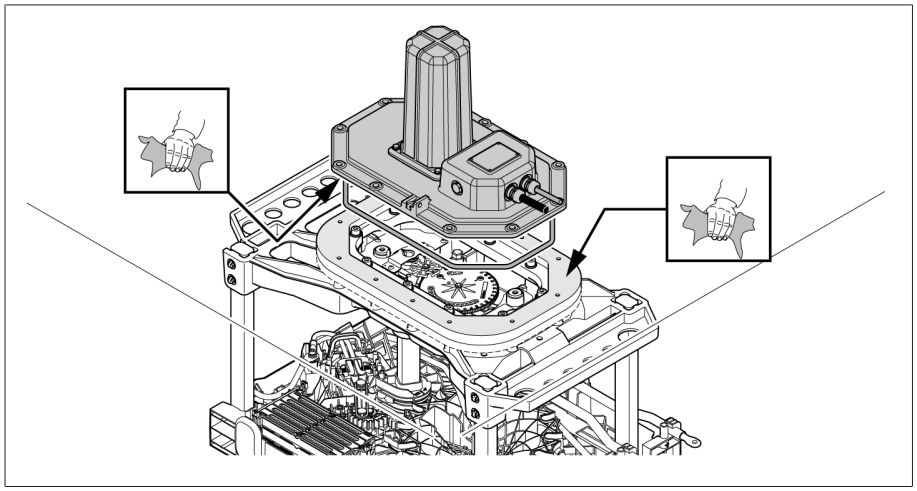


Figure 76: Cleaning sealing surfaces

3. Align the motor shaft so that the feather key is in line with the triangle mark on the motor plate.

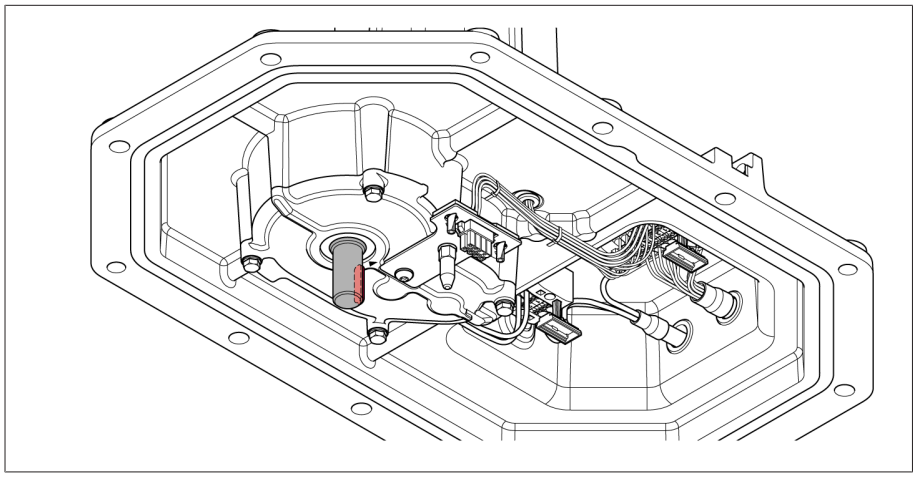


Figure 77: Triangle mark

4. When doing so, ensure that the shaft does not become damaged or scratched during alignment.

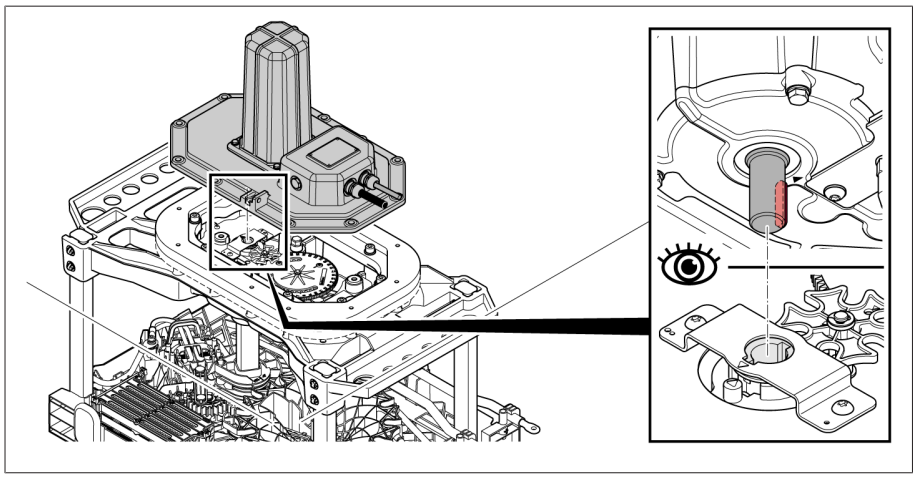


Figure 78: Aligning the motor shaft

5. Put the motor cover on so that the feather key on the drive shaft of the motor-drive unit slides into the keyway on the drive shaft of the on-load tap-changer in the sealing module and that at the same time the centering bolt in the motor plate engages in the locating hole provided for this in the sealing module. This ensures that the plugs of the micro-switch wiring in the motor cover and the sealing module are aligned.

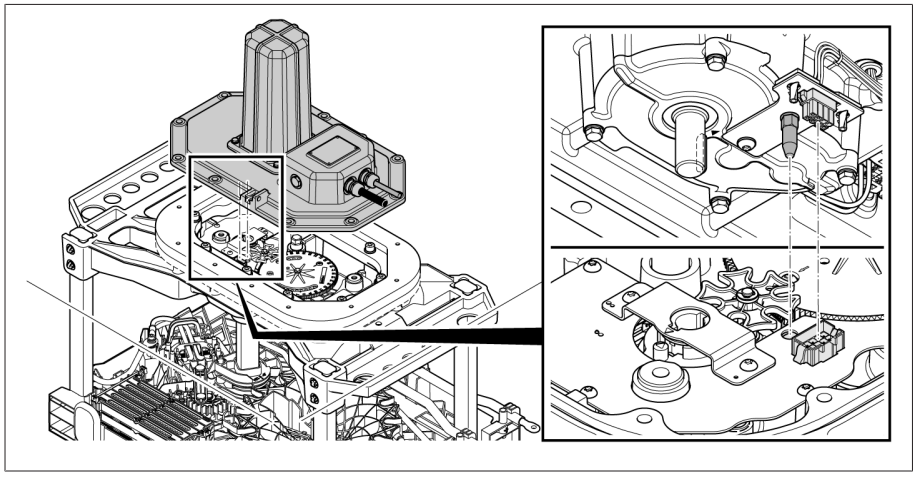


Figure 79: Centering

6. Screw the motor cover to the mounting flange, tightening the screws crosswise. To do so, use the screws and washers provided by Maschinenfabrik Reinhausen.

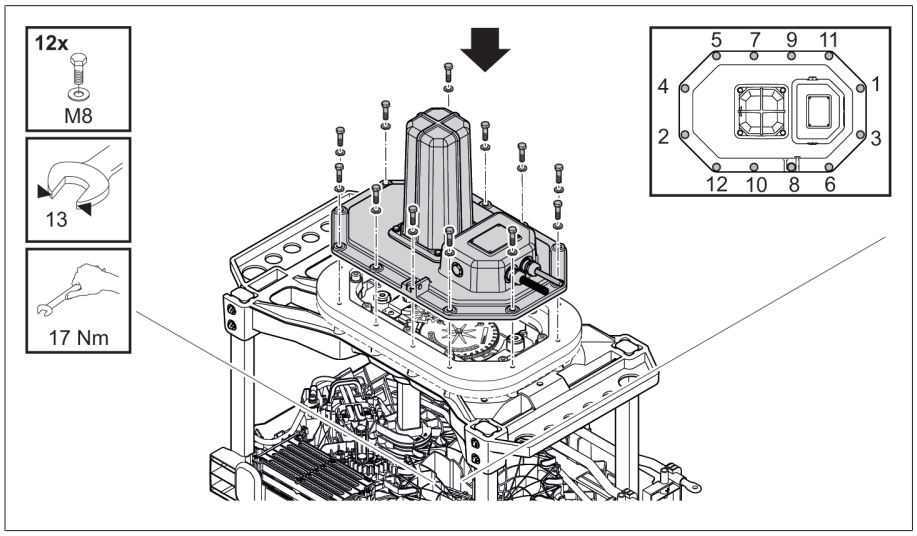


Figure 80: Screwing the motor cover on

5.3.10.3 Fitting the control cabinet

› Fit the ETOS® ID or Tap Changer Control control cabinet to the transformer in accordance with the corresponding Maschinenfabrik Reinhausen operating instructions.

5.3.10.4 Connecting the on-load tap-changer to the control cabinet

› Connect the control cabinet in accordance with the corresponding Maschinenfabrik Reinhausen ETOS® ID or Tap Changer Control operating instructions.

5.3.10.5 Electrically connecting the control cabinet and motor cover

› Connect the motor cover and ETOS® ID or Tap Changer Control control cabinet electrically in accordance with the corresponding Maschinenfabrik Reinhausen operating instructions.

5.3.10.6 Performing homing

› Perform a homing operation in accordance with the ETOS® ID or the Tap Changer Control user guide.

Homing is not necessary with on-load tap-changers that have the AutoSync function. The control system can automatically detect the position of the on-load tap-changer via the built-in sensor.

NOTICE

Damage to the on-load tap-changer.

Without homing, the on-load tap-changer may become damaged.

› After homing, do not separate the motor cover from the sealing module again.

› Repeat homing if the motor cover is separated from the sealing module.

5.3.10.7 Driving to the pass-through position

For the transformer ratio test, DC resistance measurement or high-voltage tests, it may be necessary to drive to the pass-through positions individually.

Further information on this is available in the chapter "Setup mode" in the Maschinenfabrik Reinhausen ETOS® ID or Tap Changer Control control cabinet operating instructions.

5.3.11 Performing the transformer ratio test

Before taking measurements, observe the section "Fitting the motor cover and control cabinet".

NOTICE

Damage to the on-load tap-changer.

Switching the on-load tap-changer via the hand crank can lead to damage to the on-load tap-changer.

- Only switch the on-load tap-changer via the control cabinet.
- Do not remove the motor cover.

NOTICE

Damage to the on-load tap-changer.

Dry tap-change operations lead to damage to the on-load tap-changer.

- If the on-load tap-changer is not yet in the oil-filled transformer tank, moisten all movable and fixed on-load tap-changer contacts with transformer oil before taking measurements.

1. Switch the on-load tap-changer into the desired operating position.
2. Perform the transformer ratio test.
3. Repeat the transformer ratio test in all operating positions.
4. Switch the on-load tap-changer into the adjustment position (see the accompanying connection diagram for the on-load tap-changer).

6 Commissioning

⚠ WARNING



Danger of explosion.

Explosive gases in the transformer, pipe system, oil conservator and at the dehydrating breather opening can deflagrate or explode and result in severe injury or death.

- Ensure that there are no ignition sources such as naked flame, hot surfaces or sparks (e.g. caused by the build-up of static charge) in the transformer's immediate surroundings during commissioning and that none occur.
- Do not operate any electrical devices (e.g. risk of sparks from impact wrench).
- Only use conductive and grounded hoses, pipes and pump equipment that are approved for flammable liquids.

⚠ WARNING



Danger of explosion!

Overloading the on-load tap-changer can lead to explosion and thus to death or serious injury.

- Ensure that the on-load tap-changer is not overloaded.
- Ensure use of the on-load tap-changer in accordance with section "Appropriate use".
- Prevent operations outside of the permitted operating conditions by taking suitable measures.
- The tap changer does not have an integrated protective device. The transformer manufacturer is responsible for the selection and dimensioning of the transformer protective devices. The protective devices must still function in the event of tap changer errors.

6.1 Commissioning the on-load tap-changer at the transformer manufacturer's site

Perform the following work and functional checks before commissioning the transformer.

6.1.1 Connecting the grounding cable

Connect the on-load tap-changer to the transformer cover via the grounding screw on the motor cover (see dimensional drawing [▶10186367]).

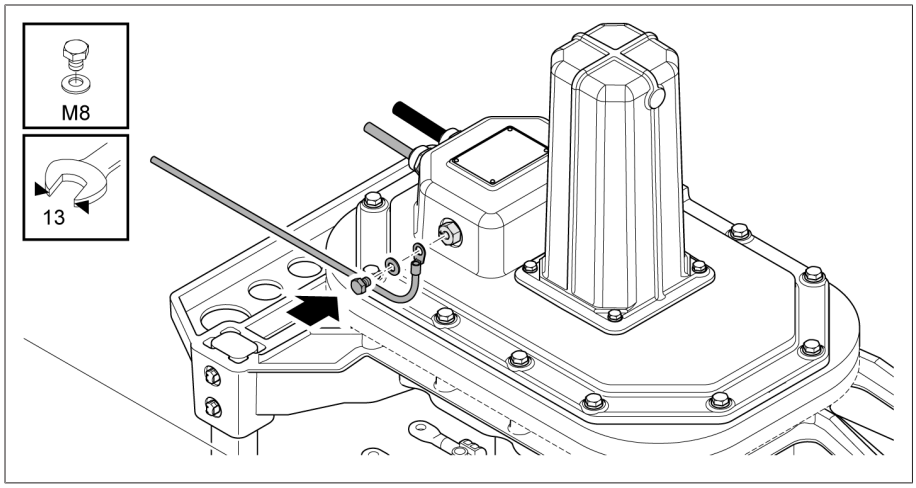


Figure 81: Grounding the on-load tap-changer

Further information on connecting the grounding cable to the control cabinet is available in the corresponding Maschinenfabrik Reinhausen ETOS® ID or Tap Changer Control operating instructions.

6.1.2 Commissioning the control cabinet

A description of how to commission the control cabinet is given in the control cabinet operating instructions.

6.1.3 Driving to the pass-through position

For the transformer ratio test, DC resistance measurement or high-voltage tests, it may be necessary to drive to the pass-through positions individually.

Further information on this is available in the chapter "Setup mode" in the Maschinenfabrik Reinhausen ETOS® ID or Tap Changer Control control cabinet operating instructions.

6.1.4 High-voltage tests on the transformer

Before performing the lightning and impulse voltage test on the transformer, the following must be ensured:
- The connecting cable between the sealing module and the control cabinet does not need to be disconnected.
- Disconnect the control cabinet from the electrical supply grid.
- Disconnect the supply cable from the control cabinet.
- Coil the supply cable and position it safely directly on the control cabinet.

Note the following points before performing high-voltage tests on the transformer:
- Ensure that the transformer is completely filled with transformer oil.
- Ensure that all protective devices for the on-load tap-changer are functioning correctly and are ready for use.
- Check that the ground connections on the control cabinet and the control cabinet fixings are free of paint.
- Only perform a high voltage test with the control cabinet door closed.
- Disconnect the sensor cable and other external connections to electronic components in the control cabinet to prevent damage through overvoltage.
- Remove the bushing adapter and fit the cap of the bushing test tap.
- When connecting the control cabinet supply voltage, only use the cable bushings provided for this in the control cabinet base.
- Guide all ground connecting leads to one central connection point (establishment of a suitable reference earth).
- Disconnect all electronic components before the high voltage test. Before a dielectric test of the wiring, remove all devices with a withstand voltage of < 1,000 V.
- Remove leads used for testing before the high voltage test, because these function as antennas.
- Wherever possible, route the measurement leads and data leads separately to the energy cables.

Contact the manufacturer if you have any questions about possible sources of danger.

6.1.5 Preparing the on-load tap-changer for transport

› Perform a homing operation in accordance with the ETOS® ID or the Tap Changer Control user guide.

Homing is not necessary with on-load tap-changers that have the AutoSync function. The control system can automatically detect the position of the on-load tap-changer via the built-in sensor.

NOTICE

Damage to the on-load tap-changer.

Without homing, the on-load tap-changer may become damaged.

› After homing, do not separate the motor cover from the sealing module again.

› Repeat homing if the motor cover is separated from the sealing module.

6.2 Transporting transformer to the operating site

NOTICE

On-load tap-changer

Damage to the on-load tap-changer.

An impermissibly high pressure load can lead to damage to the sealing points and deformations on the on-load tap-changer.

› Ensure that the maximum permissible pressure of 1.5 bar is not exceeded at the on-load tap-changer.

› Move the on-load tap-changer to its adjustment position (see supplied connection diagram).

NOTICE

Control cabinet

Damage to the device!

Damage to the device due to condensate in the control cabinet.

› Always keep the control cabinet tightly closed.

› In the event of downtimes of more than 8 weeks prior to initial commissioning or an operational interruption of more than 2 weeks, connect and operate the anti-condensation heater in the control cabinet. If this is not possible, place a sufficient amount of desiccant (silicon-free) in the control cabinet.

› If the control cabinet is to be dismantled for transportation, dismantle the control cabinet in accordance with the corresponding operating instructions and transport it to the installation location in the MR delivery packaging.

6.3 Commissioning transformer at operating site

6.3.1 Commissioning the control cabinet

A description of how to commission and test the control cabinet is given in the ETOS control cabinet operating instructions.

6.3.2 Checking the function of the Buchholz relay

Check that the Buchholz relay is functioning correctly in accordance with the corresponding operating instructions before commissioning the transformer

6.3.3 Securing the motor cover with a padlock

⚠ WARNING



Danger of explosion!

Operating the hand crank with the transformer energized can lead to death, serious injury and material damage due to overload of the on-load tap-changer.

- Never actuate the on-load tap-changer with the hand crank if the transformer is energized.
- De-energize the transformer before using the hand crank.
- Attach the padlock to prevent unauthorized opening. The padlock is not included in the MR scope of delivery.

- Secure the motor cover with a padlock before you commission the transformer.

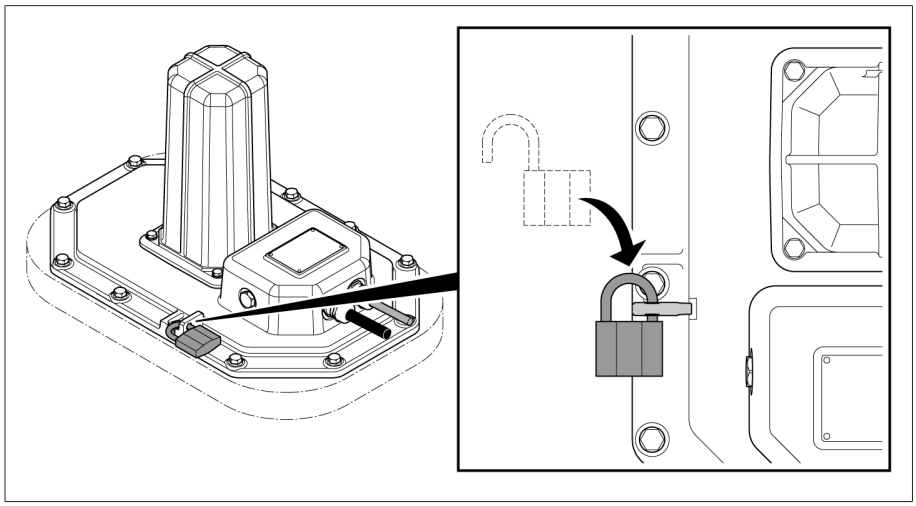


Figure 82: Securing the motor cover with a padlock

6.3.4 Commissioning the transformer

- ✓ The signaling contact for falling below the minimum insulating fluid fill level in the transformer oil conservator is looped into the tripping circuit of the circuit breaker.
- ✓ The Buchholz relay and additional protective devices are looped into the circuit breaker's tripping circuit.
- ✓ The control cabinet and all protective devices are functioning correctly and are ready for use.
- ✓ The transformer is completely filled with insulating fluid.

⚠ WARNING



Electric shock

Remaining in the danger zone can result in death or severe injuries

- Ensure that nobody is in the danger zone when engaging the transformer circuit breaker.

1. Switch on the transformer.
2. *NOTICE!* Inrush current impulses can be significantly greater than the transformer rated current and may lead to current paths with asymmetrical or non-sinusoidal curve shapes and, as a result, overload the on-load tap-changer during the diverter switch operation. Only perform tap-change operations – whether under no load or under load conditions – once the inrush current impulse has subsided.

7 Fault elimination

⚠ WARNING



Danger of explosion.

Explosive gases under the sealing module can deflagrate or explode and result in severe injury or death.

- Ensure that there are no ignition sources such as open flames, hot surfaces or sparks (e.g. caused by the build-up of static charge) in the immediate surroundings and that none occur.
- De-energize all auxiliary circuits before removing the maintenance opening.
- Do not operate any electrical devices during the work (e.g. risk of sparks from impact wrench).

NOTICE

Damage to the on-load tap-changer and transformer.

If the transformer Buchholz relay or other protective devices trip, this can indicate damage to the on-load tap-changer and transformer. The transformer must not be energized without being inspected first.

- Check the on-load tap-changer and the transformer when the Buchholz relay or other protective devices have tripped.
- Do not use the equipment again until you are sure there is no damage to the on-load tap-changer or transformer.

NOTICE

Damage to the device!

Damage to the device due to condensate in the control cabinet.

- Always keep the control cabinet tightly closed.
- In the event of downtimes of more than 8 weeks prior to initial commissioning or an operational interruption of more than 2 weeks, connect and operate the anti-condensation heater in the control cabinet. If this is not possible, place a sufficient amount of desiccant (silicon-free) in the control cabinet.

The table below is intended to assist with detecting and, where possible, remedying faults.

For more information, please consult the operating instructions of the respective protective device.

In the event of faults on the on-load tap-changer and motor-drive unit which cannot be easily corrected on site or if the protective relay or additional protective devices have been tripped, please inform your authorized MR representative, the transformer manufacturer or contact MR directly.

Maschinenfabrik Reinhausen GmbH
Technical Service
Postfach 12 03 60
93025 Regensburg
Germany
Phone: +49 94140 90-0

Fax: +49 9 41 40 90-7001
E-mail: service@reinhausen.com
Internet: www.reinhausen.com

Fault description	Measure
On-load tap-changer not changing tap position (sluggishness, Raise keys / Lower keys not working, no audible diverter switch action)	Contact Maschinenfabrik Reinhausen.
No change in voltage on the transformer despite change in position on motor-drive unit	Contact Maschinenfabrik Reinhausen.
Tap position indicator on motor-drive unit and on-load tap-changer different	Carry out a homing operation in accordance with the control cabinet operating instructions.
Warning or tripping of Buchholz relay on transformer	Contact the transformer manufacturer and, if necessary, Maschinenfabrik Reinhausen.
Deviation from desired value when measuring winding resistance of the transformer	Contact the transformer manufacturer and, if necessary, Maschinenfabrik Reinhausen, and provide measured values.
Deviation from desired value during dissolved gas analysis (transformer oil)	Contact the transformer manufacturer and, if necessary, Maschinenfabrik Reinhausen, and provide measured values.
Deviation from desired value during transformer ratio test	Contact the transformer manufacturer and, if necessary, Maschinenfabrik Reinhausen, and provide measured values.
Deviation from limit value for insulating fluids	Change insulating fluid, check oil conservator breather.

Table 7: Fault elimination

7.1 Tripping of the Buchholz relay

⚠ WARNING



Danger of explosion.

Explosive gases in the Buchholz relay can deflagrate or explode and result in death or severe injuries.

- Wait 15 minutes after switching off the transformer before beginning further work on the Buchholz relay so that the gases can dissipate.
- Ensure that there are no ignition sources such as open flames, hot surfaces or sparks (e.g. caused by the build-up of static charge) in the immediate surroundings and that none occur.
- De-energize all auxiliary circuits before beginning work.
- Do not operate any electrical devices during the work (e.g. risk of sparks from impact wrench).

⚠ WARNING



Danger of death or serious injury!

Danger of severe injury or death if the on-load tap-changer and the transformer are insufficiently tested.

- It is imperative that you contact Maschinenfabrik Reinhausen to check the on-load tap-changer and the transformer if the Buchholz relay has tripped.
- Only use the equipment again when you are sure there is no damage to the on-load tap-changer or the transformer.

When the circuit breakers have been tripped by the Buchholz relay, proceed as follows:

1. Establish the time of tripping.
2. Determine the operating position of the on-load tap-changer.
3. Switch off the control cabinet via the supply fuse.
4. Check the on-load tap-changer. If insulating fluid is leaking, close the oil conservator stop valve immediately.

5. Read the Buchholz relay operating instructions.

7.2 Operating the on-load tap-changer using the hand crank

⚠ WARNING



Danger of explosion!

Operating the hand crank with the transformer energized can lead to death, serious injury and material damage due to overload of the on-load tap-changer.

- Never actuate the on-load tap-changer with the hand crank if the transformer is energized.
- De-energize the transformer before using the hand crank.

In error-free operation, the on-load tap-changer is only switched via the electronic control unit in the control cabinet.

However, if there is an urgent need to perform a tap-change operation while the electronic control unit is not functioning correctly, you can switch the on-load tap-changer using the hand crank.

1. De-energize and lock the transformer and the control cabinet to prevent unintentional restart.
2. Remove the padlock from the drive.

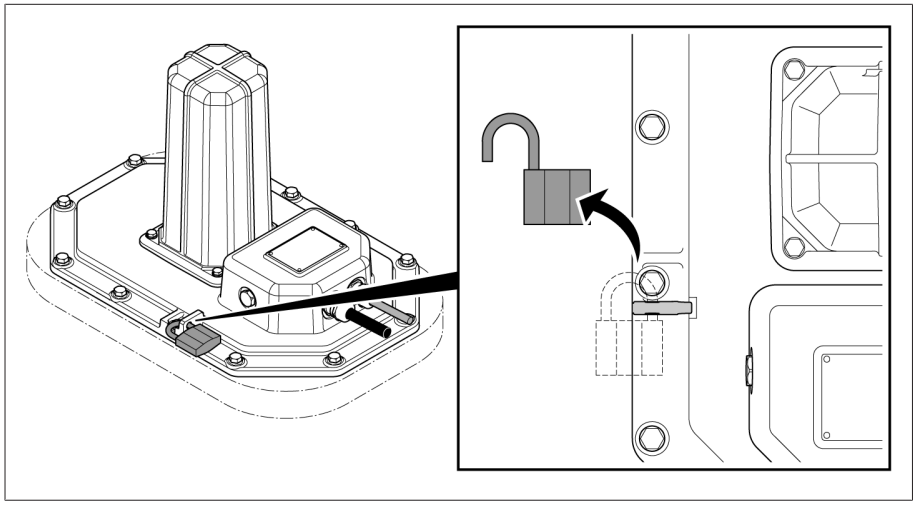


Figure 83: Removing the padlock

3. Remove the screws from the motor cover and mounting flange.

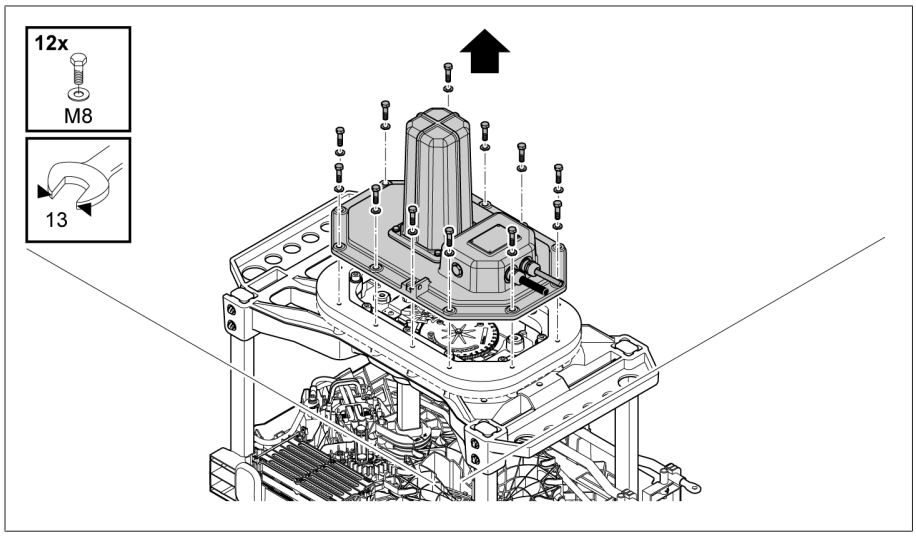


Figure 84: Removing the sealing module cover

4. Detach and remove the motor cover. To simplify removal of the motor cover, 4 tapped holes are provided in the motor cover. By screwing in screws, the motor can be pulled out evenly.

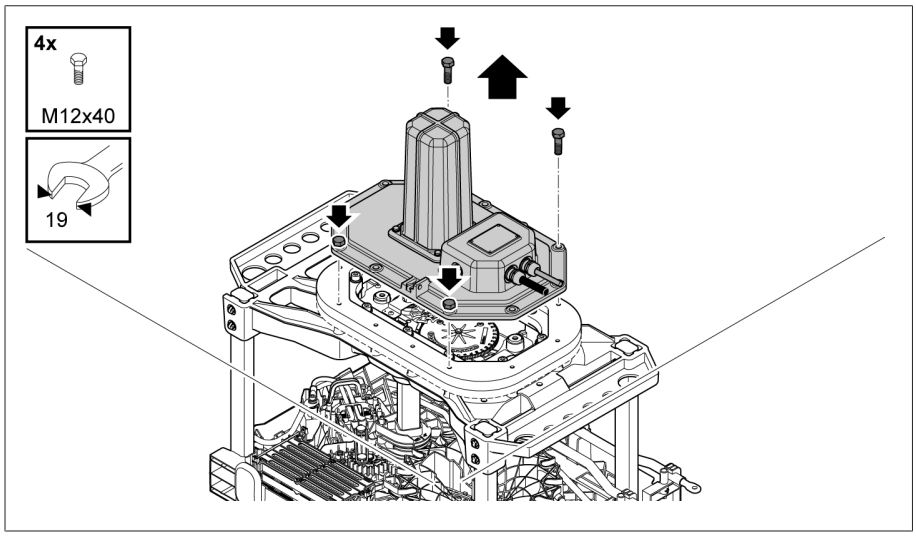


Figure 85: Pulling the motor cover off

5. Remove the hand crank adapter installed in the sealing module and insert it into the hand crank aperture.

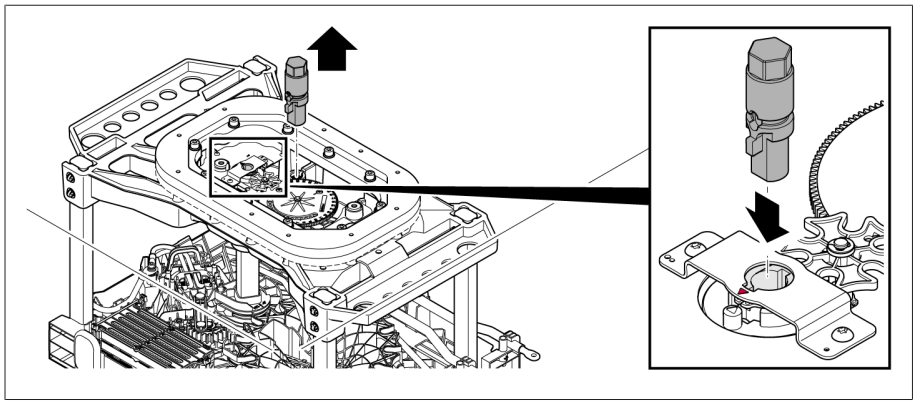


Figure 86: Hand crank adapter

6. Turn the hand crank adapter using a suitable tool (wrench size 17) until the desired operating position is reached and the tap-change operation has been ended properly. *NOTICE!* If there is a blockage or the hand crank is difficult

to move, stop turning the hand crank adapter and contact Maschinenfabrik Reinhausen GmbH. When the tap-change operation has been properly completed, the positions of the feather key groove and the red triangle will match and the corresponding operating position will be indicated on the dial.

7. Remove the hand crank adapter and replace it in the mounting bracket.

7.3 Mounting the motor cover

1. Clean the sealing surface of the motor cover and the mounting flange. Check that the o-ring gasket sits cleanly in the motor cover.

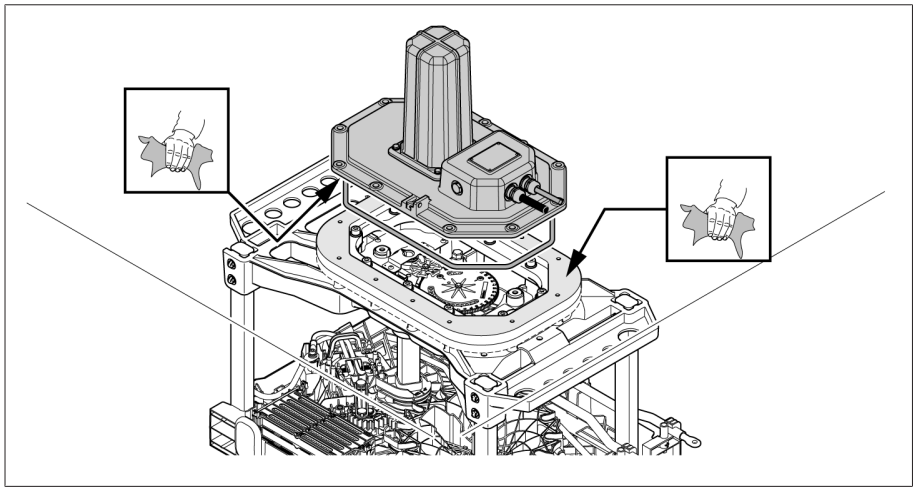


Figure 87: Cleaning sealing surfaces

2. Align the motor shaft so that the feather key is in line with the triangle mark on the motor plate.

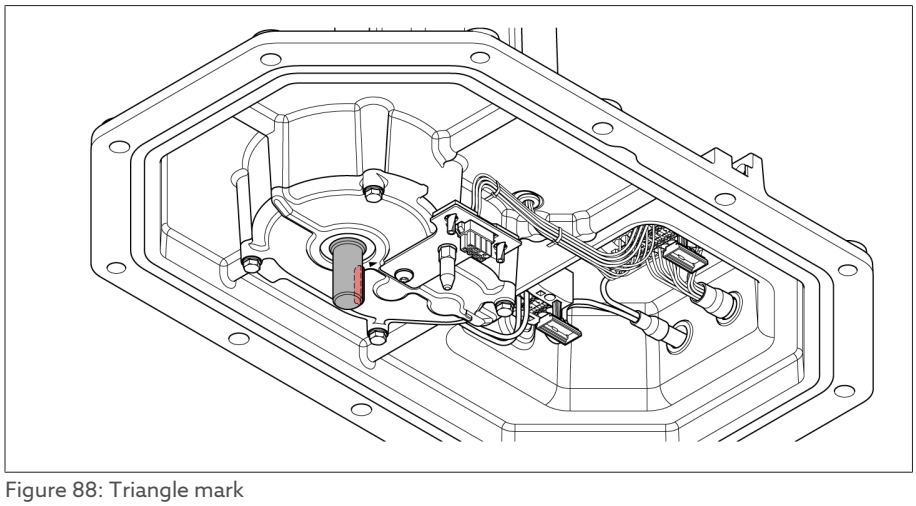


Figure 88: Triangle mark

3. When doing so, ensure that the shaft does not become damaged or scratched during alignment.

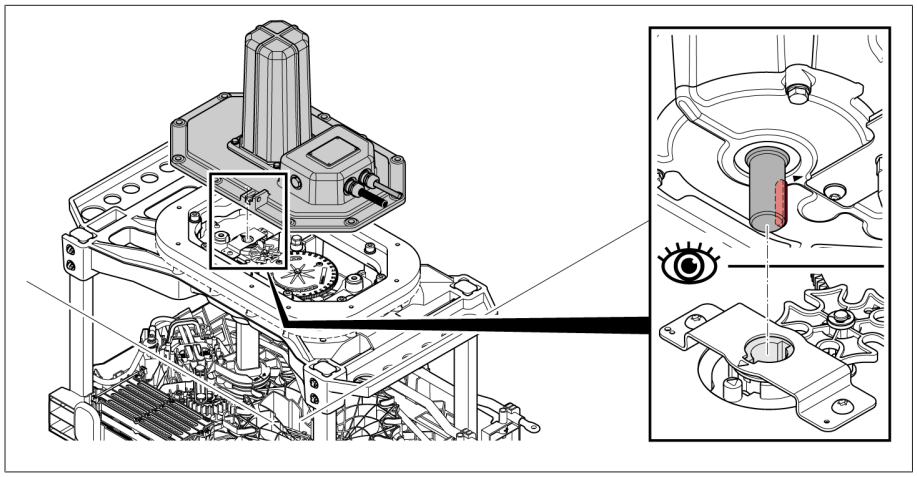


Figure 89: Aligning the motor shaft

4. Put the motor cover on so that the feather key on the drive shaft of the motor-drive unit slides into the keyway on the drive shaft of the on-load tap-changer in the sealing module and that at the same time the centering bolt in the motor plate engages in the locating hole provided for this in the sealing module. This ensures that the plugs of the micro-switch wiring in the motor cover and the sealing module are aligned.

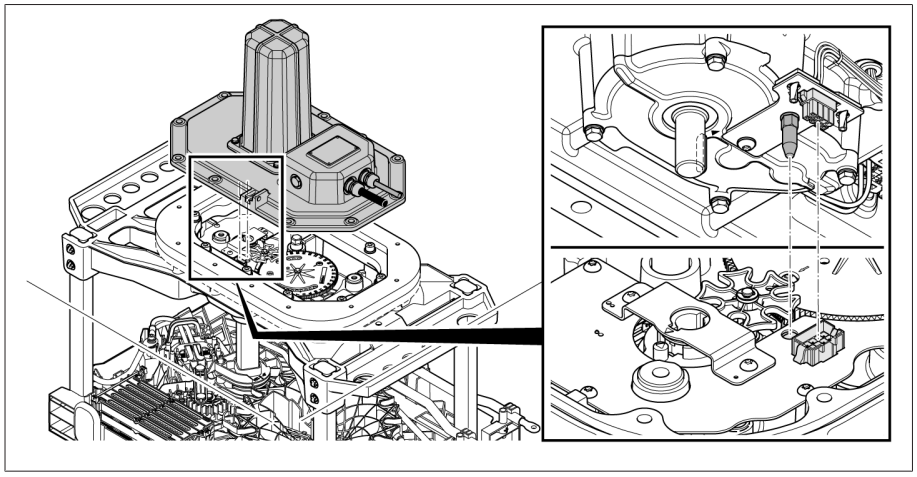


Figure 90: Centering

5. Screw the motor cover to the mounting flange, tightening the screws crosswise. To do so, use the screws and washers provided by Maschinenfabrik Reinhausen.

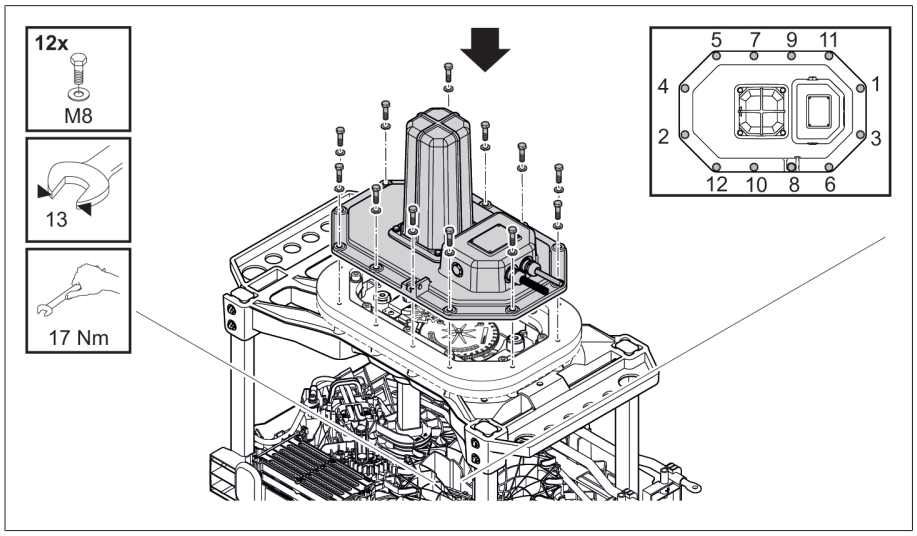


Figure 91: Screwing on the drive

6. Secure the motor cover with a padlock to prevent unauthorized opening.

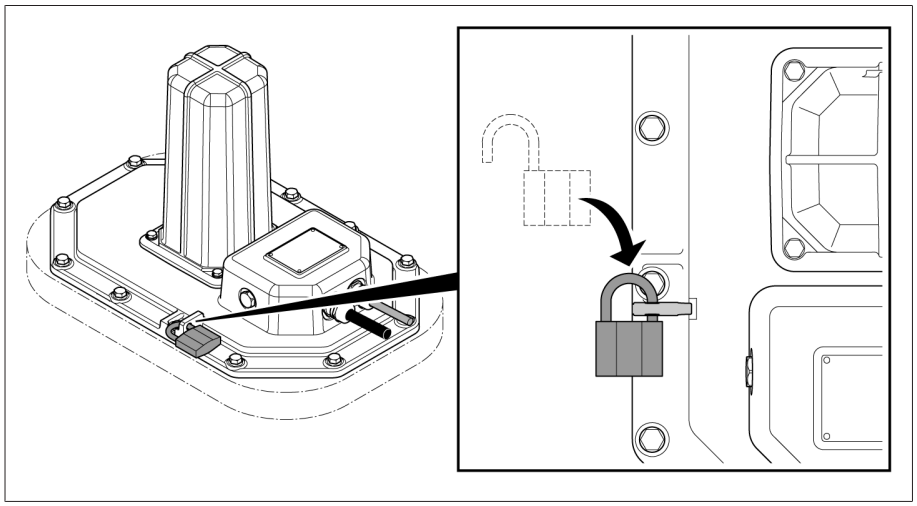


Figure 92: Securing the motor cover with a padlock

7.4 Performing homing

› Perform a homing operation in accordance with the ETOS® ID or the Tap Changer Control user guide.

Homing is not necessary with on-load tap-changers that have the AutoSync function. The control system can automatically detect the position of the on-load tap-changer via the built-in sensor.

NOTICE

Damage to the on-load tap-changer.

Without homing, the on-load tap-changer may become damaged.

› After homing, do not separate the motor cover from the sealing module again.

› Repeat homing if the motor cover is separated from the sealing module.

8 Disposal

For disposal, observe the national requirements applicable in the country of use.

If you have any questions about disassembly and disposal, please contact Maschinenfabrik Reinhausen GmbH's Technical Service department.

8.1 SVHC information in accordance with the REACH regulation

This product complies with the provisions of European Regulation 1907/2006/EC dated December 18, 2006 on the Registration, Evaluation, Authorization and Restriction of Chemicals (REACH).

The following components of the product contain > 0.1% [w/w] of the SVHC substance lead (CAS no. 7439-92-1):

- Aluminum alloy
- Brass alloy
- Machining steel
- Standard parts with a low property class

9 Technical data

The key technical data for the on-load tap-changer is summarized in this chapter.

Further information on the selection of on-load tap-changers in general can be found in the chapters "Tap-changer designation", "Electrical properties" and "Selecting the on-load tap-changer" in the technical data TD61.

9.1 On-load tap-changer properties

On-load tap-changer	VI III 400 Y			VI III 400 D		
Number of phases and application	3, neutral point			3, at any point in the winding		
Maximum rated through-current I_{rm} [A]	400					
Rated short-time current [kA]	5					
Rated duration of short-circuits [s]	3					
Rated peak withstand current [kA]	12.5					
Maximum rated step voltage U_{irm} [V]	1500					
Step capacity P_{stm} [kVA]	600					
Rated frequency [Hz]	50...60					
Highest voltage for equipment U_m [kV] [1]	40 [2]	76 [3]	123	40 [2]	76 [3]	123/76
Rated lightning impulse withstand voltage [kV, 1.2|50 µs]	200	350	550	200	350	550
Rated short-duration power frequency withstand voltage [kV, 50 Hz, 1 min]	70	140	230	70	140	230

Table 8: Electrical data

[1] In accordance with IEC 60214-1: Highest effective value for phase-to-phase voltage in a three-phase system for which an on-load tap-changer is designed with respect to its insulation.

[2] Highest voltage for equipment Um 40 kV. Test voltages in accordance with IEC 60214-1:2014 for type series Um=36 kV.

[3] Highest voltage for equipment Um 76 kV. Test voltages in accordance with IEC 60214-1:2014 for type series Um=72.5 kV.

On-load tap-changer	VI III 400 Y	VI III 400 D
Max. number of operating positions	Without change-over selector: 14 With change-over selector: 27	
Weight without insulating fluid [kg]	Max. 150	
The on-load tap-changer is vacuum-tight		

Table 9: Mechanical data

9.2 Permissible ambient conditions

Air temperature during operation [°C]	- 25...+ 55
Transport temperature, storage temperature [°C]	- 40...+ 70
Max. drying temperature [°C]	+125
Insulating fluid	– Unused insulating oils derived from petroleum products[1] in accordance with IEC 60296 and ASTM D3487 (equivalent standards on request) – Unused insulating oils derived from other virgin hydrocarbons in accordance with IEC 60296, or blends of these oils with petroleum products[1] in accordance with IEC 60296, ASTM D3487 or equivalent standards on request – Alternative insulating fluids, such as natural and synthetic esters or silicone oils, on request [1] Gas-to-liquid oils (GTL oils) are understood in this context to be petroleum products
Temperature of the insulating fluid in operation	
– Mineral insulating oil (IEC 60296) [°C]	-25...+105 (up to +115 when the transformer is in emergency operation)[1]
– Synthetic ester fluid (IEC 61099) [°C]	On request
– Natural ester fluid (IEC 62770) [°C]	On request
Permissible absolute pressure during operation [bar]	0.7...1.5
Installation height of the oil conservator [m]	Max. 5
Installation altitude above sea level [m]	Max. 2,000[2]
Degree of protection	IP66 in accordance with IEC 60529

Table 10: Permissible ambient conditions

[1] Arctic version down to -40 °C available on request (fixed operation without switching operations)

[2] For altitudes >2,000 m, a minimum height must be observed for the oil conservator. (Further information is available in the "Installation altitude above sea level" chapter in the TD61 Technical Data.)

9.3 Limit values for dielectric strength and water content of insulating fluids

Limit values for insulating fluids in accordance with IEC 60296	U_d	H_2O
When commissioning the transformer for the first time	> 60 kV/2.5 mm	< 12 ppm
During operation	> 30 kV/2.5 mm	< 30 ppm
After maintenance	> 50 kV/2.5 mm	< 15 ppm

Table 11: Limit values for insulating fluids in accordance with IEC 60296, dielectric strength measured in accordance with IEC 60156, water content measured in accordance with IEC 60814

Limit values for natural esters in accordance with IEC 62770	U_d	H_2O
When commissioning the transformer for the first time	> 60 kV/2.5 mm	≤ 100 ppm
During operation	> 30 kV/2.5 mm	≤ 200 ppm
After maintenance	> 50 kV/2.5 mm	≤ 100 ppm

Table 12: Limit values for natural esters in accordance with IEC 62770, dielectric strength measured in accordance with IEC 60156, water content measured in accordance with IEC 60814

Limit values for synthetic esters in accordance with IEC 61099	U_d	H_2O
When commissioning the transformer for the first time	> 60 kV/2.5 mm	≤ 100 ppm
During operation	> 30 kV/2.5 mm	≤ 400 ppm
After maintenance	> 50 kV/2.5 mm	≤ 150 ppm

Table 13: Limit values for synthetic esters in accordance with IEC 61099, dielectric strength measured in accordance with IEC 60156, water content measured in accordance with IEC 60814

9.4 Electrical properties

This chapter contains information about the electrical properties.

9.4.1 Step capacity diagram

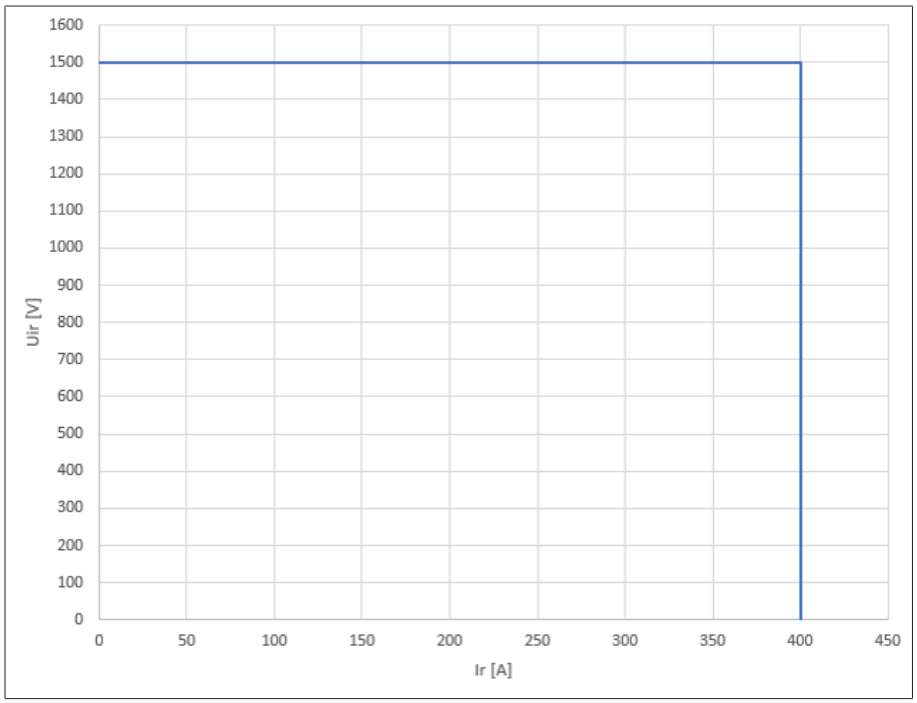


Figure 93: Step capacities (rated voltage Uir, rated through-current Ir)

9.4.2 Recovery voltage and breaking current

During its switching operation, the tap winding is briefly electrically isolated from the main winding by the reversing change-over selector or coarse change-over selector. It then adopts a potential resulting from the voltages of the adjacent windings and coupling capacity for these windings or grounded parts.

This potential shift of the tap winding produces corresponding voltages between the deactivating change-over selector contacts because one contact is always connected to the tap winding and the other contact is always connected to the main winding. This voltage is known as the recovery voltage U_W.

When separating the change-over selector contacts, a capacitive current has to be interrupted. This current depends on the aforementioned coupling capacities of the tapped winding. This current is known as the breaking current I_S.

The recovery voltage U_W and breaking current I_S may result in impermissible discharge events on the change-over selector. The permissible range of recovery voltage U_W and breaking current I_S can be seen in the following diagrams. For more information about recovery voltage and breaking current, see Technical Data TD61 – General Part.

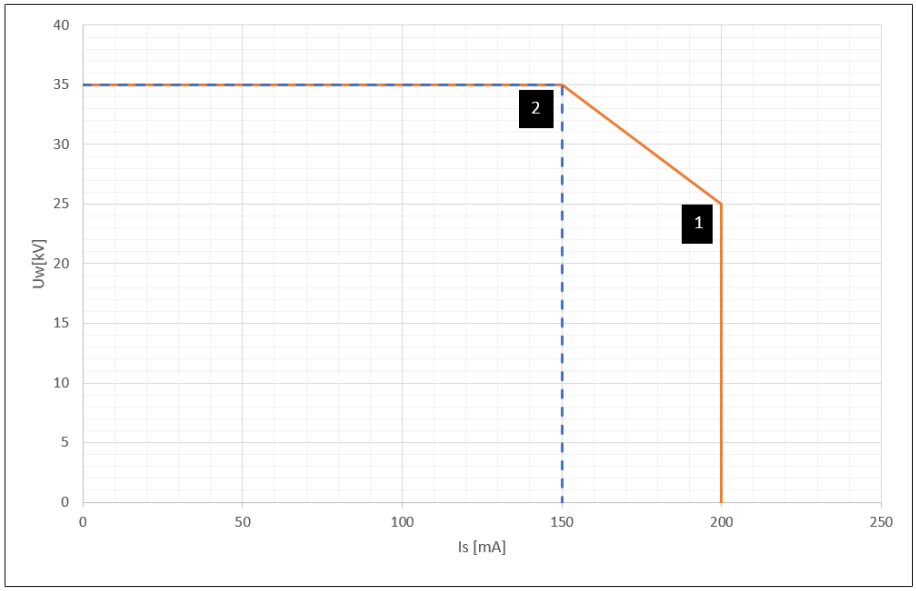


Figure 94: Recovery voltage without tie-in resistor

1	Mineral insulating oil in accordance with IEC 60296
2	Alternative insulating fluids in accordance with IEC 61099 or IEC 62770

For versions with tie-in resistor, the tie-in resistor will be delivered separately. The transformer manufacturer is responsible for installation in the transformer and connection to the tap changer.

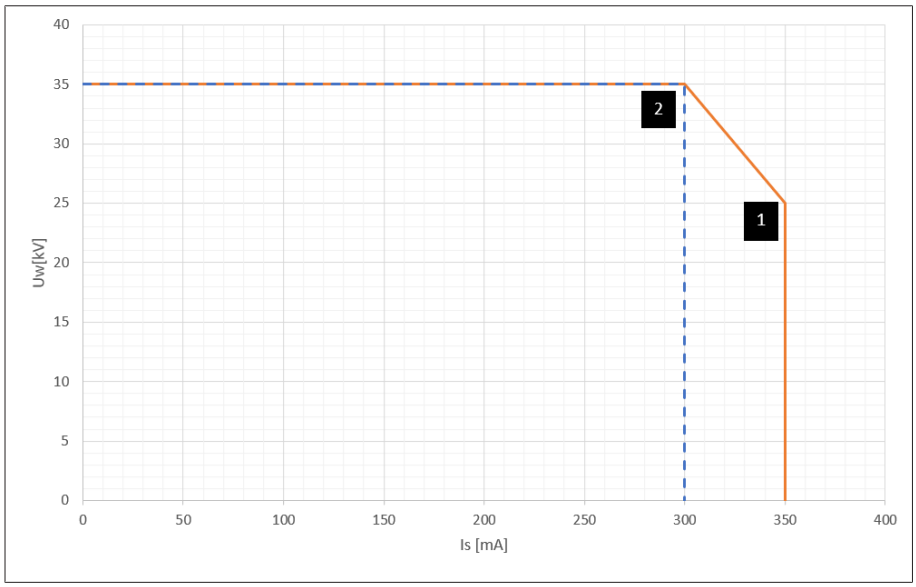


Figure 95: Recovery voltage with tie-in resistor

1	Mineral insulating oil in accordance with IEC 60296
2	Alternative insulating fluids in accordance with IEC 61099 or IEC 62770

9.4.3 Insulation

The stated rated withstand voltages for the insulation arrangement apply to new, perfectly dried insulation in prepared transformer oil (at an ambient temperature of at least 10 °C).

The following details are needed to select an on-load tap-changer:
- Maximum mains-frequency operating voltages
- Test AC voltages during transformer test
- Impulse voltages during transformer test (lightning surge, switching surge, wave isolated in back and wave isolated in front)

The transformer manufacturer is responsible for the correct choice of rated withstand voltages depending on insulation coordination at the operating site. The necessary rated withstand voltages should be observed for the various insulation distances:
- Insulation to ground
- With multi-phase types: Insulation between phases
- Insulation between a phase's contacts

The details required depend on the type of regulation (e.g. basic connection of tapped winding with on-load tap-changers) and the tap-changer type.

9.4.4 Permitted voltage stresses

This section describes the permissible voltage stress of the on-load tap-changer.

When selecting the on-load tap-changer, you must check that the highest stresses do not exceed the related rated withstand voltages at the insulation distances.

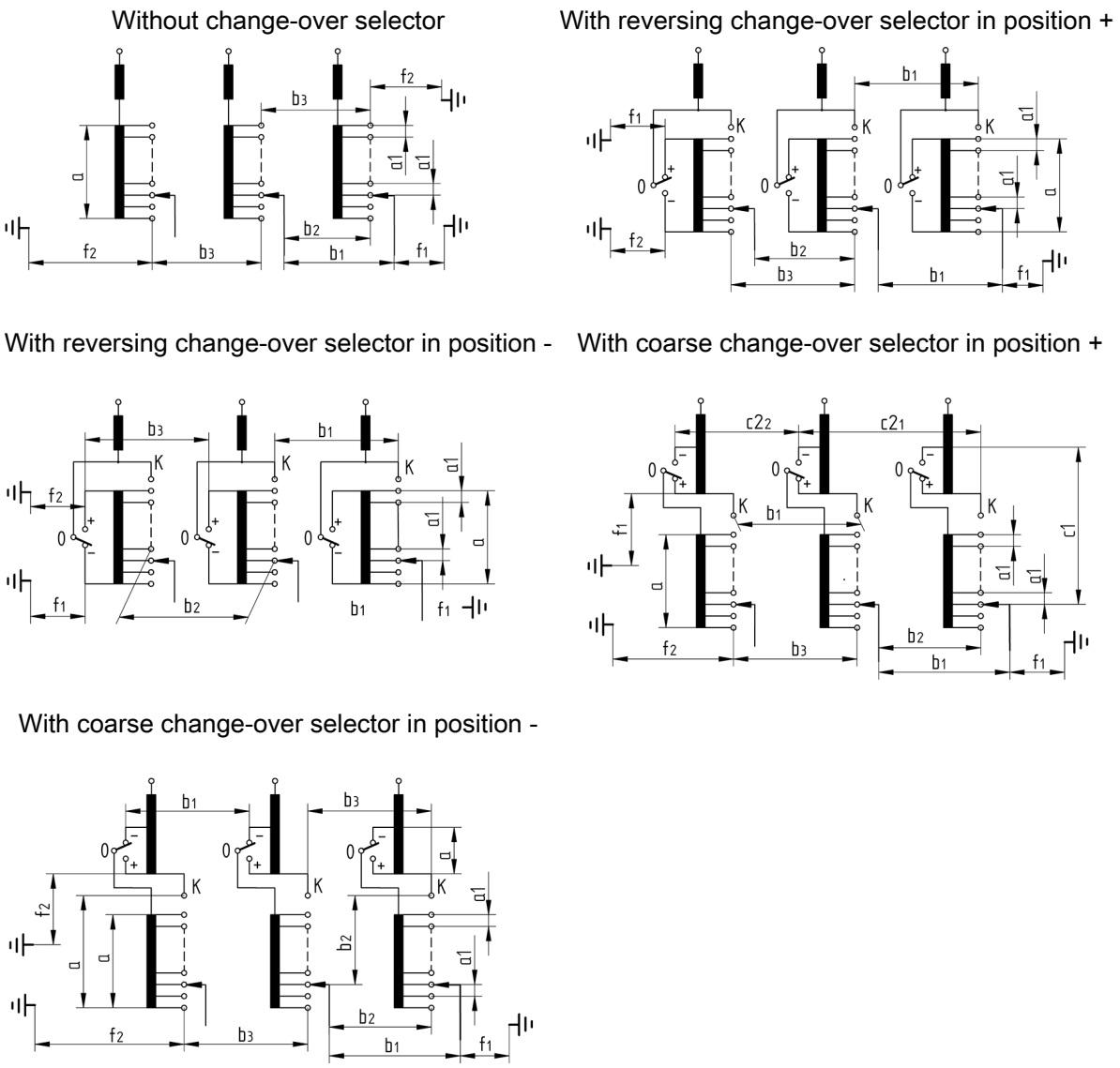


Figure 96: Insulation distances

a1	= between tap selector contacts of the winding of one tap position (connected or not connected)
a	= between the beginning and end of a tap winding and, for a model with a coarse winding, also between the beginning and end of a coarse winding. Note for coarse tap selector connection in the (-) change-over selector position: When stressed with impulse voltage, the permissible withstand voltage "a" between the end of a coarse winding connected with the K tap selector contact and the tap selector contact at the end of the tap winding of the same phase must be adhered to.
b	= between the tap selector contacts of different phases and between change-over selector contacts of different phases that are connected with the beginning/end of a step of the tap winding or with a tap selector contact The permissible voltages between the contacts for tap-changers for delta connections depend on the positions of the change-over selector and tap selector. The different permissible withstand voltages b1, b2 or b3 must therefore be noted for an insulation distance: b_1 = between selected contacts for different phases b_2 = from the selected contact of one phase to non-selected contacts of other phases b_3 = between non-selected contacts for different phases

f	= between tap selector contacts and (+) change-over selector contacts to ground Different values apply for the delta connection version: f_1 = between take-off lead and ground and in (+) position of the change-over selector between (+) change-over selector contacts and ground f_2 = between non-selected tap selector contacts and in (-) position of the change-over selector between (+) contacts and ground
Additionally for coarse tap selector connection in (+) position of the change-over selector:	
c1	= from one (-) change-over selector contact to the take-off lead of the same phase
c2	= between (-) change-over selector contacts of different phases and from one (-) change-over selector contact to one (+) change-over selector contact of another phase For tap-changers, different permissible values should be noted for the delta connection: $c2_1$ = between a (-) change-over selector contact and a (+) change-over selector contact of another phase $c2_2$ = between (-) change-over selector contacts of different phases

Voltage waveform	VI-Y Um=40kV				VI-D Um=40kV				VI-Y Um=76kV			
Insulation distances	LI	LIC	SI	AC	LI	LIC	SI	AC	LI	LIC	SI	AC
a1	100	110	80	20	100	110	80	20	100	110	80	20
a	200	220	160	50	200	220	160	50	200	220	160	50
b	200	220	160	85	-	-	-	-	200	220	160	85
b1	-	-	-	-	200	220	160	70	-	-	-	-
b2	-	-	-	-	250	275	200	80	-	-	-	-
b3	-	-	-	-	300	330	240	90	-	-	-	-
f1	200	220	160	70	200	220	160	70	350	385	280	140
f2	250	275	200	80	250	275	200	80	490	539	392	165
c1	400	440	320	100	400	440	320	100	400	440	320	100
c2	300	330	240	90	-	-	-	-	350	385	280	140
$c2_1$	-	-	-	-	250	275	200	80	-	-	-	-
$c2_2$	-	-	-	-	300	330	240	90	-	-	-	-

Table 14: Rated insulation level

Voltage waveform	VI-D Um=76kV				VI-Y Um=123kV				VI-D Um=123/76kV			
Insulation distances	LI	LIC	SI	AC	LI	LIC	SI	AC	LI	LIC	SI	AC
a1	100	110	80	20	100	110	80	20	100	110	80	20
a	200	220	160	50	200	220	160	50	200	220	160	50
b	-	-	-	-	200	220	160	85	-	-	-	-
b1	350	385	280	140	-	-	-	-	350	385	280	140
b2	490	539	392	165	-	-	-	-	490	539	392	165
b3	520	572	416	180	-	-	-	-	520	572	416	180
f1	350	385	280	140	550	605	460	230	550	605	460	230
f2	490	539	392	165	650	715	520	275	650	715	520	275
c1	400	440	320	100	400	440	320	100	400	440	320	100
c2	-	-	-	-	350	385	280	140	-	-	-	-

Voltage waveform	VI-D Um=76kV				VI-Y Um=123kV				VI-D Um=123/76kV			
Insulation distances	LI	LIC	SI	AC	LI	LIC	SI	AC	LI	LIC	SI	AC
$c2_1$	490	539	392	165	-	-	-	-	490	539	392	165
$c2_2$	520	572	416	180	-	-	-	-	520	572	416	180

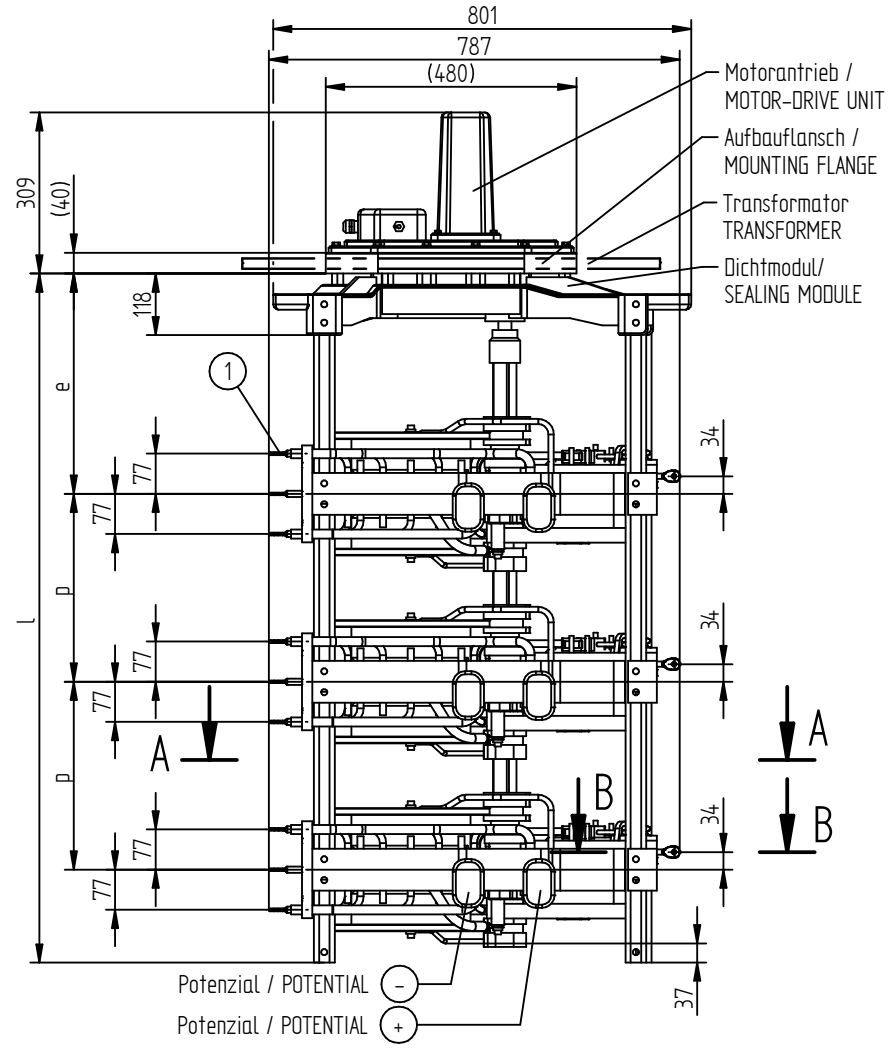
Table 15: Rated insulation level

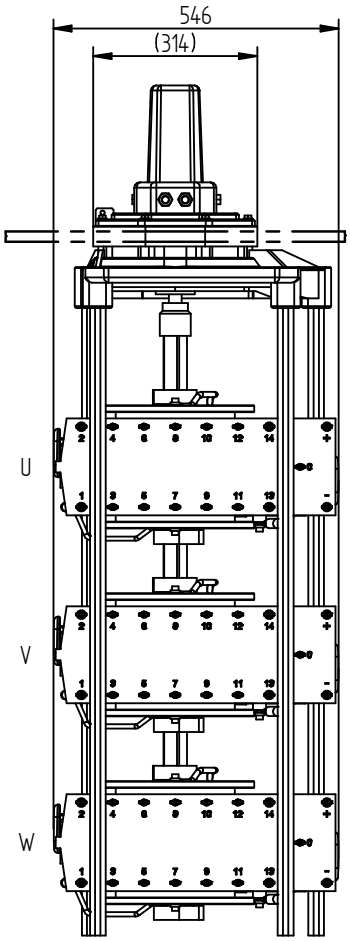
LI	Full wave lightning impulse (kV, 1.2/50 µs)
LIC	Chopped wave lightning impulse (kV, 1.2/50/3 µs)
SI	Switching impulse voltage (kV, 250/2500 µs)
AC	Applied voltage (kV, 50 Hz, 1 min)

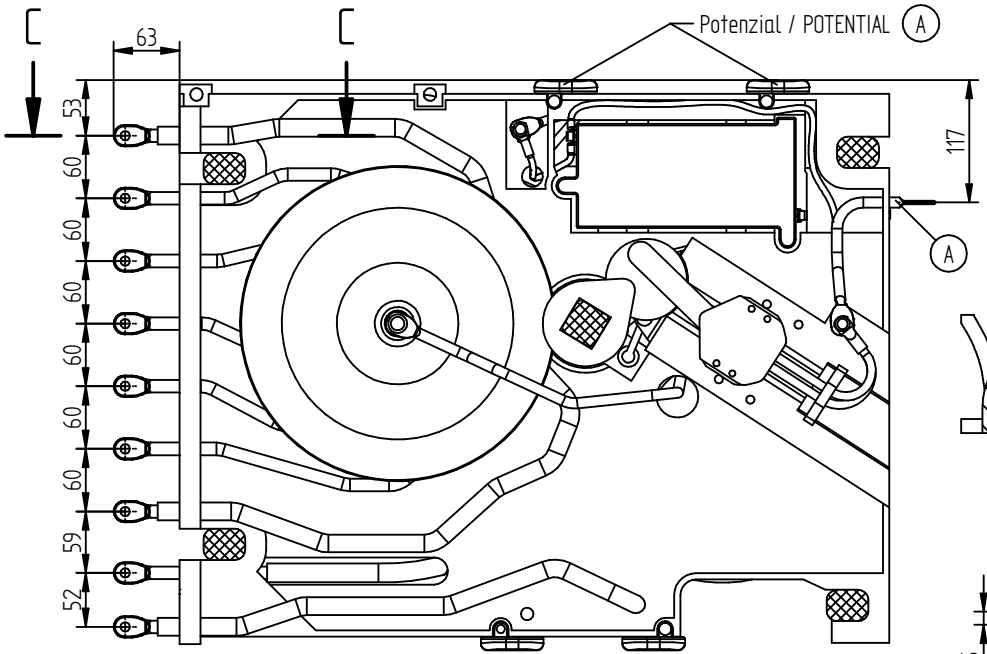
Table 16: Abbreviations for the rated insulation level

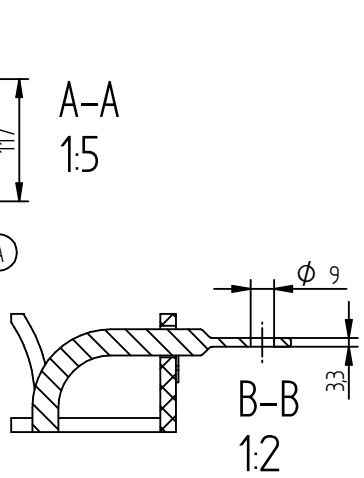
10 Drawings

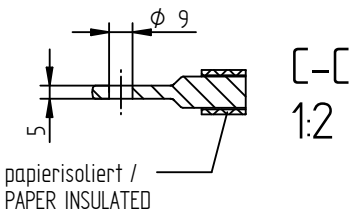
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Betriebsstellungen / SERVICE POSITIONS max. 27			
Um in kV	e	p	l
	Maße / DIMENSIONS in mm		
40 – Y	422	320	1240
40 – D	422	360	1320
76 – Y	532	320	1350
76 – D	532	420	1550
123 – Y	637	320	1455
123/76 – D	637	420	1655

(1) – Anschlusskontakt / TERMINAL

(A) – Ableitkontakt / OUTPUT TERMINAL

Gewicht max. ca. 150 kg
Verbindlich für die Bezeichnung der Anschlusskontakte und Phasen ist das für den Auftrag gültige Ausführungsschaltbild
WEIGHT MAX. APPROX 150 KG
THE CONNECTION DIAGRAM APPLICABLE TO THE ORDER IS BINDING FOR THE DESIGNATION OF THE TERMINALS AND PHASES.

	NAME	DATE
DFTR.	FELDMEIERS	10.02.2025
CHKD.	MUENZBERG	10.02.2025
STAND	KLEYN	11.02.2025

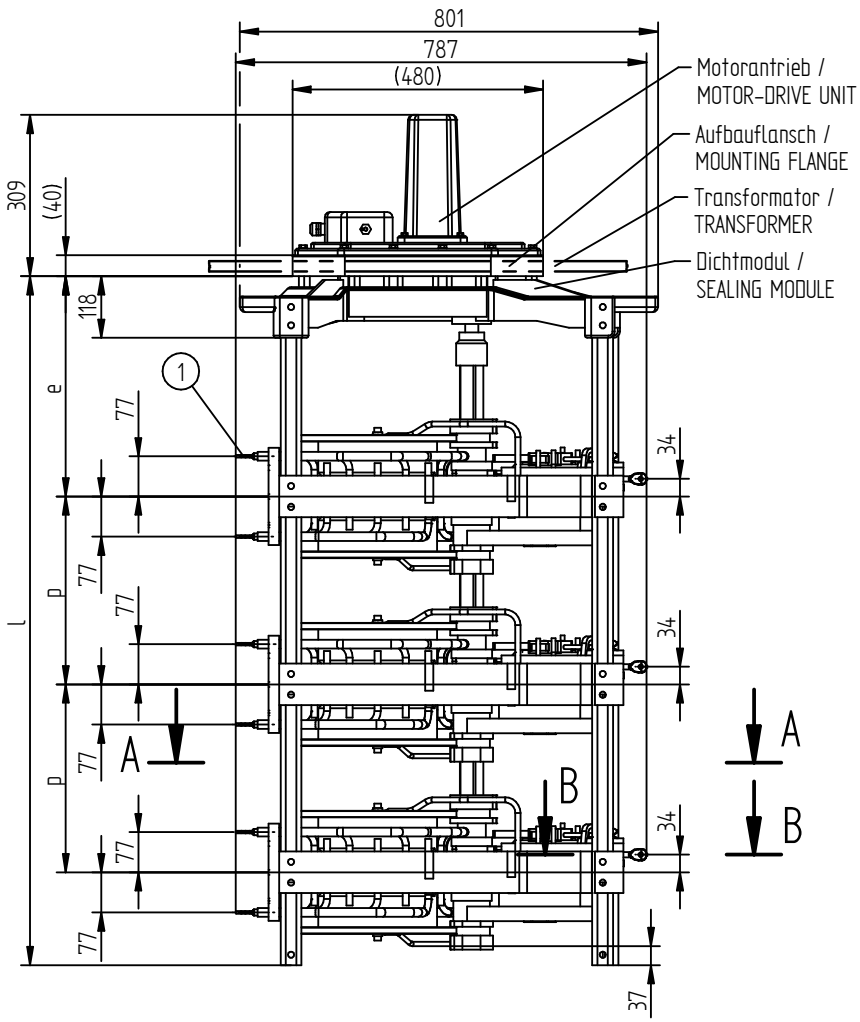
DOCUMENT NO. SED 11719280 000 00
CHANGE NO. 1133298
SCALE 1:10

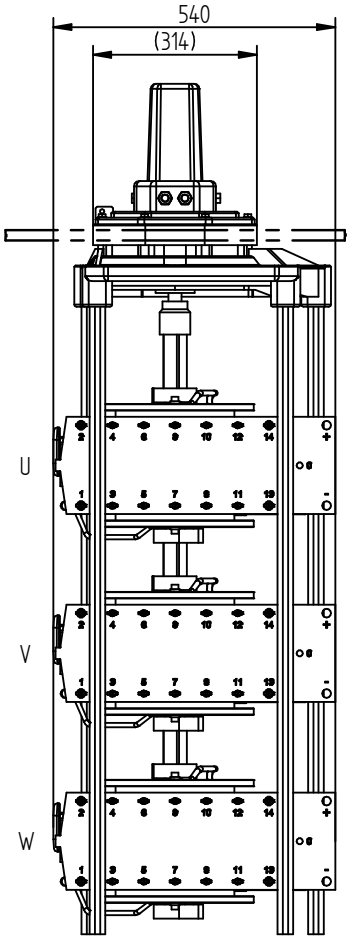
DIMENSION IN mm EXCEPT AS NOTED

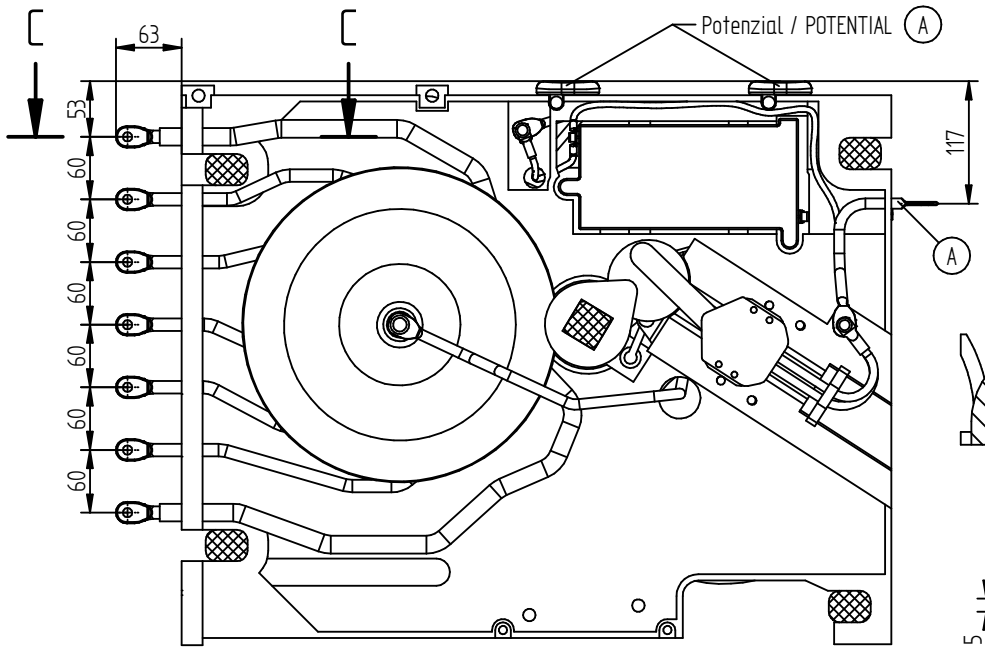
ON-LOAD TAP-CHANGER VACUTAP® VI
VI III 40-123kV WITH CHANGE-OVER SELECTOR
DIMENSION DRAWING

SERIAL NUMBER –
MATERIAL NUMBER 101617390M
SHEET 1 / 1
A3

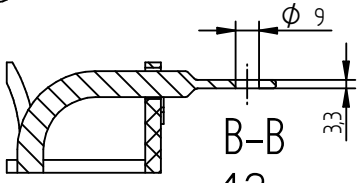
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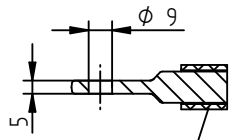




A–A
1:5



B–B
1:2



C–C
1:2

papierisoliert / PAPER INSULATED

Betriebsstellungen / SERVICE POSITIONS max. 14			
Um in kV	e	p	l
	Maße / DIMENSIONS in mm		
40 – Y	422	320	1240
40 – D	422	360	1320
76 – Y	532	320	1350
76 – D	532	420	1550
123 – Y	637	320	1455
123/76 – D	637	420	1655

(1) – Anschlusskontakt / TERMINAL

(A) – Ableitkontakt / OUTPUT TERMINAL

Gewicht max. ca. 150 kg
Verbindlich für die Bezeichnung der Anschlusskontakte und Phasen ist das für den Auftrag gültige Ausführungsschaltbild
WEIGHT MAX. APPROX 150 KG
THE CONNECTION DIAGRAM APPLICABLE TO THE ORDER IS BINDING FOR THE DESIGNATION OF THE TERMINALS AND PHASES.

	NAME	DATE
DFTR.	FELDMEIERS	10.02.2025
CHKD.	MUENZBERG	10.02.2025
STAND	KLEYN	11.02.2025

DOCUMENT NO. SED 11719198 000 00
CHANGE NO. 1133298
SCALE 1:10

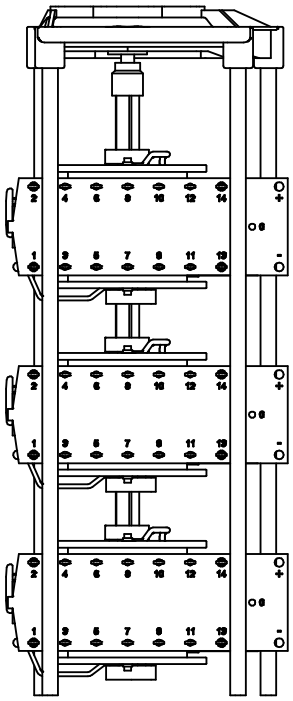
DIMENSION IN mm EXCEPT AS NOTED

ON-LOAD TAP-CHANGER VACUTAP® VI
VI III 40-123kV WITHOUT CHANGE-OVER SELECTOR
DIMENSION DRAWING

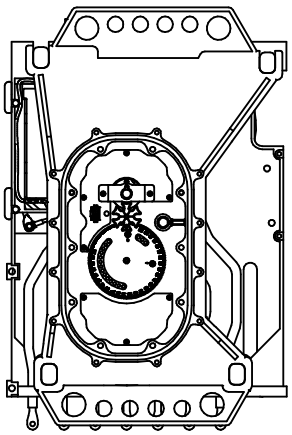
SERIAL NUMBER –
MATERIAL NUMBER 101617410M
SHEET 1 / 1
A3

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Laststufenschalter /
ON-LOAD TAP-CHANGER



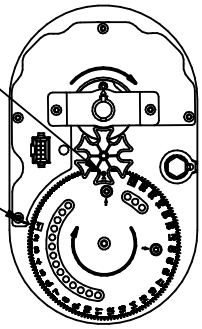
Laststufenschalter Kopf /
ON-LOAD TAP-CHANGER HEAD



1:5

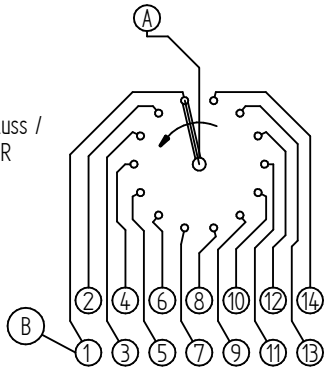
Justiermarkierung /
ADJUSTMENT MARKING

Stellungsanzeige /
TAP POSITION INDICATOR



14140

Wählerebene mit Kundenanschluss /
SELECTOR PLANE WITH CUSTOMER CONNECTION



	DATE	NAME	DOCUMENT NO.	
DFTR.	15.07.2025	FELDMEIERS	SED 117222200 000 01	
EHKO.	15.07.2025	MUENZBERG	CHANGE NO.	SCALE
STAND	16.07.2025	KLEYN	1121934	1:10

(A) = Ableitkontakt / OUTPUT TERMINAL

(B) = Betriebsstellung 1 – Justierstellung / SERVICE POSITION 1 – ADJUSTMENT POSITION

alle Ansichten in Justierstellung dargestellt / ALL VIEWS ARE SHOWN IN ADJUSTMENT POSITION
Verbindungsbrücken optional bestellbar / CONNECTION BRIDGES AVAILABLE ON REQUEST

Verbindlich für die Bezeichnung der Anschlusskontakte und Phasen ist das für den Auftrag gültige Ausführungsschaltbild /
THE CONNECTION DIAGRAM APPLICABLE TO THE ORDER IS BINDING FOR THE DESIGNATION OF TERMINALS AND PHASES

DIMENSION IN mm EXCEPT AS NOTED

MR

ON-LOAD TAP-CHANGER VACUTAP® VI
CONNECTIONS WITHOUT CHANGE-OVER SELECTOR
ADJUSTMENT PLAN

SERIAL NUMBER
–

MATERIAL NUMBER
101624330M

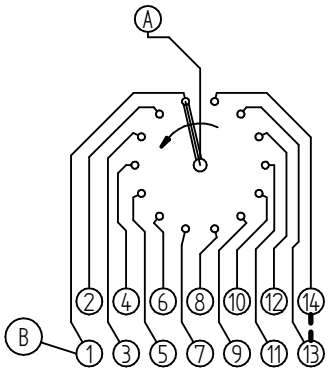
SHEET
1 / 2

A3

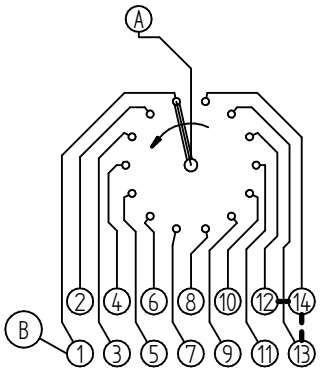
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Ausführbare Schaltungen ohne Vorwähler / REALIZABLE CONNECTIONS WITHOUT CHANGE-OVER SELECTOR

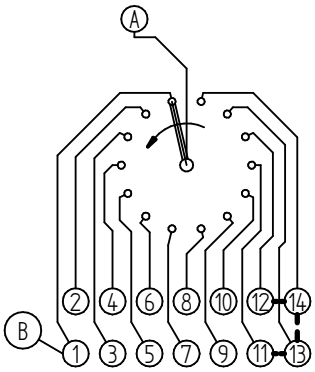
14130



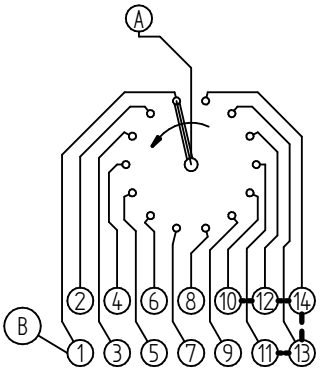
14120



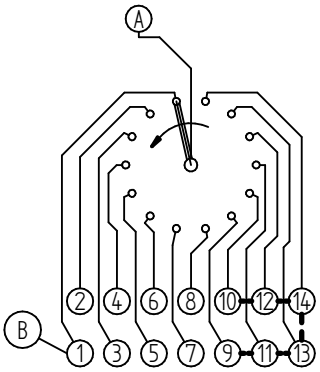
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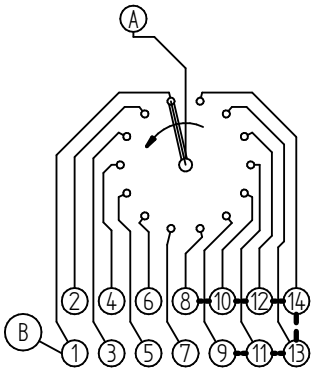
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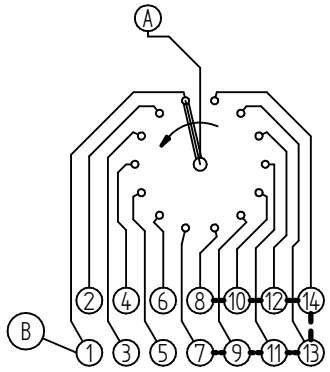
14090



14080

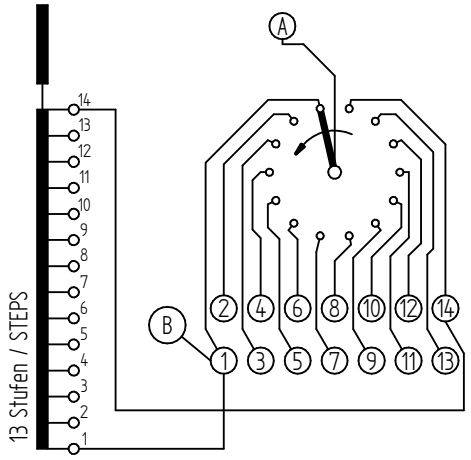


14070



Schaltung / CONNECTION	Stufen der Regelwicklung / STEPS OF TAP WINDING
14140	13
14130	12
14120	11
14110	10
14100	9
14090	8
14080	7
14070	6

Beispielschaltbild für 14140 / EXAMPLE CONNECTION DIAGRAMM FOR 14140



	DATE	NAME	DOCUMENT NO.	
DFTR.	15.07.2025	FELDMEIERS	SED 117222200 000 01	
CHKD.	15.07.2025	MUENZBERG	CHANGE NO.	SCALE
STAND	16.07.2025	KLEYN	1121934	1:10

DIMENSION IN mm EXCEPT AS NOTED

MR

ON-LOAD TAP-CHANGER VACUTAP® VI
CONNECTIONS WITHOUT CHANGE-OVER SELECTOR
ADJUSTMENT PLAN

SERIAL NUMBER
–

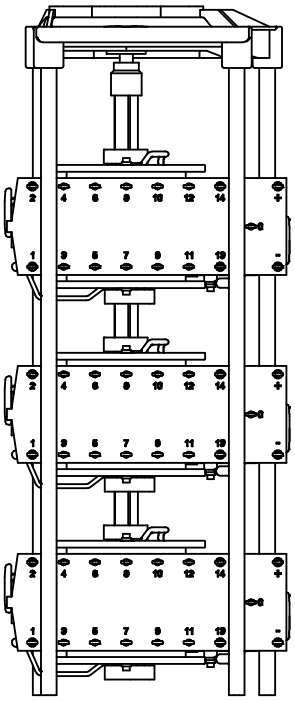
MATERIAL NUMBER
101624330M

SHEET
2 / 2

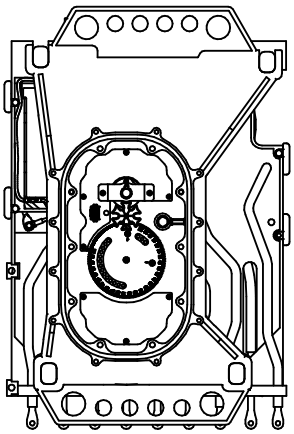
A3

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Laststufenschalter /
ON-LOAD TAP-CHANGER



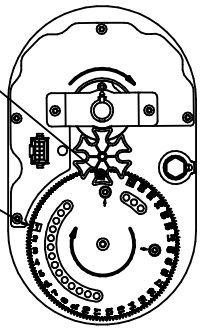
Laststufenschalter Kopf /
ON-LOAD TAP-CHANGER HEAD



1:5

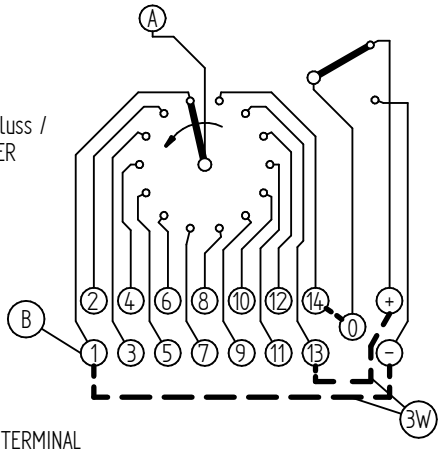
Justiermarkierung /
ADJUSTMENT MARKING

Stellungsanzeige /
TAP POSITION INDICATOR

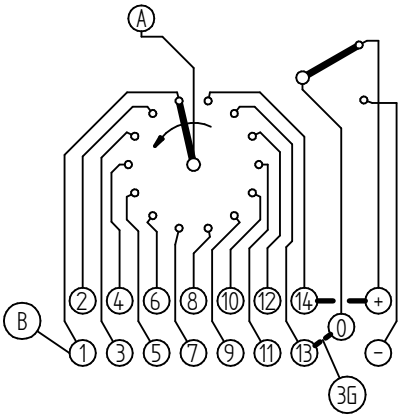


Wählerebene mit Kundenanschluss /
SELECTOR PLANE WITH CUSTOMER CONNECTION

14271W / 14273W



14271G / 14273G



(A) = Ableitkontakt / OUTPUT TERMINAL

(B) = Betriebsstellung 1 – Justierstellung / SERVICE POSITION 1 – ADJUSTMENT POSITION

(3W) = 3W-Verbindungsbrücke nur bei 3W-Ausführung / 3W-CONNECTION BRIDGES ONLY FOR 3W-VERSIONS

(3G) = 3G-Verbindungsbrücke nur bei 3G-Ausführung / 3G-CONNECTION BRIDGES ONLY FOR 3G-VERSIONS

alle Ansichten in Justierstellung dargestellt / ALL VIEWS ARE SHOWN IN ADJUSTMENT POSITION
Verbindungsbrücken optional bestellbar / CONNECTION BRIDGES AVAILABLE ON REQUEST

Verbindlich für die Bezeichnung der Anschlusskontakte und Phasen ist das für den Auftrag gültige Ausführungsschaltbild /
THE CONNECTION DIAGRAM APPLICABLE TO THE ORDER IS BINDING FOR THE DESIGNATION OF TERMINALS AND PHASES

	DATE	NAME	DOCUMENT NO.	
DFTR.	15.07.2025	FELDMEIERS	SED 117222223 000 01	
CHKD.	15.07.2025	MUENZBERG	CHANGE NO.	SCALE
STAND	16.07.2025	KLEYN	1121934	1:10

DIMENSION IN mm EXCEPT AS NOTED

MR

ON-LOAD TAP-CHANGER VACUTAP® VI
CONNECTIONS WITH CHANGE-OVER SELECTOR
ADJUSTMENT PLAN

SERIAL NUMBER
–

MATERIAL NUMBER
101863740M

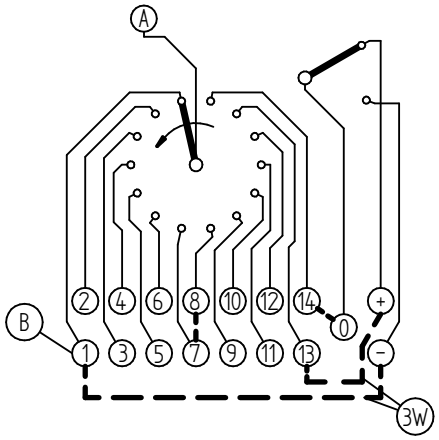
SHEET
1 / 5

A3

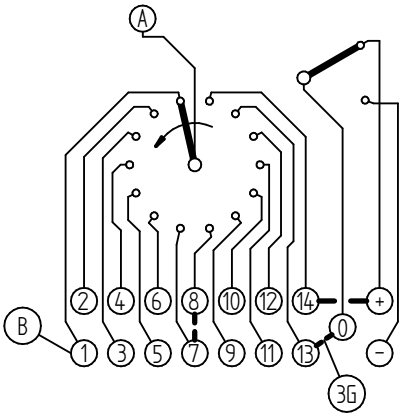
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Ausführbare Schaltungen mit Vorwähler / REALIZABLE CONNECTIONS WITH CHANGE-OVER SELECTOR

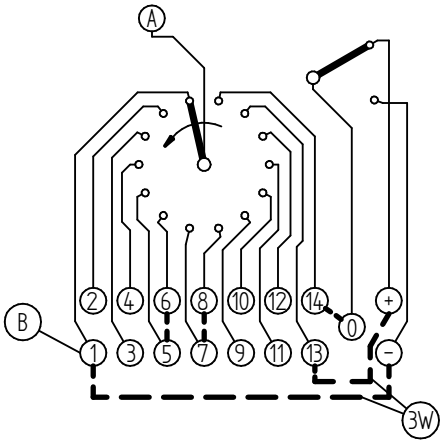
14251W / 14253W



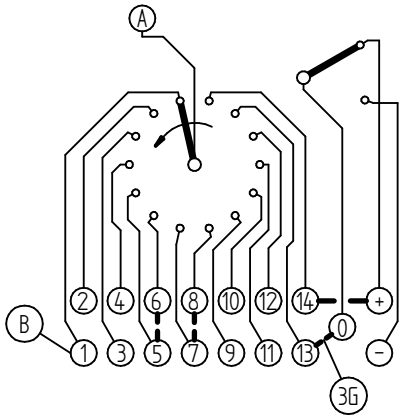
14251G / 14253G



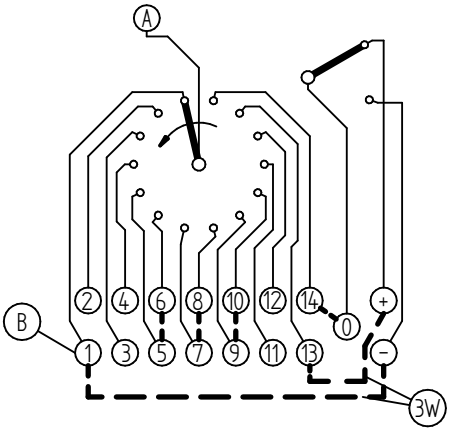
14231W / 14233W



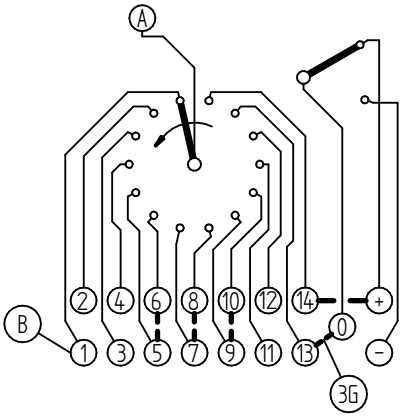
14231G / 14233G



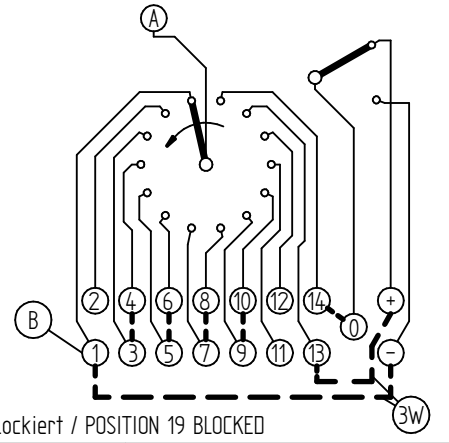
14211W / 14213W



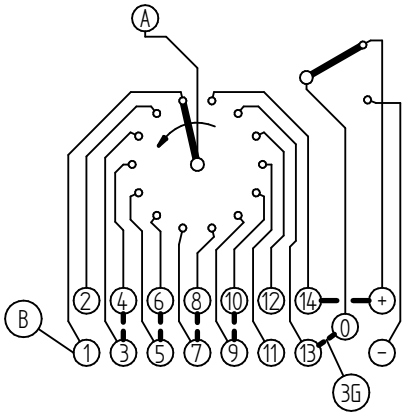
14211G / 14213G



14191W / 14193W / 14181W* / 14183W*



14191G / 14193G / 14181G* / 14183G*



* Position 19 blockiert / POSITION 19 BLOCKED

	DATE	NAME	DOCUMENT NO.	
DFTR.	15.07.2025	FELDMEIERS	SED 117222223 000 01	
CHKD.	15.07.2025	MUENZBERG	CHANGE NO.	SCALE
STAND	16.07.2025	KLEYN	1121934	1:10

DIMENSION IN mm EXCEPT AS NOTED

MR

ON-LOAD TAP-CHANGER VACUTAP® VI
CONNECTIONS WITH CHANGE-OVER SELECTOR
ADJUSTMENT PLAN

SERIAL NUMBER
–

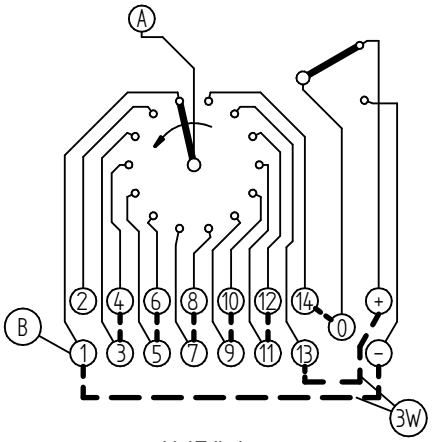
MATERIAL NUMBER 101863740M

SHEET 2 / 5

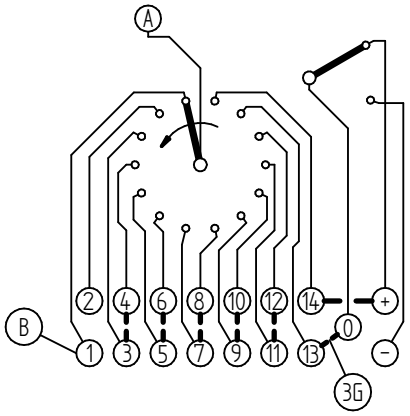
A3

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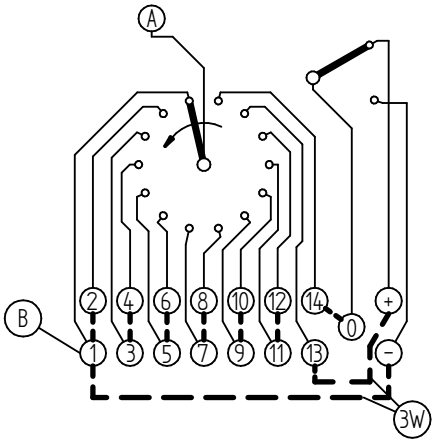
14171W / 14173W



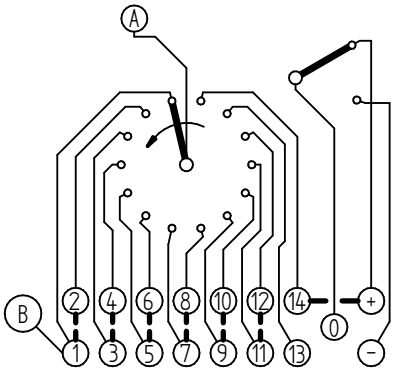
14171G / 14173G



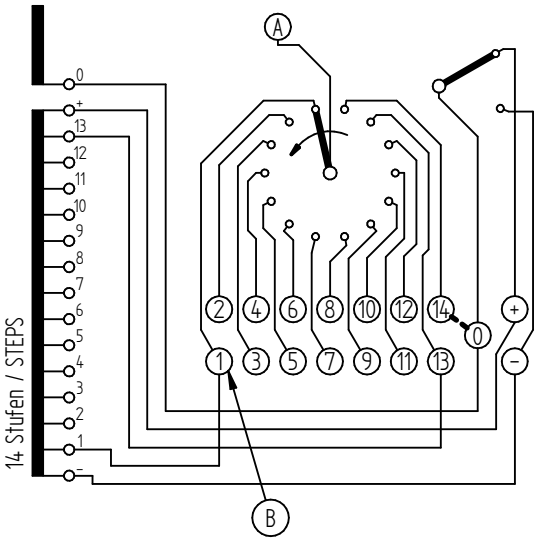
14151W



14151G



Beispielschaltbild für 14271W /
EXAMPLE CONNECTION DIAGRAMM FOR 14271W



	DATE	NAME	DOCUMENT NO.	
DFTR.	15.07.2025	FELDMEIERS	SED 117222223 000 01	
EHKO.	15.07.2025	MUENZBERG	CHANGE NO.	SCALE
STAND	16.07.2025	KLEYN	1121934	1:10

DIMENSION IN mm EXCEPT AS NOTED

MR

ON-LOAD TAP-CHANGER VACUTAP® VI
CONNECTIONS WITH CHANGE-OVER SELECTOR
ADJUSTMENT PLAN

SERIAL NUMBER
–

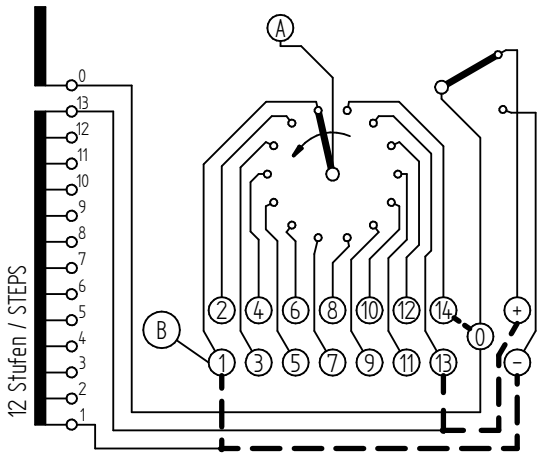
MATERIAL NUMBER
101863740M

SHEET
3 / 5

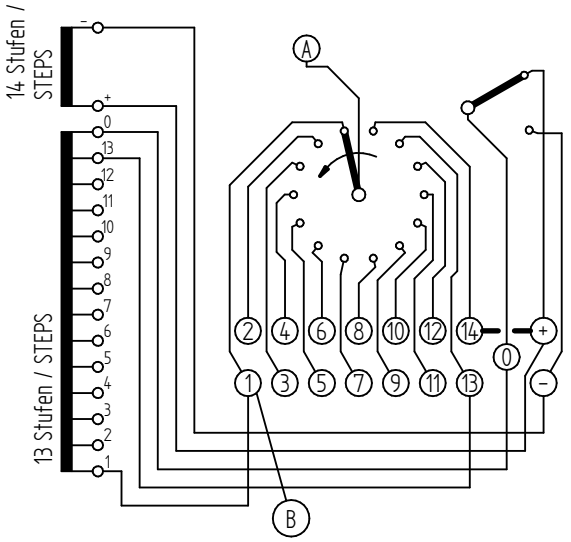
A3

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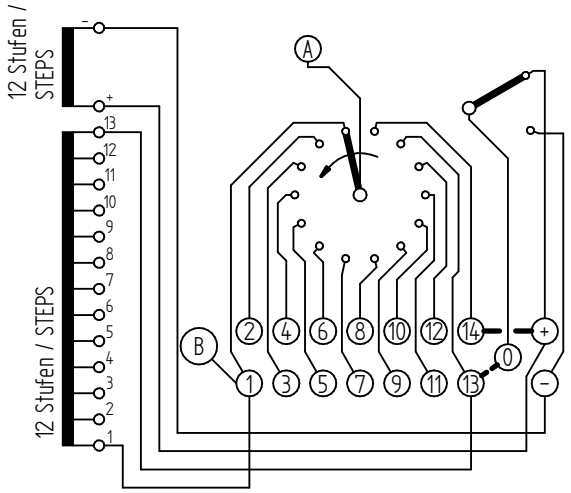
Beispielschaltbild für 14273W /
EXAMPLE CONNECTION DIAGRAMM FOR 14273W



Beispielschaltbild für 14271G /
EXAMPLE CONNECTION DIAGRAMM FOR 14271G



Beispielschaltbild für 14273G /
EXAMPLE CONNECTION DIAGRAMM FOR 14273G



	DATE	NAME	DOCUMENT NO.	
DFTR.	15.07.2025	FELDMEIERS	SED 117222223 000 01	
EHKO.	15.07.2025	MUENZBERG	CHANGE NO.	SCALE
STAND	16.07.2025	KLEYN	1121934	1:10

DIMENSION IN mm EXCEPT AS NOTED

MR

ON-LOAD TAP-CHANGER VACUTAP® VI
CONNECTIONS WITH CHANGE-OVER SELECTOR
ADJUSTMENT PLAN

SERIAL NUMBER
–

MATERIAL NUMBER 101863740M

SHEET 4 / 5

A3

Schaltung / CONNECTION	Stufen der Regelwicklung / STEPS OF TAP WINDING	Stufen der Grobstufen-Wicklung / STEPS OF COARSE STEP WINDING
14271W	14	
14251W	13	
14231W	12	
14211W	11	
14191W	10	
14181W	10	
14171W	9	
14151W	8	
14273W	12	
14253W	11	
14233W	10	
14213W	9	
14193W	8	
14183W	8	
14173W	7	
14271G	13	14
14251G	12	13
14231G	11	12
14211G	10	11
14191G	9	10
14181G	9	10
14171G	8	9
14151G	7	8
14273G	12	12
14253G	11	11
14233G	10	10
14213G	9	9
14193G	8	8
14183G	8	8
14173G	7	7

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	DATE	NAME	DOCUMENT NO.	
DFTR.	15.07.2025	FELDMEIERS	SED 117222223 000 01	
EHKO.	15.07.2025	MUENZBERG	CHANGE NO.	SCALE
STAND	16.07.2025	KLEYN	1121934	1:10

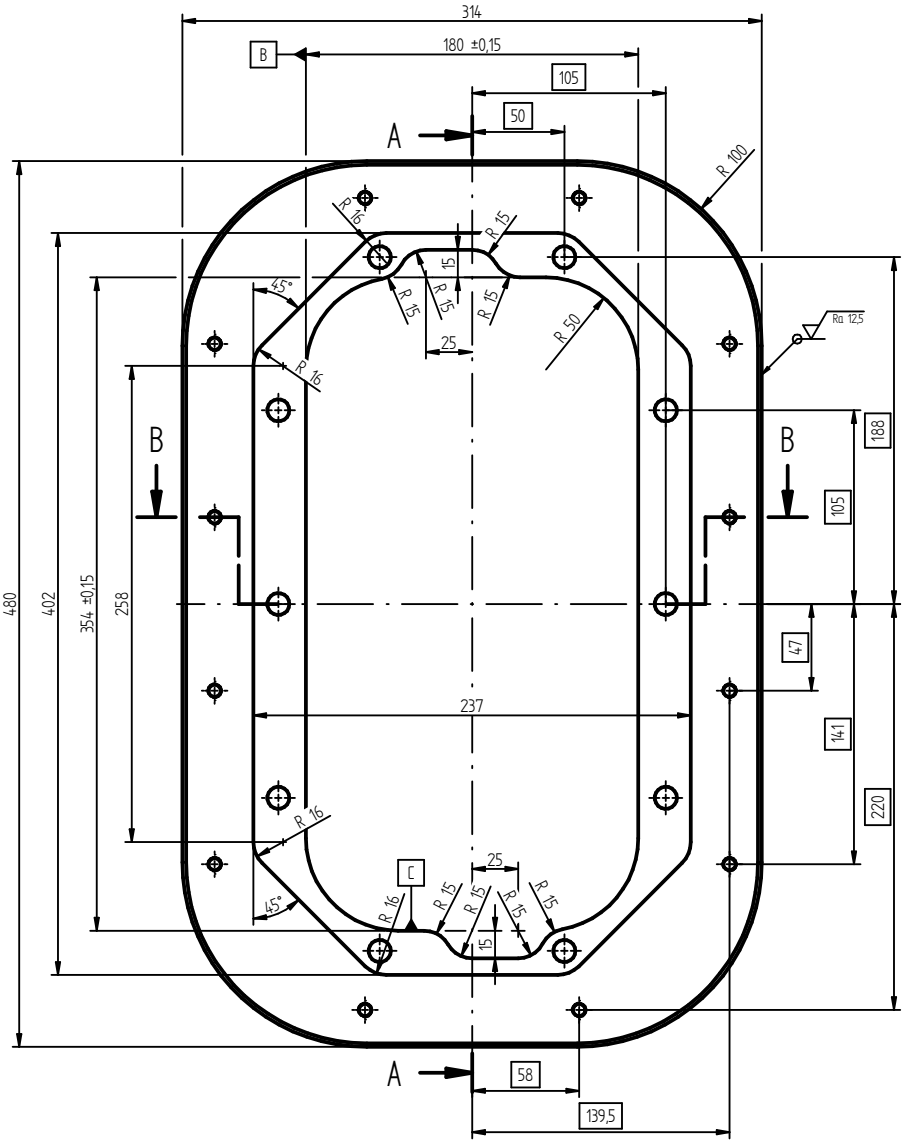
DIMENSION IN mm EXCEPT AS NOTED

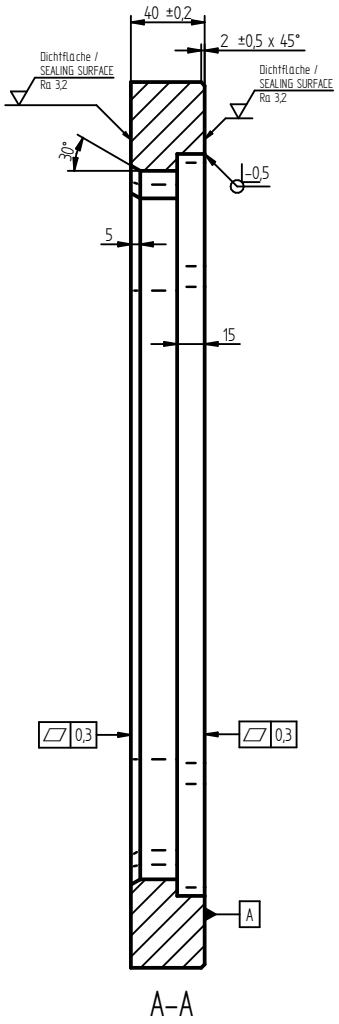
MR

ON-LOAD TAP-CHANGER VACUTAP® VI
CONNECTIONS WITH CHANGE-OVER SELECTOR
ADJUSTMENT PLAN

SERIAL NUMBER
–
MATERIAL NUMBER 101863740M
SHEET 5 / 5
A3

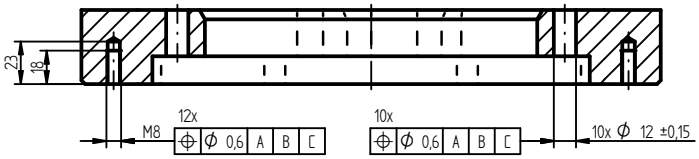
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A–A

B–B



Werkstoff: Stahl (z.B. S235JR EN 10025-2) /
MATERIAL: STEEL (FOR EXAMPLE S235JR EN10025-2)

Tolerierung / TOLERANCING ISO 8015
Allgemeintoleranz / GENERAL TOLERANCE ISO 2768:1989 -mH
Oberflächenangaben / SURFACE INDICATION ISO 1302:2002
Werkstückkanten / WORKPIECE EDGES ISO 13715

+0,50 −0,50

Ra 6,3 | Ra 12,5 | Dichtfläche / SEALING SURFACE Ra 3,2

	DATE	NAME	DOCUMENT NO.	
DFTR.	13.02.2025	FELDMEIERS	SED 11884721 000 00	
CHKD.	13.02.2025	MUENZBERG	CHANGE NO.	SCALE
STAND	13.02.2025	KLEYN	1133298	1:2

DIMENSION IN mm EXCEPT AS NOTED

ON-LOAD TAP-CHANGER VACUTAP® VI
DIMENSIONS MOUNTING FLANGE
DIMENSION DRAWING

SERIAL NUMBER –

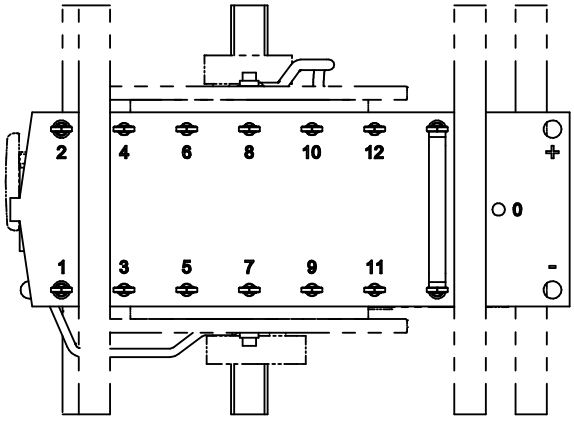
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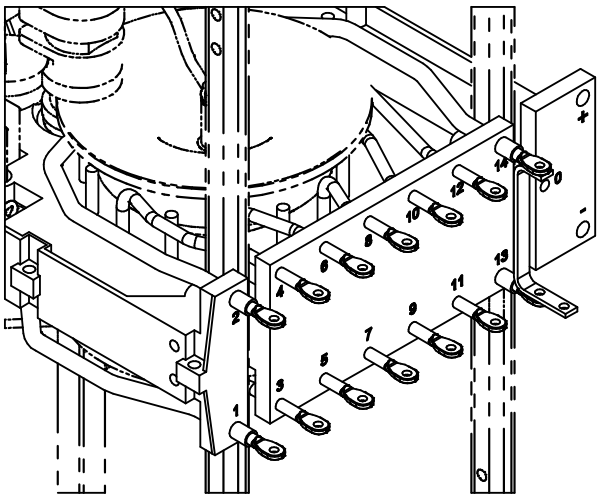
SHEET 1 / 1

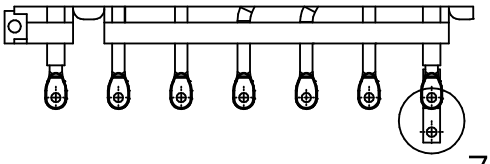
A2

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Ausführung / DESIGN: 14130

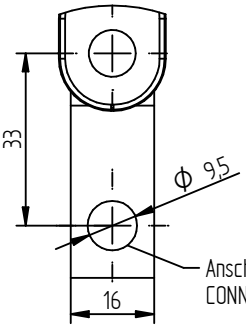






Z

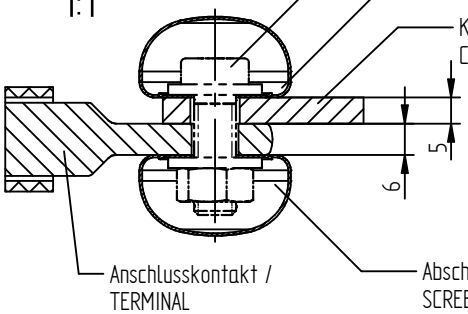
Z
1:1



Anschluss Trafoleitung /
CONNECTION TRANSFORMER LEAD

Befestigung Brücken schematische Darstellung / FASTENING BRIDGES SCHEMATIC VIEW

1:1



Befestigungsmaterial / FASTENING MATERIALS M8 (1)
Abschirmkappe / SCREENING CAP M8 (2)
Kontaktbrücke / CONTACT BRIDGE (2)
Anschlusskontakt / TERMINAL
Abschirmkappe / SCREENING CAP M8 (2)

(1) Befestigungsmaterial gehört nicht zum Lieferumfang. Jede Verschraubung durch geeignete Maßnahmen gegen Lösen und Setzen sichern. / FASTENING MATERIAL IS NOT INCLUDED IN THE SCOPE OF DELIVERY. TAKE SUITABLE MEASURES TO SECURE EACH SCREW CONNECTION AGAINST LOOSENING AND SETTING.

(2) auf Anfrage verfügbar / AVAILABLE ON REQUEST

Schraubenverbindungen nicht dargestellt, da nicht im Lieferumfang enthalten.
SCREW CONNECTIONS NOT PICTURED BECAUSE NOT INCLUDED IN THE SCOPE OF DELIVERY

	DATE	NAME	DOCUMENT NO.	
DFTR.	11.02.2025	FELDMEIERS	SED 117194.24 000 00	
EHKO.	11.02.2025	MUENZBERG	CHANGE NO.	SCALE
STAND	11.02.2025	KLEYN	1133298	1:5

DIMENSION IN mm EXCEPT AS NOTED

MR

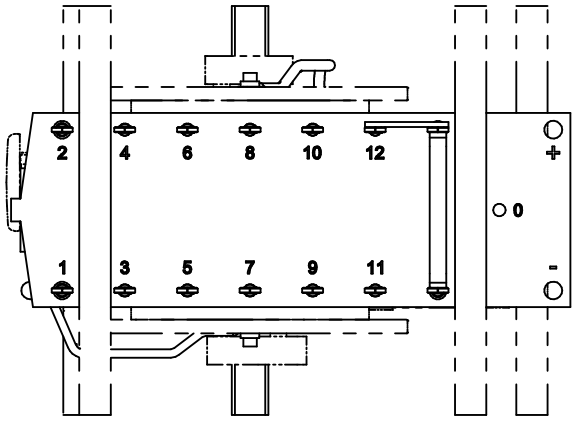
ON-LOAD TAP-CHANGER VACUTAP® VI
TAP SELECTOR BRIDGES WITHOUT CHANGE-OVER SELECTOR
DIMENSION DRAWING

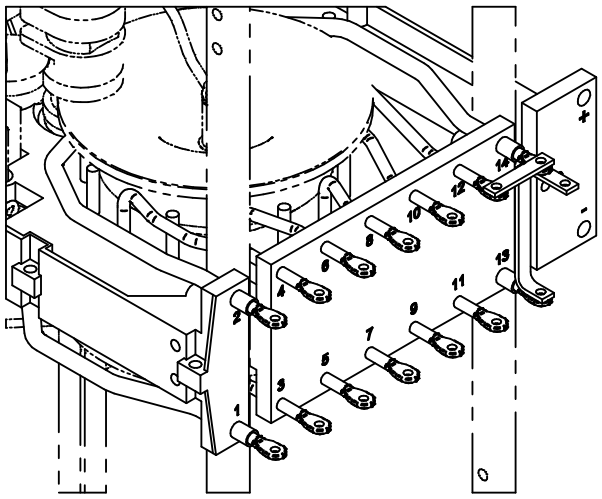
SERIAL NUMBER –
MATERIAL NUMBER 101624300M
SHEET 1 / 3

A3

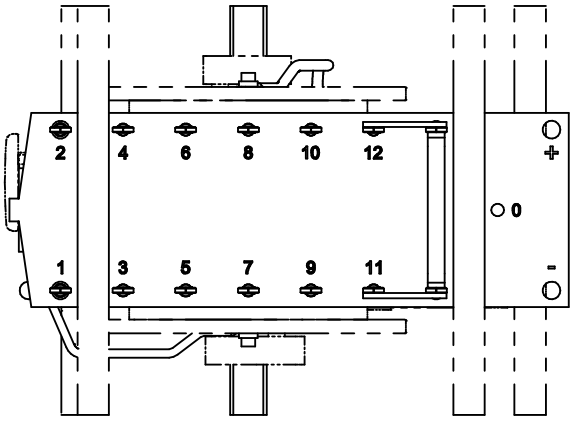
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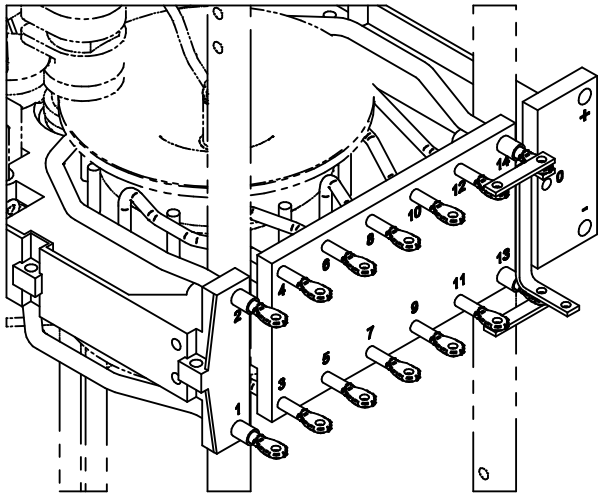
Ausführung / DESIGN: 14120



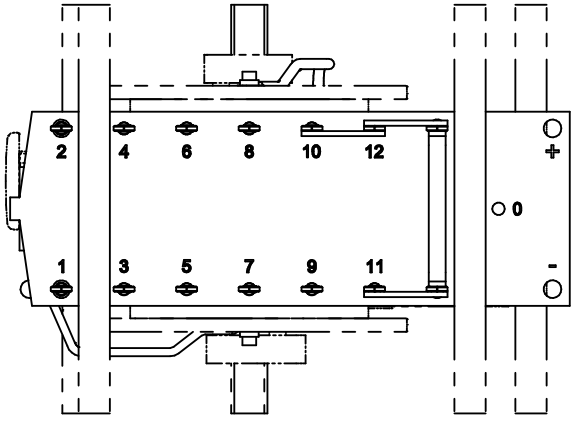


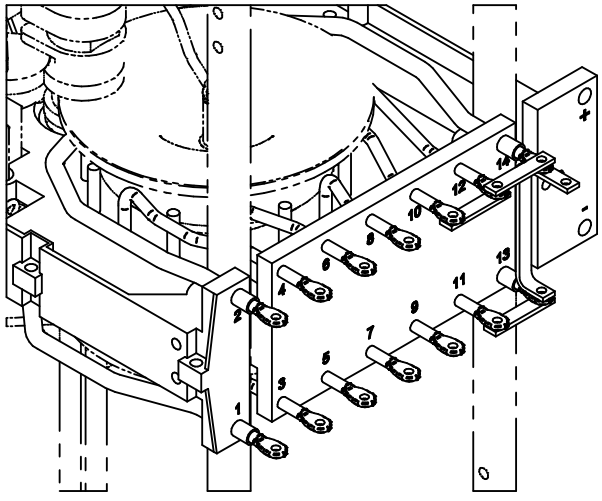
Ausführung / DESIGN: 14110





Ausführung / DESIGN: 14100





	DATE	NAME	DOCUMENT NO.	
DFTR.	11.02.2025	FELDMEIERS	SED 117719424 000 00	
CHKD.	11.02.2025	MUENZBERG	CHANGE NO.	SCALE
STAND	11.02.2025	KLEYN	1133298	1:5

DIMENSION IN mm EXCEPT AS NOTED

MR

ON-LOAD TAP-CHANGER VACUTAP® VI
TAP SELECTOR BRIDGES WITHOUT CHANGE-OVER SELECTOR
DIMENSION DRAWING

SERIAL NUMBER
–

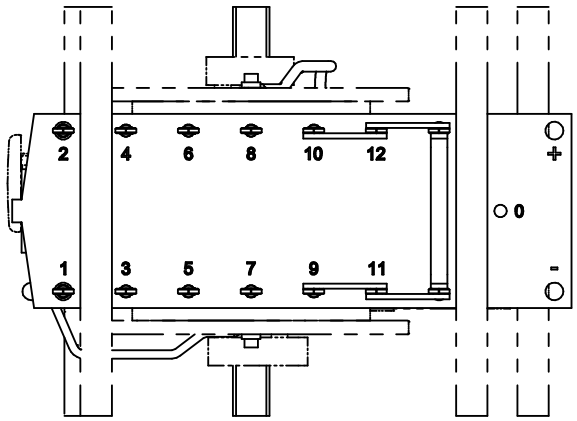
MATERIAL NUMBER 101624300M

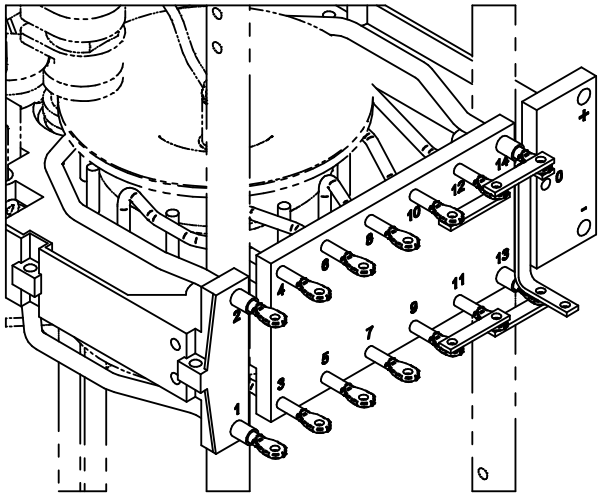
SHEET 2 / 3

A3

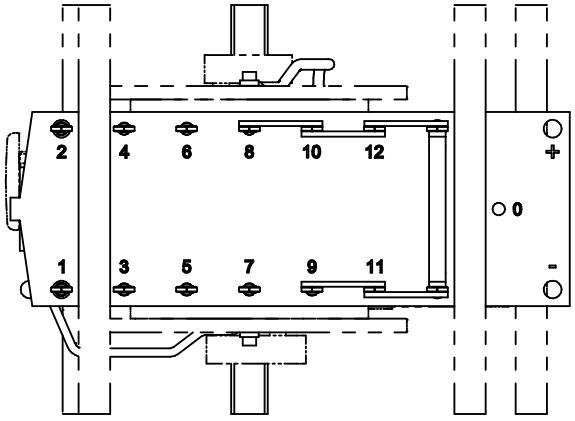
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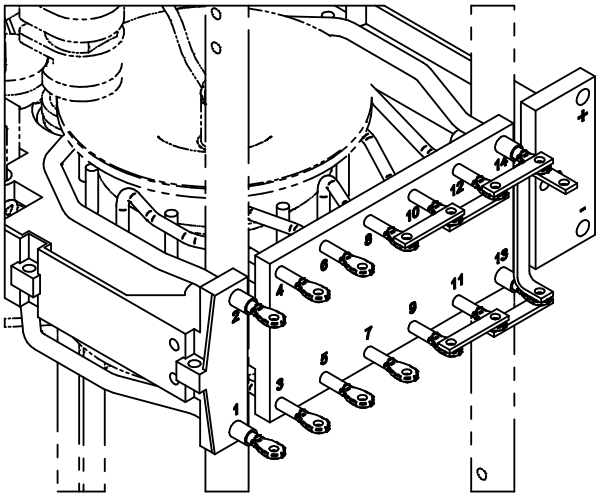
Ausführung / DESIGN: 14090



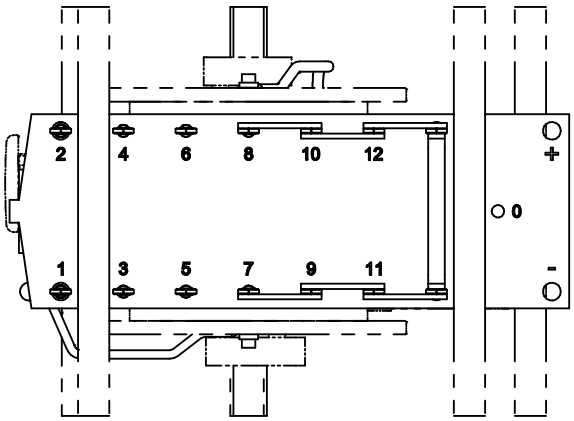


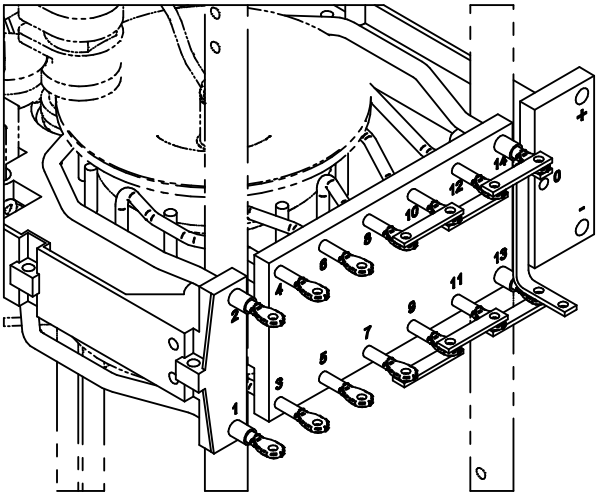
Ausführung / DESIGN: 14080





Ausführung / DESIGN: 14070





	DATE	NAME	DOCUMENT NO.	
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CHKD.	11.02.2025	MUENZBERG	CHANGE NO.	SCALE
STAND	11.02.2025	KLEYN	1133298	1:5

DIMENSION IN mm EXCEPT AS NOTED

MR

ON-LOAD TAP-CHANGER VACUTAP® VI
TAP SELECTOR BRIDGES WITHOUT CHANGE-OVER SELECTOR
DIMENSION DRAWING

SERIAL NUMBER
–

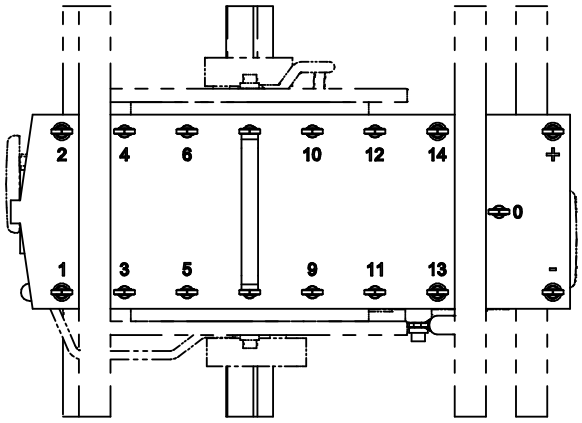
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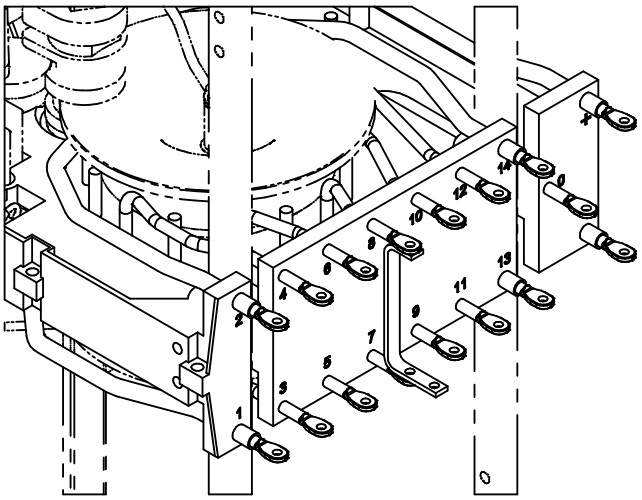
SHEET 3 / 3

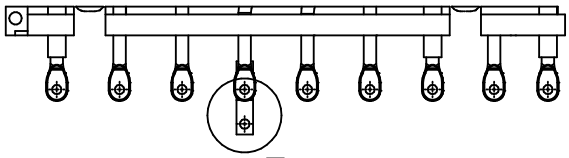
A3

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Ausführung / DESIGN: 14251W / 14251G / 14253W / 14253G

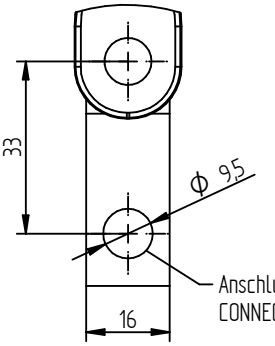






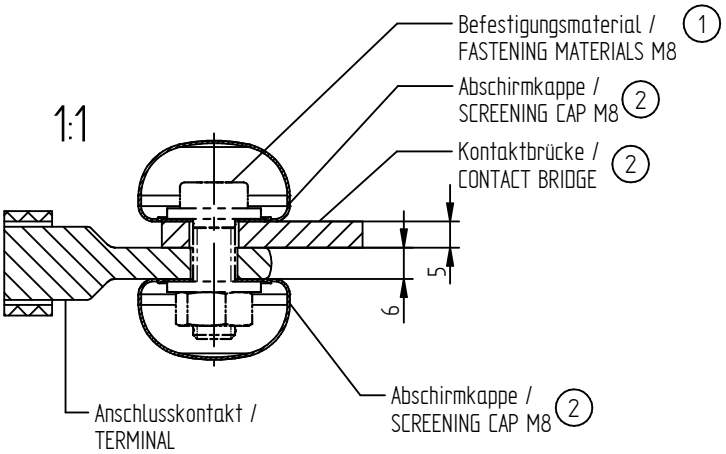
Z

Z
1:1



Befestigung Brücken schematische Darstellung / FASTENING BRIDGES SCHEMATIC VIEW

1:1



(1) Befestigungsmaterial gehört nicht zum Lieferumfang. Jede Verschraubung durch geeignete Maßnahmen gegen Lösen und Setzen sichern. / FASTENING MATERIAL IS NOT INCLUDED IN THE SCOPE OF DELIVERY.
TAKE SUITABLE MEASURES TO SECURE EACH SCREW CONNECTION AGAINST LOOSENING AND SETTING.

(2) auf Anfrage verfügbar / AVAILABLE ON REQUEST

Schraubenverbindungen nicht dargestellt, da nicht im Lieferumfang enthalten.
SCREW CONNECTIONS NOT PICTURED BECAUSE NOT INCLUDED IN THE SCOPE OF DELIVERY

Verbindungsbrücken für Ausführung 3W (1/- und 13/+) sind nicht im Lieferumfang enthalten /
CONNECTING BRIDGES FOR DESIGN 3W (1/- und 13/+) ARE NOT INCLUDED IN THE SCOPE OF DELIVERY

	DATE	NAME	DOCUMENT NO.	
DFTR.	11.02.2025	FELDMEIERS	SED 117194577 000 00	
CHKD.	11.02.2025	MUENZBERG	CHANGE NO.	SCALE
STAND	11.02.2025	KLEYN	133298	1:5

DIMENSION IN mm EXCEPT AS NOTED

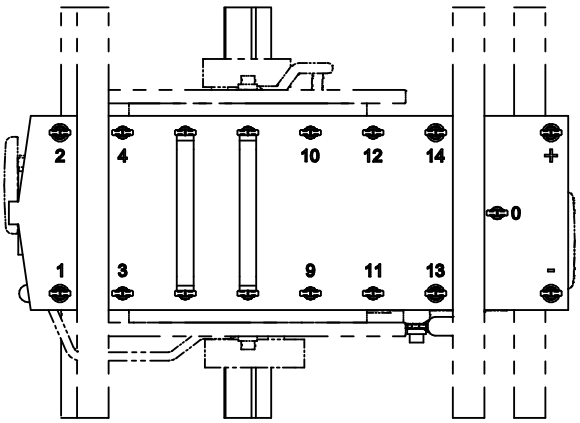
ON-LOAD TAP-CHANGER VACUTAP® VI
TAP SELECTOR BRIDGES WITH CHANGE-OVER SELECTOR
DIMENSION DRAWING

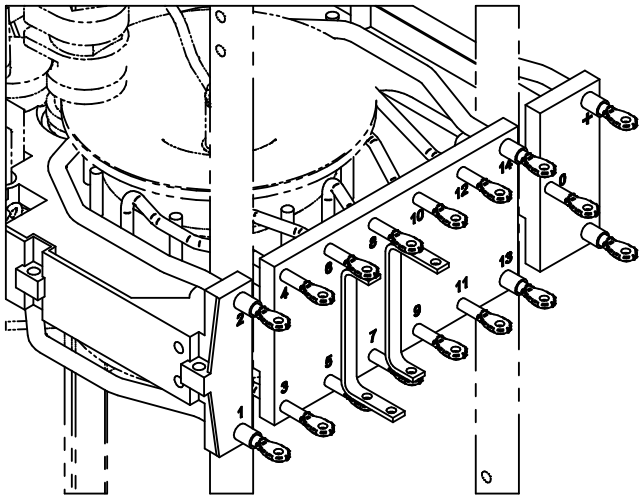
SERIAL NUMBER
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MATERIAL NUMBER 101863690M
SHEET 1 / 3

A3

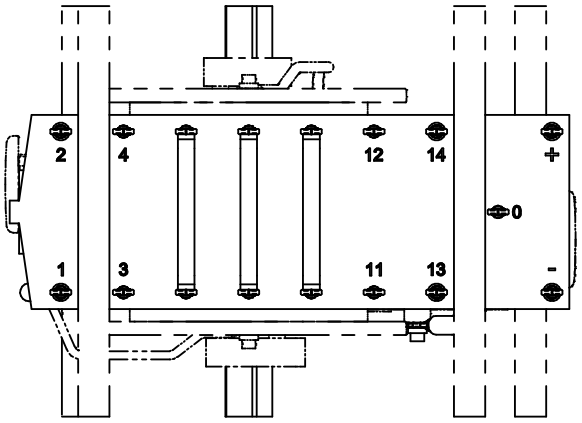
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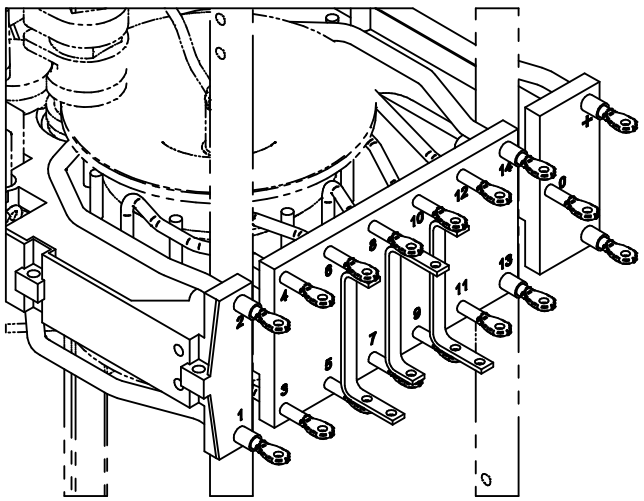
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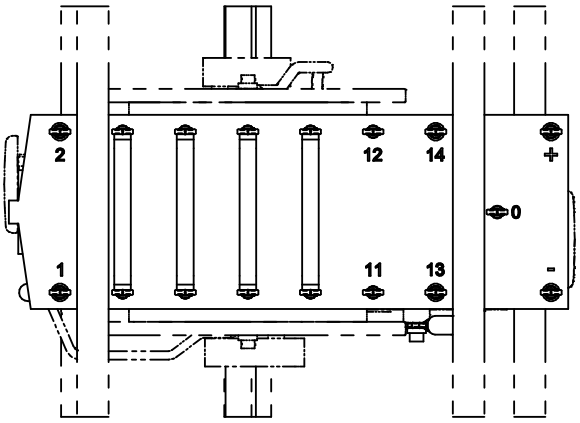


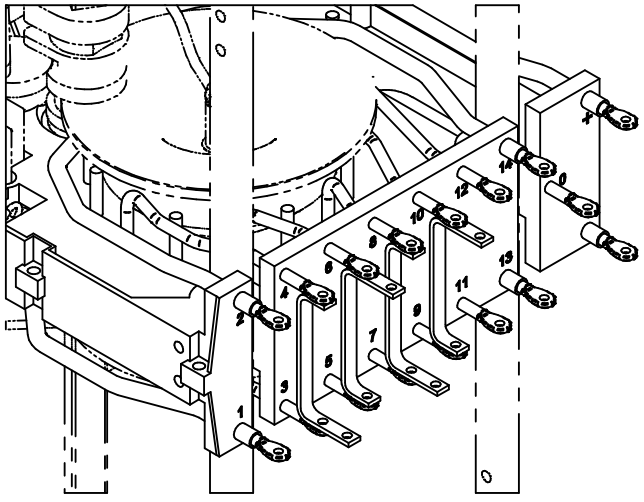
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Ausführung / DESIGN: 14191W / 14191G / 14193W / 14193G 14181W / 14181G / 14183W / 14183G





	DATE	NAME	DOCUMENT NO.	
DFTR.	11.02.2025	FELDMEIERS	SED 117194577 000 00	
CHKD.	11.02.2025	MUENZBERG	CHANGE NO.	SCALE
STAND	11.02.2025	KLEYN	1133298	1:5

DIMENSION IN mm EXCEPT AS NOTED

MR

ON-LOAD TAP-CHANGER VACUTAP® VI
TAP SELECTOR BRIDGES WITH CHANGE-OVER SELECTOR
DIMENSION DRAWING

SERIAL NUMBER
–

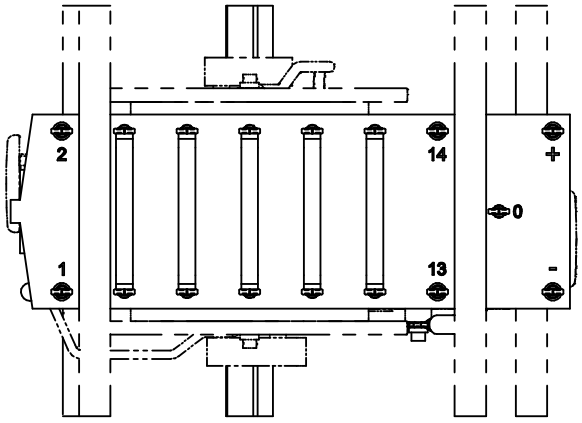
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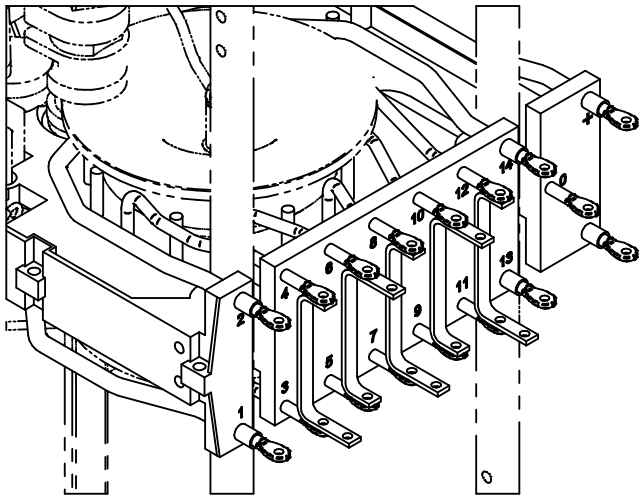
SHEET 2 / 3

A3

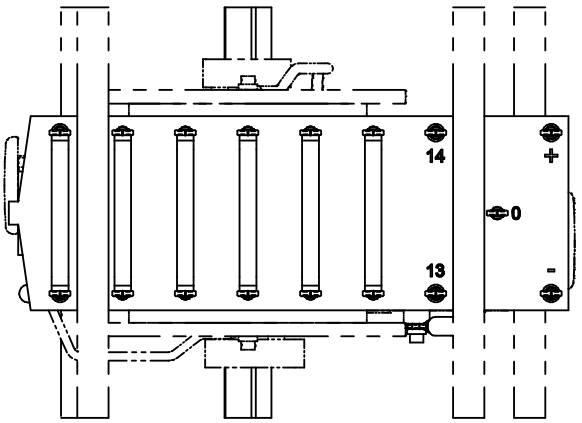
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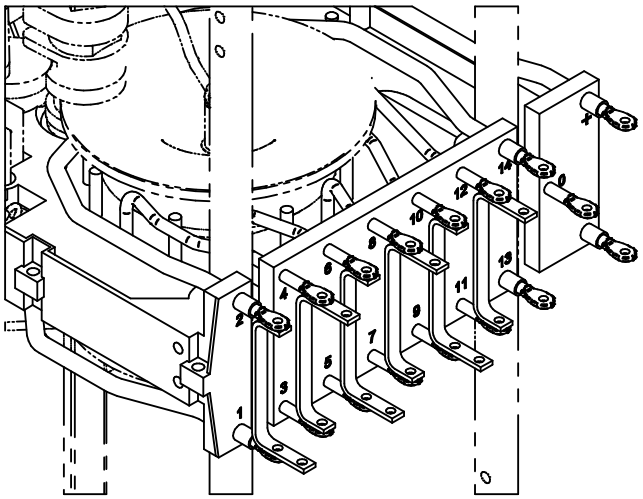
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Ausführung / DESIGN: 14151W / 14151G





	DATE	NAME	DOCUMENT NO.	
DFTR.	11.02.2025	FELDMEIERS	SED 117194577 000 00	
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STAND	11.02.2025	KLEYN	1133298	1:5

DIMENSION IN mm EXCEPT AS NOTED

MR

ON-LOAD TAP-CHANGER VACUTAP® VI
TAP SELECTOR BRIDGES WITH CHANGE-OVER SELECTOR
DIMENSION DRAWING

SERIAL NUMBER
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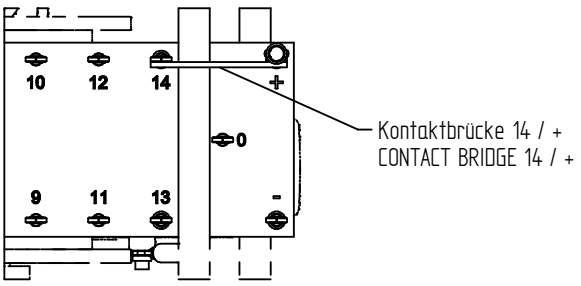
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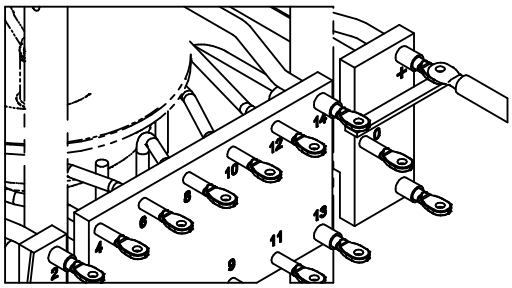
SHEET 3 / 3

A3

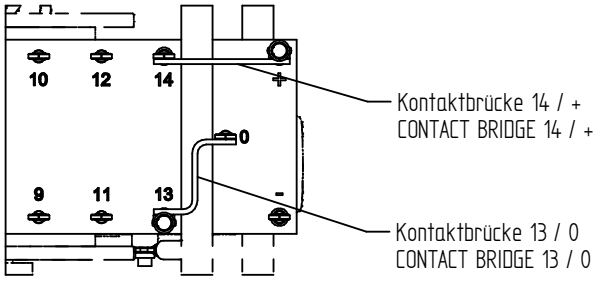
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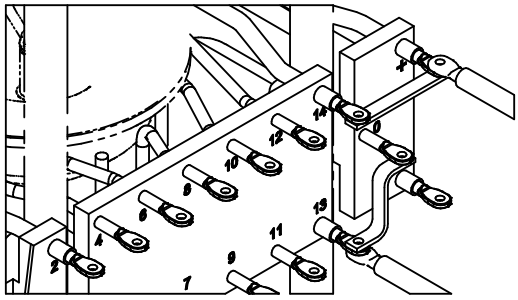
Ausführung / DESIGN: 1G



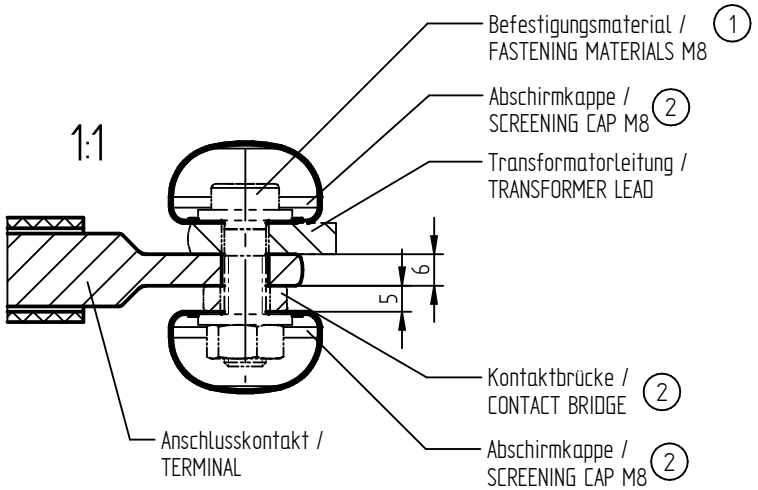


Ausführung / DESIGN: 3G





Befestigung Brücken schematische Darstellung / FASTENING BRIDGES SCHEMATIC VIEW



(1) Befestigungsmaterial gehört nicht zum Lieferumfang. Jede Verschraubung durch geeignete Maßnahmen gegen Lösen und Setzen sichern. / FASTENING MATERIAL IS NOT INCLUDED IN THE SCOPE OF DELIVERY.
TAKE SUITABLE MEASURES TO SECURE EACH SCREW CONNECTION AGAINST LOOSENING AND SETTING.

(2) auf Anfrage verfügbar / AVAILABLE ON REQUEST

Schraubenverbindungen nicht dargestellt, da nicht im Lieferumfang enthalten.
SCREW CONNECTIONS NOT PICTURED BECAUSE NOT INCLUDED IN THE SCOPE OF DELIVERY

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DFTR.	11.02.2025	FELDMEIERS	SED 117222167 000 00	
CHKD.	11.02.2025	MUENZBERG	CHANGE NO.	SCALE
STAND	11.02.2025	KLEYN	1133298	1:5

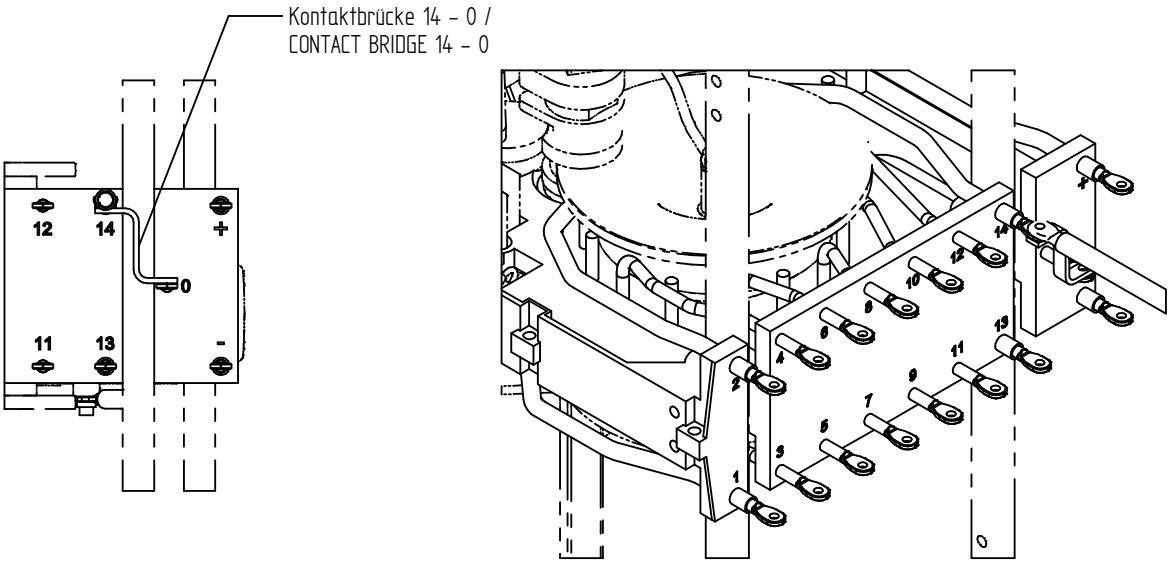
DIMENSION IN mm EXCEPT AS NOTED

MR

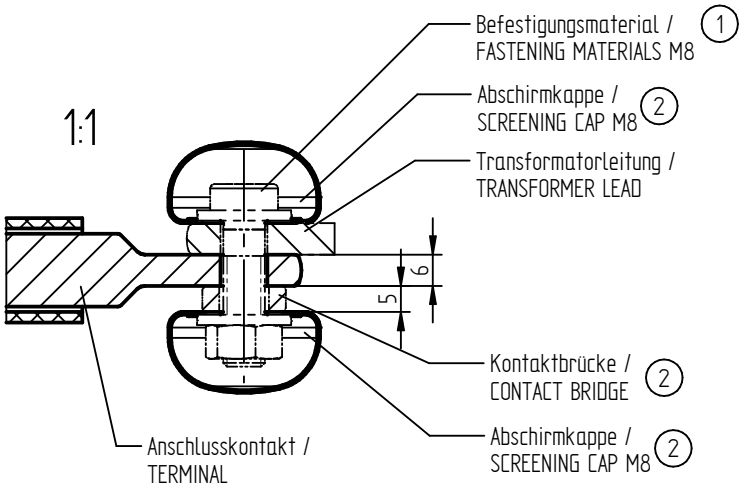
Laststufenschalter VACUTAP® VI
BRIDGES CHANGE-OVER SELECTOR 1G/3G
DIMENSION DRAWING

SERIAL NUMBER
–
MATERIAL NUMBER 101863720M
SHEET 1 / 1
A3

Ausführung / DESIGN: 1W/3W



Befestigung Brücken schematische Darstellung / FASTENING BRIDGES SCHEMATIC VIEW



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DFTR.	11.02.2025	FELDMEIERS	SED 117722081 000 00	
CHKD.	11.02.2025	MUENZBERG	CHANGE NO.	SCALE
STAND	11.02.2025	KLEYN	1133298	1:5

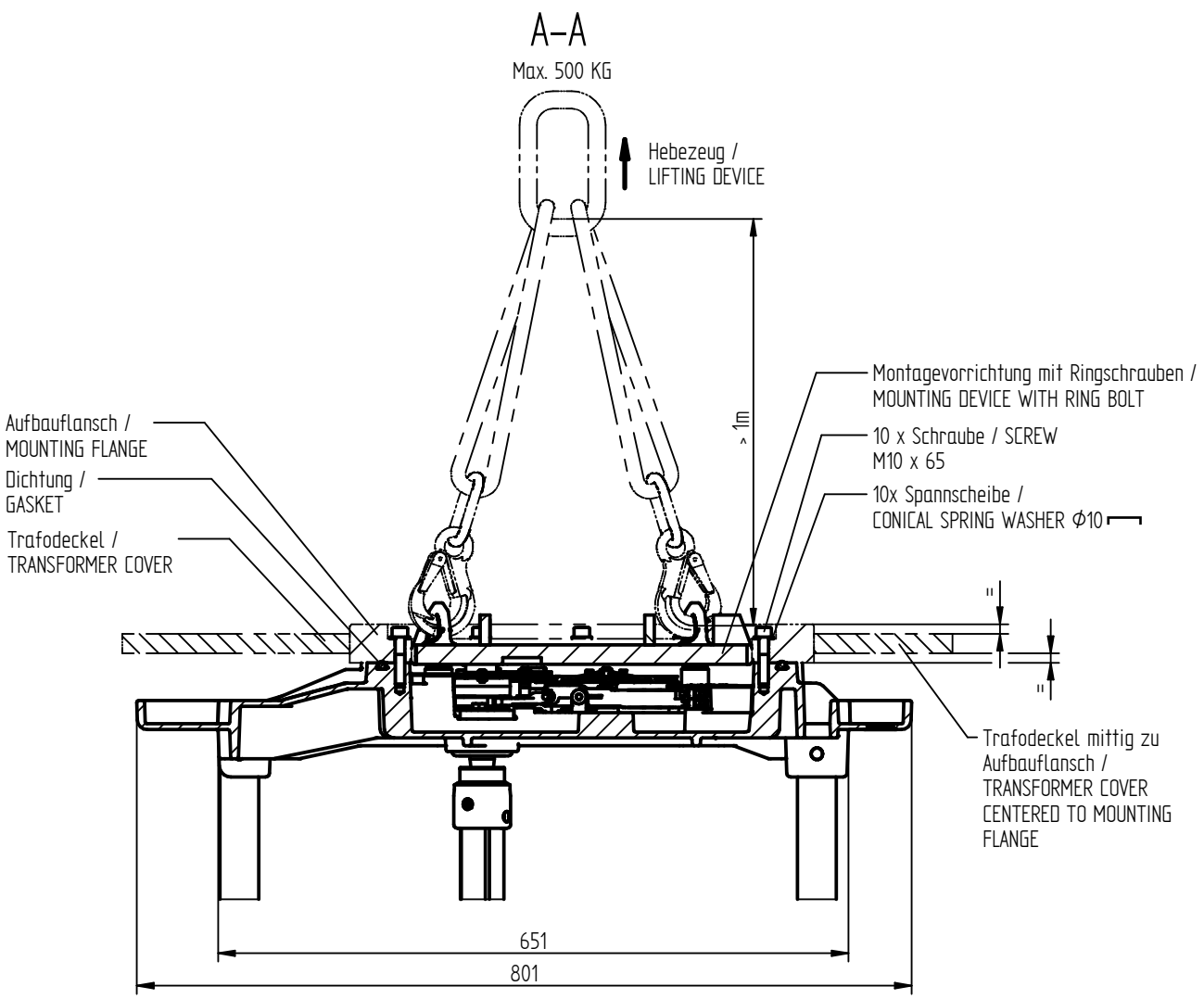
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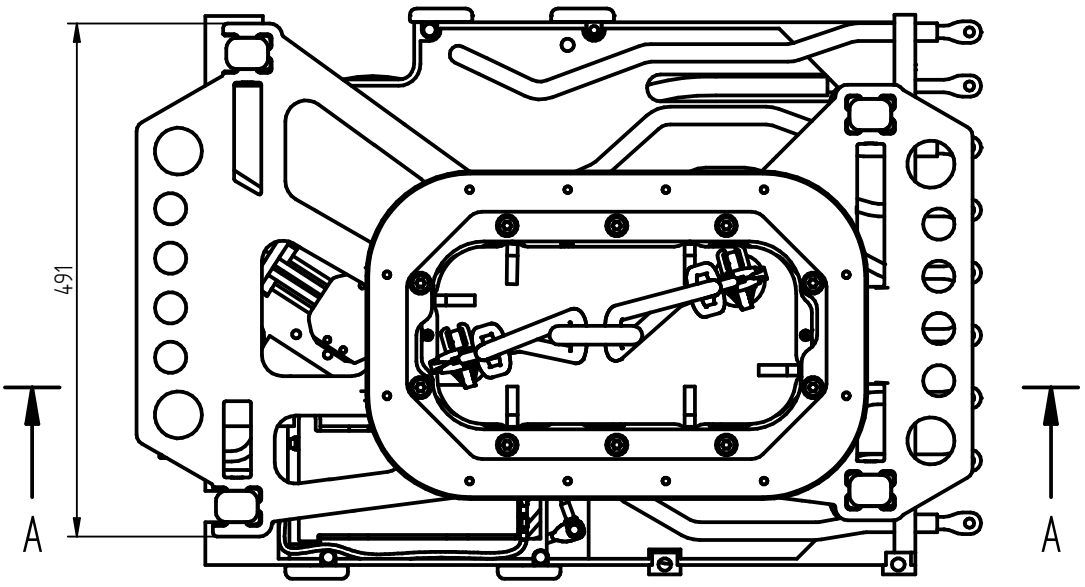
MR

ON-LOAD TAP-CHANGER VACUTAP® VI
BRIDGES CHANGE-OVER SELECTOR 1W/3W
DIMENSION DRAWING

SERIAL NUMBER
–
MATERIAL NUMBER 101863730M
SHEET 1 / 1
A3

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STAND.	11.02.2025	KLEYN	1133298	1:5

DIMENSION IN mm EXCEPT AS NOTED

MR

ON-LOAD TAP-CHANGER VACUTAP® VI
INSTALLATION STANDARD DESIGN
DIMENSION DRAWING

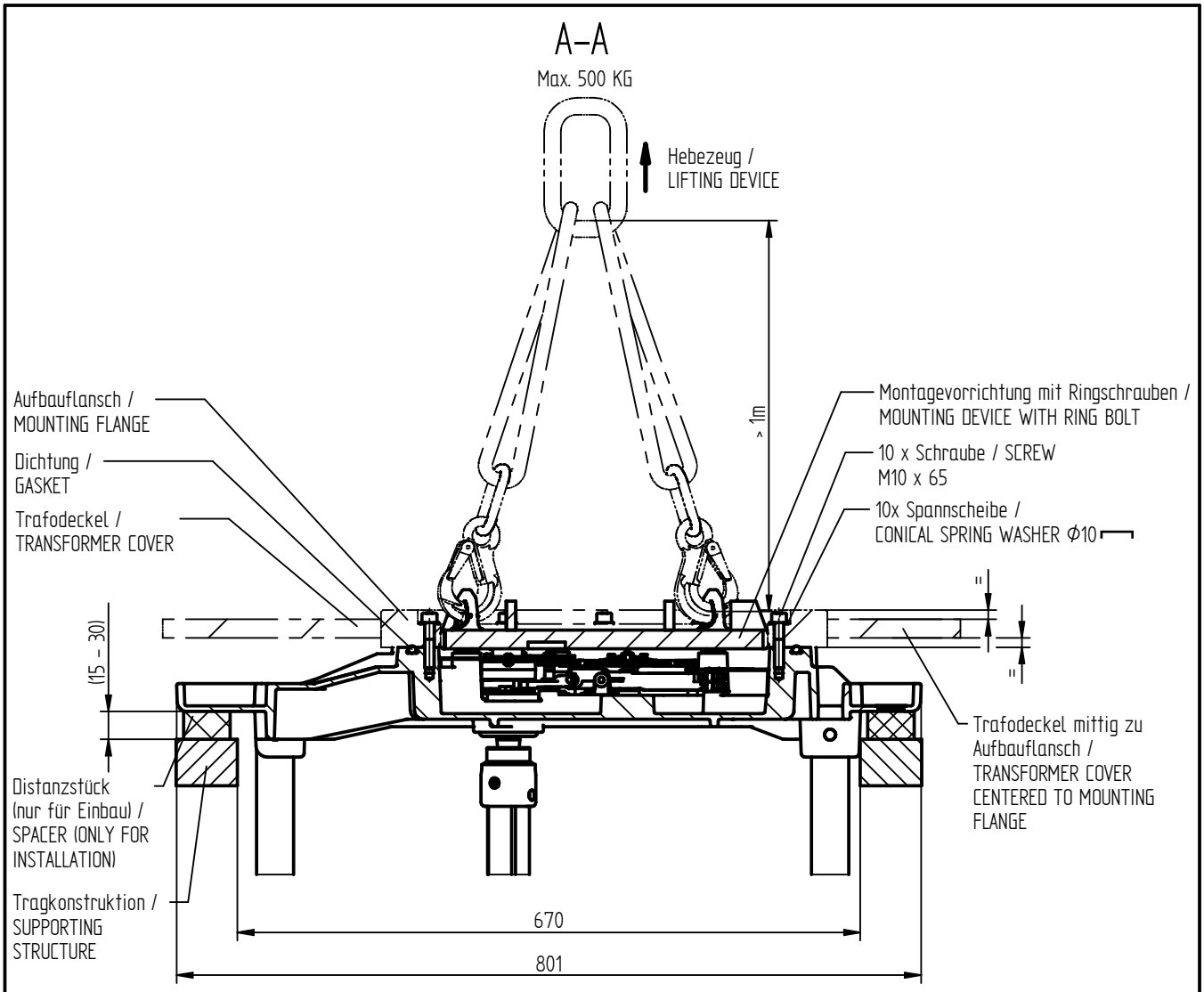
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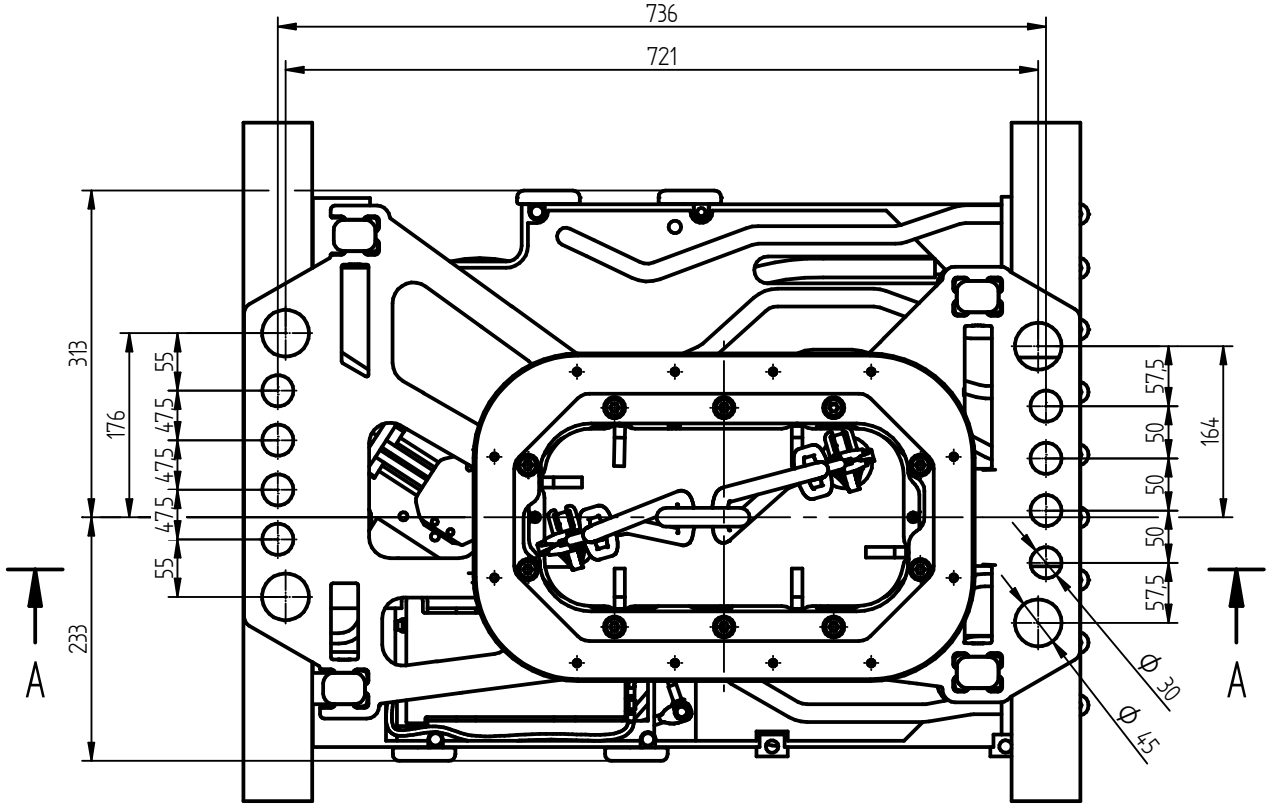
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SHEET
1 / 1

A3

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STAND	11.02.2025	KLEYN	1133298	1:5

DIMENSION IN mm EXCEPT AS NOTED

MR

ON-LOAD TAP-CHANGER VACUTAP® VI
INSTALLATION BELL-TYPE TANK
DIMENSION DRAWING

SERIAL NUMBER
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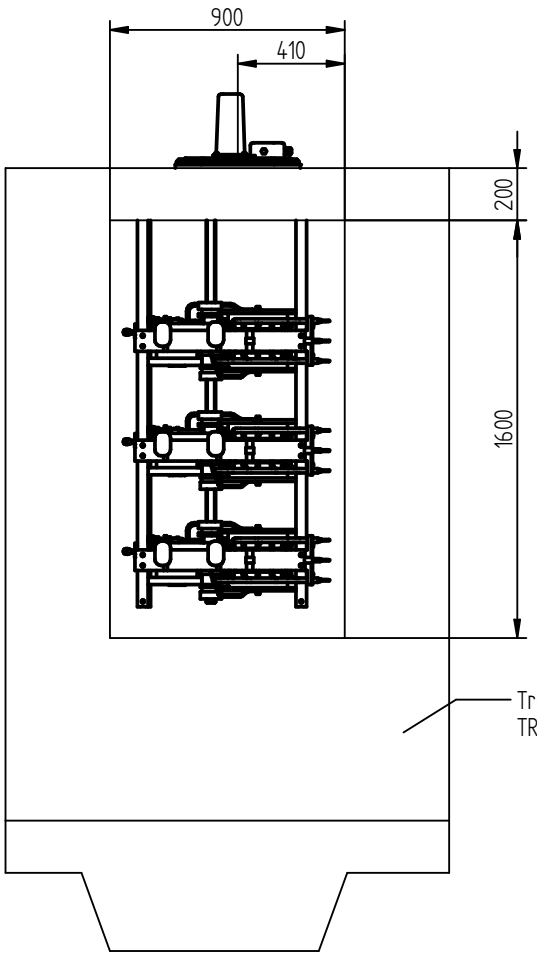
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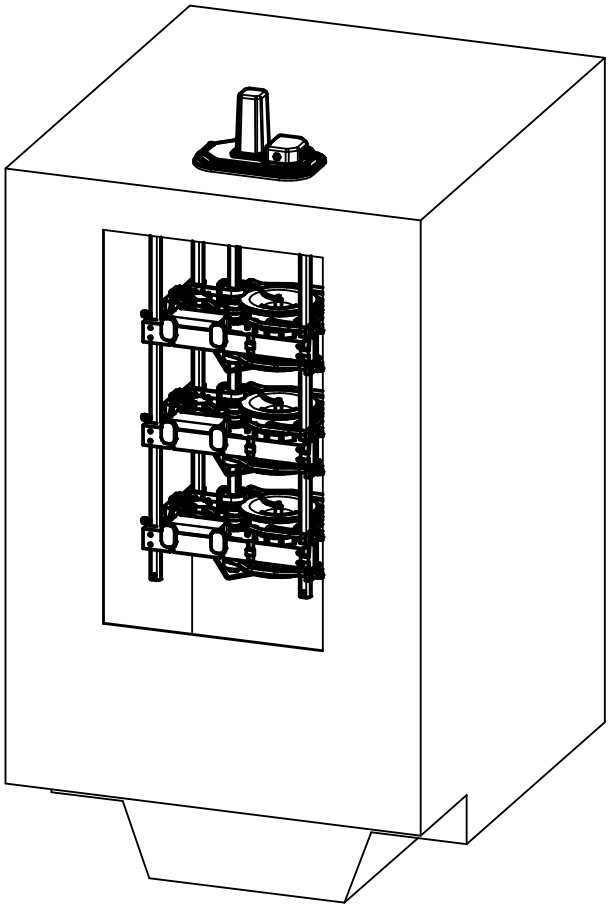
SHEET
1 / 1

A3

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RECOMMENDED MINIMUM DIMENSIONS FOR FRONTAL ACCESS OPENING





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STAND.	11.02.2025	KLEYN	1133298	1:20

DIMENSION IN mm EXCEPT AS NOTED

MR

ON-LOAD TAP-CHANGER VACUTAP® VI
DIMENSIONS ACCESS OPENING
DIMENSION DRAWING

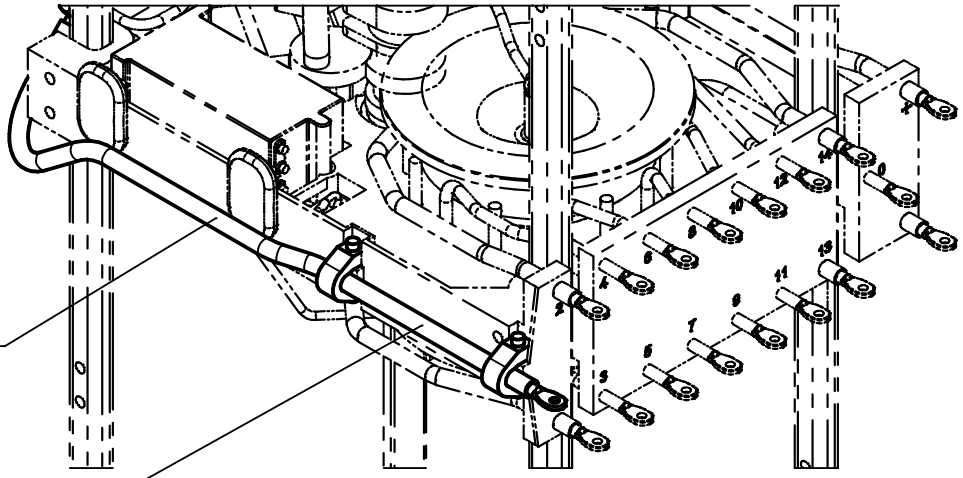
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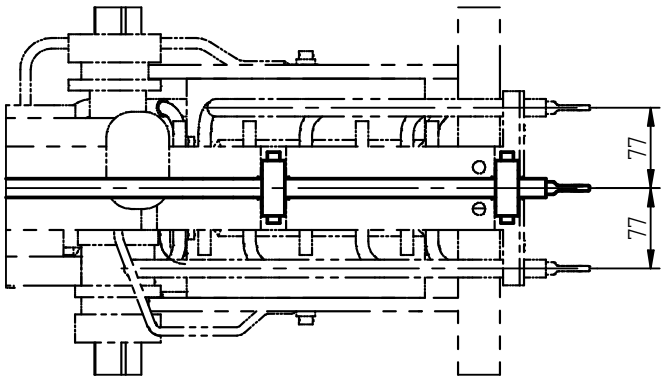
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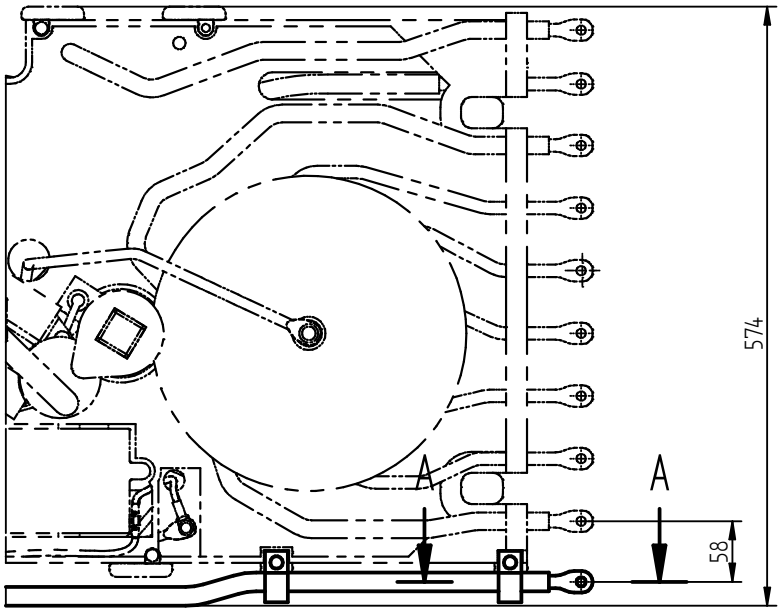
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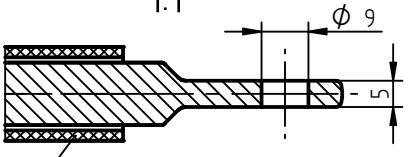
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A–A
1:1



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PAPER INSULATED

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DFTR.	11.02.2025	FELDMEIERS	SED 11723123 000 00	
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STAND	11.02.2025	KLEYN	1133298	1:5

DIMENSION IN mm EXCEPT AS NOTED

MR

ON-LOAD TAP-CHANGER VACUTAP® VI
OPTIONAL TAKE-OFF LEAD IN FRONT
DIMENSION DRAWING

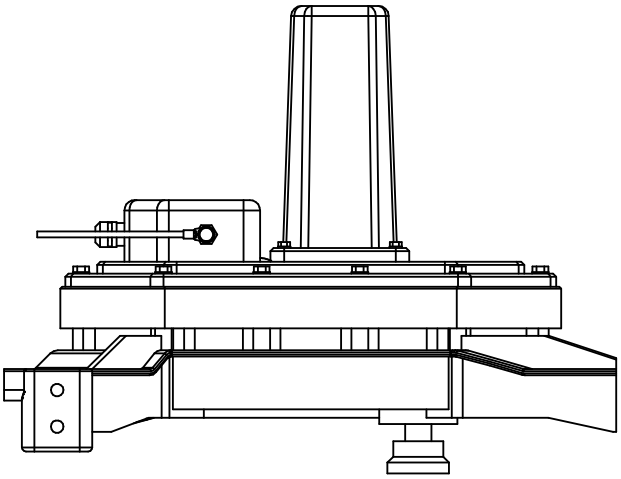
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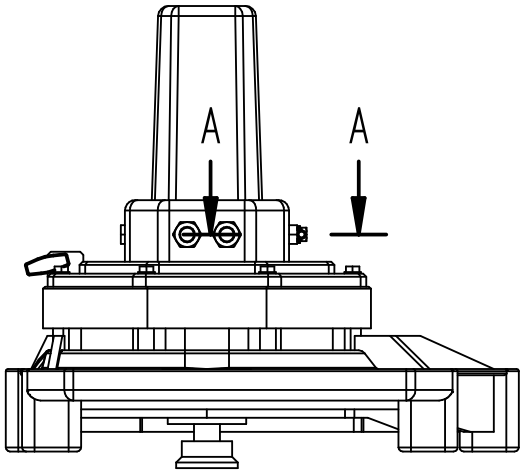
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1 / 1

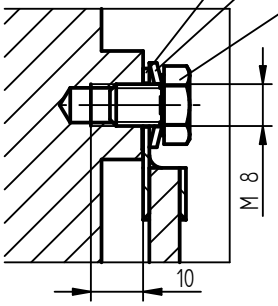
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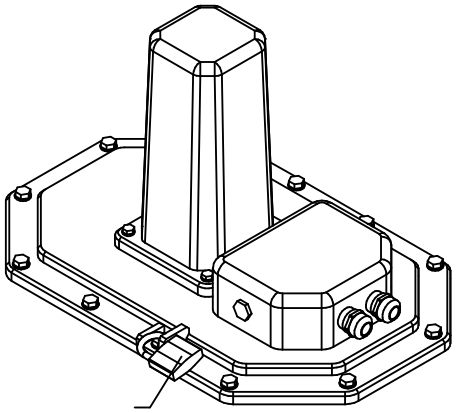
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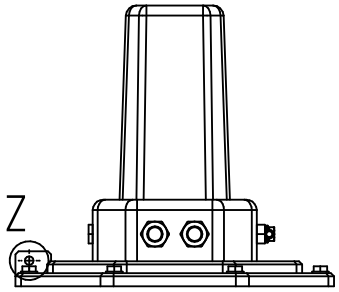
Leitung kundenseitig /
LEAD PROVIDED BY CUSTOMER

Scheibe kundenseitig /
WASHER PROVIDED BY CUSTOMER

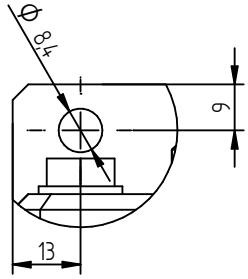
Erdungsschraube kundenseitig /
GROUNDING SCREW PROVIDED BY CUSTOMER



Vorhängeschloss
(nicht im MR-Lieferumfang enthalten) /
PADLOCK (NOT INCLUDED
IN THE MR-SCOPE OF DELIVERY)



Z
1:1



WARNUNG!
EXPLOSIONSGEFAHR!

Eine Betätigung der Handkurbel bei unter Spannung stehendem Transformator kann zu Tod, schweren Verletzungen und Sachschäden durch Überlastung des Laststufenschalters führen.

- **Niemals** diesen Laststufenschaltertyp bei unter Spannung stehendem Transformator
- mit der Handkurbel betätigen!
- Transformator vor Betätigung der Handkurbel spannungsfrei schalten!
  Vorhängeschloss zum Schutz vor unbefugtem Öffnen anbringen (Vorhängeschloss ist **nicht** im MR-Lieferumfang enthalten)

WARNING!
DANGER OF EXPLOSION!

OPERATING THE HAND CRANK WITH THE TRANSFORMER ENERGIZED CAN LEAD TO DEATH, SERIOUS INJURY AND MATERIAL DAMAGE DUE TO OVERLOAD OF THE ON-LOAD TAP-CHANGER

- **NEVER** ACTUATE THIS TYPE OF ON-LOAD TAP-CHANGER WITH THE HAND CRANK IF
- THE TRANSFORMER IS ENERGIZED!
- DE-ENERGIZE THE TRANSFORMER BEVORE USING THE HAND CRANK!
  ATTACH **PADLOCK** TO PROTECT AGAINST UNAUTHORIZED OPENING (PADLOCK IS **NOT** INCLUDED IN THE MR SCOPE OF DELIVERY)

	DATE	NAME	DOCUMENT NO.	
DFTR.	10.02.2025	FELDMEIERS	SED 11722515 000 00	
CHKD.	10.02.2025	MUENZBERG	CHANGE NO.	SCALE
STAND	11.02.2025	KLEYN	1133298	1:5

DIMENSION IN mm EXCEPT AS NOTED

ON-LOAD TAP-CHANGER VACUTAP® VI
PROTECTION / GROUNDING MOTOR COVER
DIMENSION DRAWING

SERIAL NUMBER
–

MATERIAL NUMBER
101863670M

SHEET
1 / 1

A3

Maschinenfabrik Reinhausen GmbH
Falkensteinstrasse 8
93059 Regensburg
Germany
+49 941 4090-0
info@reinhausen.com
reinhausen.com

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We reserve the right to make changes without notice.
12219500/01 EN - VACUTAP® VI Installation and commissioning instructions - 01/26
Maschinenfabrik Reinhausen GmbH 2026

THE POWER BEHIND POWER.